

Quantum Field Theory

Natural Unit

c = speed of light, $\hbar$ = Planck constant
Special Relativity Quantum mechanics } These are ugly numbers

But the number means nothing without unit

Also the number can change depending on the system of units

eg. $c = 3 \times 10^8 \text{ m/s} = 10^{10} \times 10^8 \text{ km/hr}$

$$1 \text{ m/s} = 1 \times 10^{-3} \cdot 10^3 \text{ m/s} = \frac{1}{10^3} \cdot 10^3 \text{ m/s} \xrightarrow{\text{km}} \text{km/s}$$
$$= 10^{-3} \text{ km/s} = 10^{-3} \times 3600 \text{ km} \xrightarrow{3600 \text{ s} \rightarrow \text{hr}} \text{hr} = 3.6 \text{ km/hr}$$

$$1 \text{ hr} = 60 \times 60 \text{ s} = 3600 \text{ s}$$

How about we use a system of unit such that the

number of $c = 1$ and c has no dimension!

This is not so strange!

Unit is a made up thing

$\Rightarrow$ All units are made up

$\Rightarrow$ We stick to the value that is a constant of the universe

$\therefore$ If we choose $c = 1$ (with no unit)

All units we use before are related in this system.

$$c = 1 = 3 \times 10^8 \text{ m/s} \quad \leftarrow \text{SI} \quad 1 \text{ m} = \frac{1}{3 \times 10^8} \text{ s}$$
$$\Rightarrow 3 \times 10^8 \text{ m} = 1 \text{ s}$$

natural unit

(in natural unit)

time $\sim$ length $\Rightarrow$ special relativity.

spacetime diagram

distance $\rightarrow x$
 $ct = t$

What about $\hbar = 1.05 \times 10^{-34} \text{ J}\cdot\text{s} \Rightarrow \hbar\omega = E$

In natural unit, we also set $\hbar = 1$ (with no unit)

$$\therefore \underbrace{1}_{\text{natural unit}} = \underbrace{1.05 \times 10^{-34}}_{\text{SI}} \text{ J}\cdot\text{s} \Rightarrow \underbrace{1 \text{ s} \sim 10^{34} \text{ J}^{-1}}_{\text{(natural unit.)}}$$

$$\downarrow \quad \quad \quad \downarrow$$

$$1 \text{ J} = 10^{-34} \text{ s}^{-1}$$

$$\Rightarrow \text{time} \sim \frac{1}{\text{Energy}} \quad \text{or} \quad \text{Energy} \sim \frac{1}{\text{time}} \sim \frac{1}{\text{length}}$$

★ All units are equivalent to energy unit

$$L\checkmark, T\checkmark, m\checkmark \rightarrow E^2 = m^2 c^4 + p^2 c^2 \Rightarrow E = mc^2$$

$$E = m$$

How do we use it?

1) In theoretical calculation, we simply drop c & $\hbar$

and if we want to communicate to other people

we can revert back

2). In actual calculation, we relate many quantities

together, Ex $1 \text{ m} = \frac{1}{3 \times 10^8} \text{ s} \Rightarrow 1 \text{ m} = 1 \times \frac{c}{c} \text{ m}$

$$\begin{aligned} \underline{\text{Ex}} \quad 1 \text{ eV} &= 1.6 \times 10^{-19} \text{ J} = 1.6 \times 10^{-19} \times \frac{1}{\hbar} \text{ J} \\ &= \frac{1.6 \times 10^{-19}}{1.05 \times 10^{-34}} \text{ s}^{-1} \\ &= 1.6 \times 10^{15} \frac{1}{\text{s}} \end{aligned}$$

$$\begin{aligned} &= 1 \times \frac{1}{3 \times 10^8 \text{ m/s}} \text{ m} = \frac{1}{3 \times 10^8} \text{ s} \\ &\frac{1}{\hbar} \rightarrow 1.05 \times 10^{-34} \text{ J}\cdot\text{s} \end{aligned}$$

$$\frac{1}{\text{J}} = 1.6 \times 10^{-19} \frac{1}{\text{eV}}$$

$$\underline{\text{Ex}} \quad 1 \text{ m} = \frac{1}{3 \times 10^8} \text{ s} = \frac{1}{3 \times 10^8} \times \frac{1}{1.05 \times 10^{-34}} \frac{1}{\text{J}} = \frac{1 \times 1.6 \times 10^{-19}}{(3 \times 10^8)(1.05 \times 10^{-34})} \frac{1}{\text{eV}}$$

$$= 0.5 \times 10^7 \frac{1}{\text{eV}}$$

$$1 \text{ m} = 1 \times \frac{c}{c} \text{ m} = \frac{1 \text{ m}}{3 \times 10^8 \text{ m/s}} = 1 \dots \dots \text{J} \rightarrow \frac{1}{\text{eV}}$$

* We also use the prefix of eV ($1 \text{ keV} = 10^3 \text{ eV}$, $1 \text{ MeV} = 10^6 \text{ eV}$, $1 \text{ GeV} = 10^9 \text{ eV}$, $1 \text{ TeV} = 10^{12} \text{ eV}$, $1 \text{ PeV} = 10^{15} \text{ eV}$, ...) ($1 \text{ meV} = 10^{-3} \text{ eV}$, $1 \mu\text{eV} = 10^{-6} \text{ eV}$)

The final aspect of natural unit is the power of dimensional analysis!

we have an advantage that everything is in eV

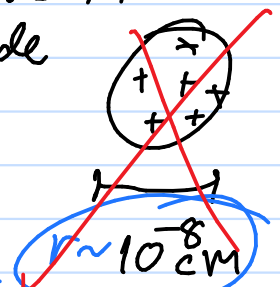
$\Rightarrow$ Often in dimensional analysis, this is very easy.

$\hookrightarrow$ We can get the answer before the actual calculation

Ex Rutherford's scattering



suppose we believe in the pudding model



$$v \sim 0.03 c$$

dimensional analysis

$\hookrightarrow$ natural unit



Δp = transferred momentum from nuclei

$$\theta \sim \frac{\Delta p}{p}$$

$$E^2 = m^2 + p^2$$

nucleus $\sim 10^{-13} \text{ cm}$

$$\boxed{\text{eV} \sim \frac{1}{r}} \sim 10^6 \text{ GeV}$$

$$p \text{ of } \alpha\text{-particle} = ? = mv = (4 \text{ GeV})(0.03)$$

$$2p + 2n = 1\alpha$$

$$m \text{ of proton} = 1 \text{ GeV}/c^2 = 1 \text{ GeV}$$

$$\therefore \theta \sim \frac{\Delta p}{p} \sim \frac{10^{-6} \text{ GeV}}{0.12 \text{ GeV}} \sim 10^{-5} \text{ rad.}$$

$$3.14 \text{ rad} = 180^\circ \Rightarrow 1 \text{ rad} \sim 60^\circ$$

If we believe in pudding model, the deflecting angle must be small

$\Rightarrow$ experiment shows the angle of $\sim 180^\circ \sim 3 \text{ rad}$

Special Relativity (SR)

→ Time is relative / not absolute

→ Fast moving object $v \sim 1$ * natural unit.

Principles of SR

1). Invariance of the speed of light: $c=1$ remains the same in any inertial frame.

2). Relativity Principle: Physical laws remain the same in any inertial frame.

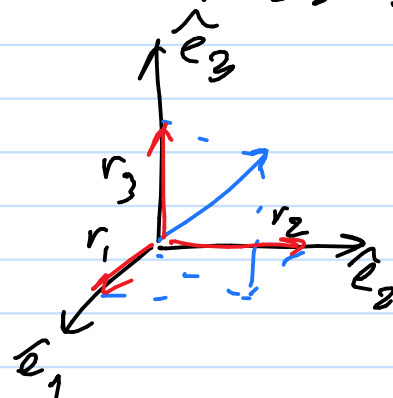
Notation in SR

3D vectors: $\vec{r}$

We can use basis $\{\hat{e}_1, \hat{e}_2, \hat{e}_3\} \Rightarrow \vec{r} = r_1 \hat{e}_1 + r_2 \hat{e}_2 + r_3 \hat{e}_3$

$\Rightarrow$ orthonormal, ... $\nearrow$

↳ Cartesian, Spherical, Cylindrical...



We can also write

$$\vec{r} = \sum_{i=1}^3 r_i \hat{e}_i$$

↳ writing this is a waste of time.!

* If we are being careful, the i index should be enough to tell that the summation is there

$\Rightarrow$ Einstein Summation Convention

$$\vec{r} = r_i \hat{e}_i$$

When we see repeating index $\Rightarrow$ Assume summation

⇒ Einstein Summation Convention

$$\vec{r} = r_i \hat{e}_i$$

When we see repeating index
⇒ Assume summation

* Note that

$$\sum_{i=1}^3 r_i \hat{e}_i = \sum_{j=1}^3 r_j \hat{e}_j$$

$$\boxed{r_i \hat{e}_i = r_j \hat{e}_j}$$

the repeating index
⇒ is called "dummy"
index

"We can always change"

* Kronecker delta δ_{ij}

$$\delta_{11} = 1, \delta_{12} = 0, \delta_{13} = 0, \delta_{22} = 1, \delta_{33} = 1, \dots = 0$$

$$\delta_{ij} = \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases}$$

$$\delta_{ii} = \sum_i \delta_{ii} = \delta_{11} + \delta_{22} + \delta_{33} = 3$$

$$\begin{aligned} \hat{e}_1 \cdot \hat{e}_3 &= 0 \\ \hat{e}_1 \cdot \hat{e}_2 &= 0 \\ \hat{e}_1 \cdot \hat{e}_1 &= 1 \end{aligned}$$

orthonormal basis

Consider the dot product of 3D vectors

$$\vec{A} = A_i \hat{e}_i, \vec{B} = B_j \hat{e}_j, \hat{e}_i \cdot \hat{e}_j = \delta_{ij}$$

$$\vec{A} \cdot \vec{B} = A_i \hat{e}_i \cdot B_j \hat{e}_j \quad (\neq A_i \hat{e}_i \cdot B_i \hat{e}_i)$$

$$= A_i B_j (\hat{e}_i \cdot \hat{e}_j) = A_i B_j \delta_{ij} = A_i \delta_{ij} B_j$$

The final form is simple

$$\vec{A} \cdot \vec{B} = \sum_i \sum_j A_i \delta_{ij} B_j = \sum_i A_i \cdot 1 \cdot B_i + \sum_i A_i \delta_{ij} B_j + \dots$$

$\xrightarrow{j=i} = A_i B_i$

We can also write this in a matrix form!

$$\vec{A} = A_i \hat{e}_i \sim \underbrace{A_i}_{\text{component}} \rightarrow \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix}$$

$i = \text{free index}$

How about

$$M_{ij} A_j = \sum_j M_{ij} A_j$$

$j = \text{dummy index}$
 $i = \text{free index}$
 $\hookrightarrow \text{we can choose}$

$$= \begin{pmatrix} M_{1j} A_j \\ M_{2j} A_j \\ M_{3j} A_j \end{pmatrix} = \begin{pmatrix} M_{11} A_1 + M_{12} A_2 + M_{13} A_3 \\ M_{21} A_1 + M_{22} A_2 + M_{23} A_3 \\ M_{31} A_1 + M_{32} A_2 + M_{33} A_3 \end{pmatrix}$$

$$= \begin{pmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix}$$

Transpose

$$A_i B_i = A_1 B_1 + A_2 B_2 + A_3 B_3 = (A_1 \ A_2 \ A_3) \begin{pmatrix} B_1 \\ B_2 \\ B_3 \end{pmatrix}$$

$$\therefore \vec{A} \cdot \vec{B} = A_i \delta_{ij} B_j = (A_1 \ A_2 \ A_3) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} B_1 \\ B_2 \\ B_3 \end{pmatrix}$$

Rotation (3D) $\rightarrow$ linear transformation

$\Rightarrow$ vector transformation such that the length of

vectors are the same.



$$\vec{A} \rightarrow \vec{A}' = \hat{R} \vec{A} = A'_i \hat{e}_i = R_{ij} A_j \hat{e}_i$$

linear operator

$$A'_i = R_{ij} A_j$$

linear combination of A_i
 $R_{11} A_1 + R_{12} A_2 + \dots = A'_1$

$$A'_i = R_{ij} A_j \Rightarrow \underbrace{\begin{pmatrix} A'_1 \\ A'_2 \\ A'_3 \end{pmatrix}}_{\substack{= A \\ \hookrightarrow \text{column vector}}} = \underbrace{\begin{pmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{pmatrix}}_{\substack{\tilde{R} \\ \hookrightarrow \text{matrix}}} \underbrace{\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix}}_{\substack{= A \\ \hookrightarrow \text{column vector}}} \\ A' = \tilde{R} A$$

Now, let's impose the invariance! Rotation $\Rightarrow$ length is the same!

$$|\vec{A}| \equiv \sqrt{\vec{A} \cdot \vec{A}} \Rightarrow \boxed{\vec{A} \cdot \vec{A} = \vec{A}' \cdot \vec{A}'} \Rightarrow \text{length is invariant}$$

$$\underbrace{(A_1 \ A_2 \ A_3)}_{A^T} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\tilde{I}} \underbrace{\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix}}_A = \underbrace{(A'_1 \ A'_2 \ A'_3)}_{\substack{\text{transpose} \\ \begin{pmatrix} A'_1 \\ A'_2 \\ A'_3 \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix}}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A'_1 \\ A'_2 \\ A'_3 \end{pmatrix}$$

$$= \underbrace{(A_1 \ A_2 \ A_3)}_{A^T} \underbrace{\begin{pmatrix} R \\ R \\ R \end{pmatrix}}_{\tilde{R}^T} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\tilde{I}} \underbrace{\begin{pmatrix} R \\ R \\ R \end{pmatrix}}_{\tilde{R}} \underbrace{\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix}}_A$$

Matrix equation:

$$\boxed{A^T \tilde{I} A} = \boxed{A^T \tilde{R}^T \tilde{I} \tilde{R} A}$$

(rotation matrix)
orthogonal matrix

$$\Rightarrow \text{The condition for rotation matrix is } \boxed{\tilde{I} = \tilde{R}^T \tilde{I} \tilde{R}} \uparrow \text{Orthogonal condition.}$$

$\Rightarrow$ The condition for rotation matrix is $\tilde{\mathbb{I}} = \tilde{R}^T \tilde{\mathbb{I}} \tilde{R}$ Orthogonal condition.

or $\tilde{\mathbb{I}} = \tilde{R}^T \tilde{R}$ because $\tilde{\mathbb{I}} \tilde{X} = \tilde{X}$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \dots \end{pmatrix} = \begin{pmatrix} \dots \end{pmatrix}$$

In the matrix multiplication,

$$\tilde{R}^T = \tilde{R}^{-1}$$

In component form, we have

$$\vec{A} \cdot \vec{A}' = \vec{A} \cdot \vec{A} \quad \& \quad A_i' = R_{ij} A_j$$

$$A_i' \delta_{ij} A_j' = A_k \delta_{kl} A_l$$

$$A_k \delta_{kl} A_l = (R_{im} A_m) \delta_{ij} (R_{jn} A_n)$$

dummy $\rightarrow k$ $\rightarrow l$

$$= A_m R_{im} \delta_{ij} R_{jn} A_n$$

dummy $\rightarrow k$ $\rightarrow l$

$$A_k \delta_{kl} A_l = A_k R_{ik} \delta_{ij} R_{jl} A_l$$

so that the indices works for both side of

Orthogonal condition: $\delta_{kl} = R_{ik} \delta_{ij} R_{jl}$

$\hookrightarrow j=i$

or simply: $\delta_{kl} = R_{ik} R_{il} \Rightarrow (\tilde{A} \tilde{B})_{kl} = A_{ki} B_{il}$

$\hookrightarrow (\tilde{R}^T)_{ki} R_{il}$

Let's go to SR Can we find the vector transformation (change of the inertial frame of reference) such that the principle of SR holds!

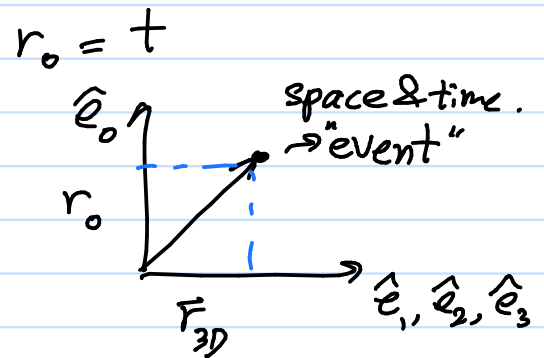
3D is not enough $\Rightarrow$ 4D (t, x, y, z)

Let's define 4D vector!

$$\vec{r} = r^0 \hat{e}_0 + r^1 \hat{e}_1 + r^2 \hat{e}_2 + r^3 \hat{e}_3$$

$$\vec{r} = r^\mu \hat{e}_\mu, \quad \mu = \textcircled{0}, 1, 2, 3$$

$\nearrow$ time.



1). we use Greek indices

$(\mu, \nu, \rho, \sigma, \kappa, \alpha, \delta, \delta, \dots)$ to represent

index in 4 dimension & Latin indices for 3D

$i, j, k, \dots = 1, 2, 3$

2). Now we have to be careful on the position of indices $\Rightarrow$ note that we use the up stair in the definition of vector $r^\mu \Rightarrow$ contravariant vector

$\hookrightarrow$ transform differently than the basis

Next, we define the "length" of vector in 4D

We cannot use the standard length for SR

$\hookrightarrow$ Euclidean Space

$\Rightarrow$ We use instead the Minkowski space with the new definition of "length"

$$\vec{r} = r^\mu \hat{e}_\mu \Rightarrow \vec{r} \cdot \vec{r} \equiv (r^0 \ r^1 \ r^2 \ r^3) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} r^0 \\ r^1 \\ r^2 \\ r^3 \end{pmatrix}$$

$$\vec{r} \cdot \vec{r} = r^\mu \underbrace{g_{\mu\nu}}_{\text{matrix}} r^\nu$$

$\hookrightarrow$ matrix

We have $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ for Euclidean

$\hookrightarrow$ object with 2 indices

we call this "metric tensor"

- Similar to the rotation that preserves the length of 3D vector
We can find the transformation that preserves the length in 4D
Minkowski
Lorentz transformation

- Suppose that the Lorentz transformation can be written as

$$r^\mu \rightarrow r'^\mu = \Lambda^\mu{}_\nu r^\nu \quad \begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix} \begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix} \quad A_i' = R_{ij} A_j = A_i$$

free indices
must be at the
same level up-up or down-down

The contraction of dummy indices
is always up-down

$$r'^\mu = \Lambda^\mu{}_\nu r^\nu$$

new length

old length

$$\text{The length } r'^\mu g_{\mu\nu} r'^\nu = r^\mu g_{\mu\nu} r^\nu$$

This is the condition for Lorentz transformation

$$r'^\mu g_{\mu\nu} r'^\nu = \left(\Lambda^\mu{}_\rho r^\rho \right) g_{\mu\nu} \left(\Lambda^\nu{}_\sigma r^\sigma \right)$$

new dummy index dummy

$$= r^\rho \left(\Lambda^\mu{}_\rho g_{\mu\nu} \Lambda^\nu{}_\sigma \right) r^\sigma$$

$$= r^\rho g_{\rho\sigma} r^\sigma$$

$\mu \rightarrow \rho$
 $\nu \rightarrow \sigma$
dummy

The Lorentz transformation requires

$$\Lambda^\mu{}_\rho g_{\mu\nu} \Lambda^\nu{}_\sigma = g_{\rho\sigma} \quad *$$

We can look at this in the matrix form

$$(r^0' \ r^1' \ r^2' \ r^3') \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} r^0' \\ r^1' \\ r^2' \\ r^3' \end{pmatrix} = \begin{pmatrix} r^0 & r^1 & r^2 & r^3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} r^0 \\ r^1 \\ r^2 \\ r^3 \end{pmatrix}$$

$$\begin{pmatrix} r^0' \\ r^1' \\ r^2' \\ r^3' \end{pmatrix} = \begin{pmatrix} \Lambda^0_0 & \Lambda^0_1 & \Lambda^0_2 & \Lambda^0_3 \\ \Lambda^1_0 & \Lambda^1_1 & - & - \\ \Lambda^2_0 & : & - & - \\ \Lambda^3_0 & : & - & - \end{pmatrix} \begin{pmatrix} r^0 \\ r^1 \\ r^2 \\ r^3 \end{pmatrix}$$

(Λ)

$$\begin{pmatrix} r^0 & r^1 & r^2 & r^3 \end{pmatrix} \begin{pmatrix} \Lambda \end{pmatrix}^T \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \Lambda \end{pmatrix} \begin{pmatrix} r^0 \\ r^1 \\ r^2 \\ r^3 \end{pmatrix}$$

In the matrix form:

$$\begin{pmatrix} \Lambda \end{pmatrix}^T \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \Lambda \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} *$$

Exercise

$$1) \begin{pmatrix} \Lambda \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & R & & \\ 0 & & & \end{pmatrix}$$

show that this is a Lorentz transf.

∴ Rotation is a part of Lorentz transf.

$$2). \begin{pmatrix} \Lambda \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \text{ where } \gamma = \frac{1}{\sqrt{1-\beta^2}}$$

⇒ show that this is a Lorentz transf.

Lorentz transf.

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & R & \\ 0 & & & \end{pmatrix} \Rightarrow \Lambda^T \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} \Lambda = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & R & \\ 0 & & & \end{pmatrix}^T \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & R & \\ 0 & & & \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \boxed{R^T \cdot \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}} & & \\ 0 & & R & \\ 0 & & & \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \boxed{R^T \cdot \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} R} & & \\ 0 & & R & \\ 0 & & & \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$R^T \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} R = (-1) R^T \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} R$$

def. of rotation $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 1 & & 0 \\ & -1 & \\ 0 & & -1 \end{pmatrix}$$

$\therefore$ rotation is a part of Lorentz transf.

* In rotation $R^T = R^{-1}$

Can we do similar thing for Λ ?

$$\tilde{R}^T \tilde{\Lambda} R = \tilde{\Lambda} \Rightarrow \tilde{R}^T \tilde{R} = \mathbb{1} \Rightarrow \boxed{\tilde{R}^T = \tilde{R}^{-1}}$$

$$\tilde{\Lambda}^T \tilde{g} \tilde{\Lambda} = \tilde{g} \Rightarrow \boxed{\tilde{\Lambda}^T = \tilde{g} \tilde{\Lambda}^{-1} \tilde{g}^{-1}}$$

$$\tilde{g} = \begin{pmatrix} 1 & & 0 \\ & -1 & \\ 0 & & -1 \end{pmatrix}$$

$\times \tilde{\Lambda}^{-1} \times \tilde{g}$

$$2) \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \tilde{\Lambda} = \text{boost}$$

$\tilde{\Lambda}^T = \tilde{\Lambda}$ Symmetric
This is an accident!

$$\tilde{\Lambda}^T \tilde{g} \tilde{\Lambda} = \tilde{g} \quad \checkmark$$

$$\begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} (\gamma \ \gamma\beta)(1 \ 0)(\gamma \ \gamma\beta) & 0 & 0 \\ (\gamma\beta \ \gamma)(0 \ -1)(\gamma\beta \ \gamma) & 0 & 0 \\ 0 & 0 & (1 \ 0)(-1 \ 0)(1 \ 0) \\ 0 & 0 & (0 \ -1)(0 \ -1)(0 \ 1) \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\gamma^2 = \frac{1}{1-\beta^2}$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}$$

We have $\begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix}$ $\gamma^2(1-\beta^2) = 1$

$$= \begin{pmatrix} \gamma & -\gamma\beta \\ \gamma\beta & -\gamma \end{pmatrix} \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} = \begin{pmatrix} \gamma^2 - \gamma^2\beta^2 & 0 \\ 0 & \gamma^2\beta^2 - \gamma^2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \checkmark$$

$-\gamma^2(1-\beta^2) = -1$

$\therefore$ Boost is another part of Lorentz transf.

The "distance" between events

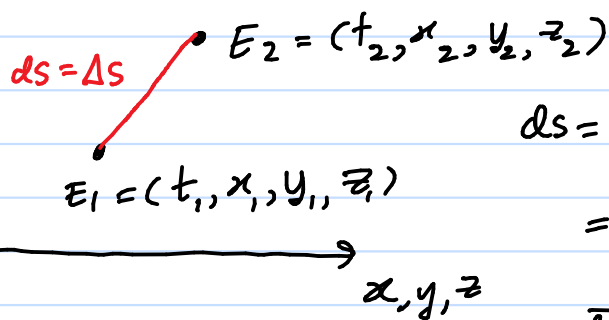
What we want is that in different inertial frames we should see the same "distance" (length) in 4D

~ Lorentz transf.

=> time is relative

3D length is also relative

t Space time diagram



$$ds = (t_2 - t_1, x_2 - x_1, y_2 - y_1, z_2 - z_1) \\ = (dt, dx, dy, dz)$$

$$\vec{ds} = dt\hat{e}_0 + dx\hat{e}_1 + dy\hat{e}_2 + dz\hat{e}_3$$

Sometimes we write $dx^\mu \sim (dt, dx, dy, dz)$

$$(\overset{''}{dx}^0, \overset{''}{dx}^1, \overset{''}{dx}^2, \overset{''}{dx}^3)$$

The distance

$$ds^2 = dx^\mu \underbrace{g_{\mu\nu}}_{\text{invariant under Lorentz transf}} dx^\nu$$

$$= (dt \ dx \ dy \ dz) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} dt \\ dx \\ dy \\ dz \end{pmatrix}$$

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2$$

which invariant under Lorentz transf.

(change in the inertial frame)

★ We can write dx_μ & $g^{\mu\nu}$

$$ds^2 = dx^\mu g_{\mu\nu} dx^\nu$$

$$= dx_\mu \underbrace{g^{\mu\nu}}_{\text{where } g^{\mu\nu} \text{ is an inverse of } g_{\mu\nu}} dx^\nu$$

We can show that Lorentz transformation preserves distance

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2$$

We can show that Lorentz transformation preserves distance

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2$$

$$ds^2 = \underline{dx^\mu} g_{\mu\nu} \underline{dx^\nu} = (\dots) \begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} \begin{pmatrix} \cdot \\ \cdot \\ \cdot \\ \cdot \end{pmatrix}$$

Lorentz transformation : $\underline{dx^{\mu'}} = \Lambda^\mu_{\ \nu} dx^\nu \Rightarrow \begin{pmatrix} \cdot \\ \cdot \\ \cdot \\ \cdot \end{pmatrix} = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} \begin{pmatrix} \cdot \\ \cdot \\ \cdot \\ \cdot \end{pmatrix}$

$$ds'^2 = dx'^\mu g_{\mu\nu} dx'^\nu \leftarrow \text{after trans.}$$

$$= (\Lambda^\mu_{\ \rho} dx^\rho) g_{\mu\nu} (\Lambda^\nu_{\ \sigma} dx^\sigma) = dx^\rho \underbrace{(\Lambda^\mu_{\ \rho} g_{\mu\nu} \Lambda^\nu_{\ \sigma})}_{g_{\rho\sigma}} dx^\sigma$$

Comparing with before trans.

$$ds^2 = \underline{dx^\mu} g_{\mu\nu} dx^\nu = dx^\rho g_{\rho\sigma} dx^\sigma \quad \checkmark$$

★ We can write dx_μ & $g^{\mu\nu}$

$$ds^2 = dx^\mu g_{\mu\nu} dx^\nu$$

$$= dx_\mu g^{\mu\nu} dx_\nu \quad \text{where } g^{\mu\nu} \text{ is an inverse of } g_{\mu\nu}$$

dummy up-down

$$* \boxed{g^{\mu\nu} g_{\nu\rho} = \delta^\mu_{\ \rho}} \Rightarrow \text{Kronecker delta (in 4D)}$$

$$\begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} \begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} = \begin{pmatrix} 1 & & & 0 \\ & 1 & & \\ & & 1 & \\ 0 & & & 1 \end{pmatrix} = \begin{cases} 1, & \mu = \rho \\ 0, & \mu \neq \rho \end{cases}$$

Also, we can write

covariant

$$\boxed{dx_\mu = g_{\mu\nu} dx^\nu}$$

$$g_{\mu\nu} = \begin{pmatrix} 1 & & 0 \\ & -1 & \\ 0 & & -1 \end{pmatrix}$$

$$dx^\mu = (dt, dx, dy, dz)$$

$$dx_\mu = g_{\mu\nu} dx^\nu = (dt, -dx, -dy, -dz)$$

$$g^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}^{-1}$$

Another form of ds^2

$$ds^2 = dx^\mu \boxed{g_{\mu\nu}} dx^\nu$$

$$= dx^\mu dx_\mu$$

dummy

dx_μ

$$dr^2 = dx^2 + dy^2 + dz^2 > 0$$

Notice that unlike the 3D space, the distance in 4D can be negative!

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2 \Rightarrow ds^2 < 0 \text{ if } dx^2 + dy^2 + dz^2 > dt^2$$

$$ds^2 = 0 \text{ if } dx^2 + dy^2 + dz^2 = dt^2$$

$\therefore$ We classify the distance into 3 types:
↳ (4-vector)

1). Null vector = dx^μ such that $ds^2 = dx^\mu dx_\mu = 0$

Suppose that we have $E_2 = (t_2, x_2, y, z)$

$$E_1 = (t_1, x_1, y, z), \quad dx^\mu = (\underbrace{t_2 - t_1}_{\Delta t}, \underbrace{x_2 - x_1}_{\Delta x}, 0, 0)$$

light cone

$$ds^2 = dx^\mu dx_\mu = dx^\mu g_{\mu\nu} dx^\nu$$

$$(\Delta t \ \Delta x \ 0 \ 0) \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \Delta t \\ \Delta x \\ 0 \\ 0 \end{pmatrix} = \Delta t^2 - \Delta x^2 = 0$$

$$(\Delta t \ \Delta x \ 0 \ 0) \begin{pmatrix} \Delta t \\ -\Delta x \\ 0 \\ 0 \end{pmatrix} = \Delta t^2 - \Delta x^2$$

$E_2 \& E_1$

is related by
the speed of light

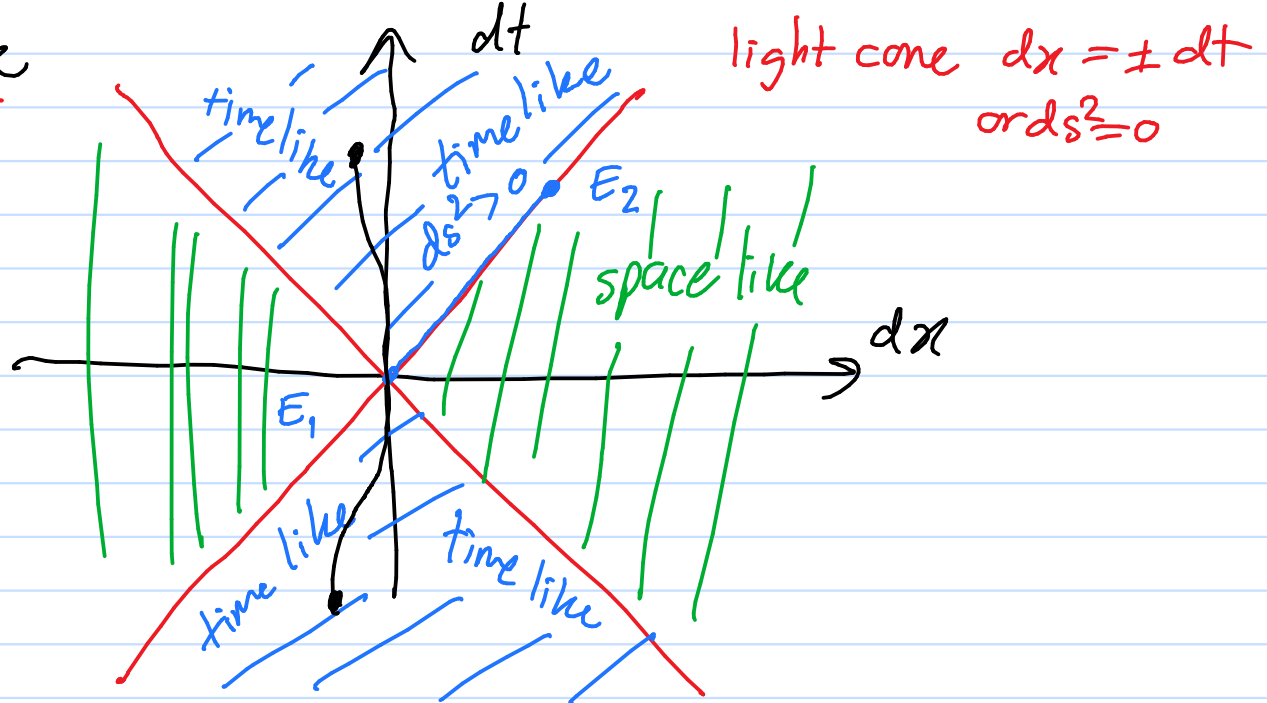
$$\Delta t^2 = \Delta x^2 \Rightarrow \frac{\Delta x}{\Delta t} = \pm 1$$

This is the limit
of information
(causality)

cause

v
 c

Space time



2). timelike vector = dx^μ such that $ds^2 > 0$

$$\Delta t^2 - \Delta x^2 > 0 \Rightarrow \Delta t^2 > \Delta x^2$$

$$\therefore \left| \frac{\Delta x}{\Delta t} \right| < 1 \Rightarrow \underline{v < 1} \Rightarrow \text{less than speed of light.}$$

(2 events that communicate within physical limit

within causality

3) spacelike vector = dx^μ such that $ds^2 < 0$

$$\Delta t^2 - \Delta x^2 < 0 \quad \Delta x^2 > \Delta t^2$$

$$\left| \frac{\Delta x}{\Delta t} \right| > 1 \Rightarrow v > 1$$

(impossible within physical limit

Example of 4 vectors

1). $dx^\mu = (dt, dx, dy, dz)$

2). 4-velocity $u^\mu = (\gamma, \gamma v_x, \gamma v_y, \gamma v_z)$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}, \quad \beta = \text{velocity} = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

3). 4-momentum $p^\mu = m u^\mu = (\gamma m, \gamma m v_x, \gamma m v_y, \gamma m v_z)$
 $= (E, p_x, p_y, p_z)$

(E, p)
 $(\gamma m c^2)$

4). Wave vector $k^\mu = (\omega, k_x, k_y, k_z)$

frequency ω wave number $= \frac{2\pi}{\lambda}$

$\sin(\omega t - kx), \quad e^{i(\omega t - kx)}$

$$k^\mu x_\mu = (\omega, k) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix} = \omega t - kx$$

5). 4-current $j^\mu = (\rho, j_x, j_y, j_z)$

Maxwell equation
is simpler using this

charge density ρ current density $\frac{I}{A}$

6). EM potential: $A^\mu = (V, \vec{A})$ $\vec{B} = \nabla \times \vec{A}$

Maxwell equation
similar to wave equation

Electric potential V Magnetic vector potential $\vec{A}$
 $\vec{E} = -\nabla V$

7). Differential operator ∂^μ

$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} \equiv \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$

down dx^μ ∂^μ contravariant

--- up
 down
 down
 up

4-velocity

Let's define the proper time $\Rightarrow$ measure by the observer that has no velocity relative to the object
 $dx = dy = dz = 0$ in this frame.

$$\therefore ds^2 = dt^2 = d\tau^2 \quad \hookrightarrow \text{proper time.}$$

We define $u^\mu = \frac{dx^\mu}{ds} = \frac{dx^\mu}{d\tau}$ $dx \neq 0, dy \neq 0, dz \neq 0$
 $\hookrightarrow$ proper time

From the time dilation effect, $dt = \gamma d\tau$

$$\therefore u^\mu = \frac{dx^\mu}{d\tau} = \left(\frac{dt}{d\tau}, \frac{dx}{d\tau}, \frac{dy}{d\tau}, \frac{dz}{d\tau} \right) \quad dt > d\tau, \quad \gamma = \frac{1}{\sqrt{1-\beta^2}} \geq 1$$

$$= \left(\gamma \frac{dt}{dt}, \gamma \underbrace{\frac{dx}{dt}}_{v_x}, \gamma \underbrace{\frac{dy}{dt}}_{v_y}, \gamma \underbrace{\frac{dz}{dt}}_{v_z} \right)$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} \\ \beta = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$= (\gamma, \gamma v_x, \gamma v_y, \gamma v_z)$$

Notice that $u^\mu u_\mu = (\gamma, \gamma v_x, \gamma v_y, \gamma v_z) \begin{pmatrix} \gamma \\ -\gamma v_x \\ -\gamma v_y \\ -\gamma v_z \end{pmatrix}$

$$= \gamma^2 - \gamma^2 v_x^2 - \gamma^2 v_y^2 - \gamma^2 v_z^2$$

$$= \gamma^2 (1 - v_x^2 - v_y^2 - v_z^2) = \gamma^2 (1 - \beta^2) = \frac{1}{1-\beta^2} \cdot (1-\beta^2)$$

$u^\mu u_\mu = 1$

 $\Rightarrow$ This is invariant under Lorentz transformation

* An object at rest, $u^\mu = (1, 0, 0, 0)$

$$v = 0 = \beta$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}$$

$$\gamma = 1 \quad v = 0$$

$$u^\mu u_\mu = 1 \cdot 1 = 1$$

4-momentum

$$m \rightarrow u^\mu$$

$$\gamma v_x, \gamma v_y, \gamma v_z$$

$$p^\mu = m u^\mu = m(\gamma, \gamma \vec{v})$$

$$p^\mu = (\gamma m, \gamma m \vec{v})$$

$$\begin{cases} \gamma m = \frac{E}{c} \\ \gamma m \vec{v} = \vec{p} \end{cases} \quad \begin{cases} E = \gamma m c^2 \\ \vec{p} = \gamma m \vec{v} \end{cases}$$

$$p^\mu p_\mu = m u^\mu \cdot m u_\mu = m^2$$

$$\boxed{p^\mu \cdot p_\mu = m^2}$$

$$p^\mu = (E, \vec{p})$$

relativistic γm
relativistic $\gamma m \vec{v}$

$$p^\mu \cdot p_\mu = (E, \vec{p}) \cdot (E, -\vec{p}) = E^2 - \vec{p} \cdot \vec{p} = E^2 - \vec{p}^2 = m^2$$

$$\therefore \boxed{E^2 = \vec{p}^2 + m^2} \quad (E^2 = \vec{p}^2 c^2 + m^2 c^4)$$

$\Rightarrow$ Einstein dispersion relation
 $\hookrightarrow \omega, k$

Particle Dynamics

$$p, E, m, v$$

3D vector

Using 4-momentum $p^\mu = (E, \vec{p}) = (\gamma m, \gamma m \vec{v})$

$$= \left(\frac{m}{\sqrt{1-v^2}}, \frac{m \vec{v}}{\sqrt{1-v^2}} \right)$$

The non-relativistic limit of E & p , $v \ll 1$

$$E = \gamma m = m(1-v^2)^{-\frac{1}{2}} = m(1 + \frac{1}{2}v^2) = m + \frac{1}{2}mv^2$$

kinetic energy

$$|\vec{p}| = \gamma m v = m v (1-v^2)^{-\frac{1}{2}}$$

$$= mv + O(v^3)$$

rest mass energy (mc^2)

1) The invariant mass $= p^2 \equiv p^\mu p_\mu = E^2 - \vec{p}^2 = m^2$

↳ Lorentz invariant! (invariant under Lorentz transf.)

2) Note that we can find the velocity of a particle by

$$E = \gamma m, \quad |\vec{p}| = \gamma m v \Rightarrow \boxed{\frac{|\vec{p}|}{E} = \frac{\gamma m v}{\gamma m} = v}$$

$$\Rightarrow \gamma = \frac{1}{\sqrt{1 - \frac{|\vec{p}|^2}{E^2}}}$$

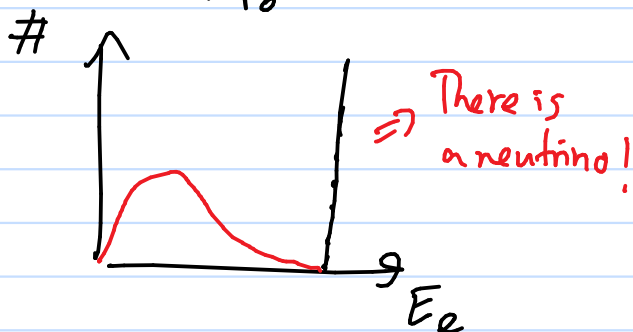
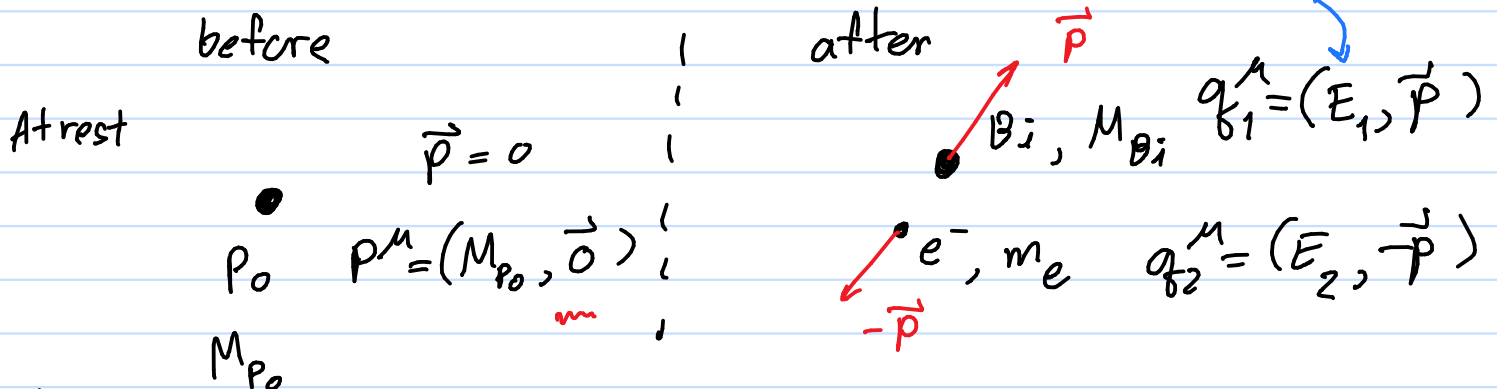
3) Conservation of Energy & Conservation of momentum

in 4-vector is simply $\boxed{\sum_i p_i^\mu \text{ before} = \sum_i p_i^\mu \text{ after}}$

$$(E_{\text{tot}}^{\text{before}}, \vec{p}_{\text{tot}}^{\text{before}}) = (E_{\text{tot}}^{\text{after}}, \vec{p}_{\text{tot}}^{\text{after}})$$

$\Rightarrow$ Conservation of 4-momentum

Example Particle decay



$$p^\mu p_\mu = M_{p_0}^2$$

$$q_1^\mu q_{1\mu} = M_{\theta_i}^2$$

$$q_2^\mu q_{2\mu} = m_e^2$$

Conservation of 4-momentum

$$p^\mu = q_1^\mu + q_2^\mu$$

Example Particle decay

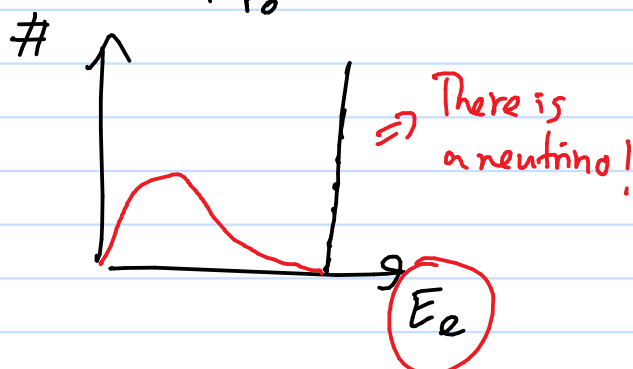
before

At rest

$$\vec{p} = 0$$

$$p_0 \quad p^\mu = (M_{p_0}, \vec{0})$$

$$M_{p_0}$$



after

$$B_i, M_{B_i} \quad q_1^\mu = (E_1, \vec{p})$$

$$e^-, m_e \quad q_2^\mu = (E_2, \vec{p})$$

$$p^\mu p_\mu = M_{p_0}^2$$

$$q_1^\mu q_{1\mu} = M_{B_i}^2$$

$$q_2^\mu q_{2\mu} = m_e^2$$

Conservation of 4-momentum

$$p^\mu = q_1^\mu + q_2^\mu$$

1) Do this straight forwardly!

$$\text{Conser. of } E : M_{p_0} = E_1 + E_2 \quad \text{--- (1)}$$

$$\text{Cons of } p : 0 = \vec{p} - \vec{p} \quad \checkmark$$

$$q_1^\mu q_{1\mu} = E_1^2 - |\vec{p}|^2 = M_{B_i}^2 \quad \text{--- (2)}$$

$$q_2^\mu q_{2\mu} = E_2^2 - |\vec{p}|^2 = m_e^2 \quad \text{--- (3)}$$

We can solve for $E_1, E_2, |\vec{p}|$

2). Or using the 4-momentum!

$$p^\mu = q_1^\mu + q_2^\mu$$

$$p^\mu - q_2^\mu = q_1^\mu$$

"dot" with it self

$$(p^\mu - q_2^\mu)(p_\mu - q_{2\mu}) = q_1^\mu q_{1\mu}$$

$$p^\mu p_\mu - p^\mu q_{2\mu} - p_\mu q_2^\mu + q_2^\mu q_{2\mu} = M_{B_i}^2$$

2). Or using the 4-momentum!

$$p^\mu = q_1^\mu + q_2^\mu$$

$$p^\mu - q_2^\mu = q_1^\mu \quad \text{"dot" with it self}$$

$$(p^\mu - q_2^\mu)(p_\mu - q_{2\mu}) = q_1^\mu q_{1\mu}$$

$$\cancel{p^\mu p_\mu} - \cancel{p^\mu q_{2\mu}} - \cancel{p_\mu q_2^\mu} + \cancel{q_2^\mu q_{2\mu}} = M_{\theta i}^2$$

$$M_{p_0}^2 - 2p^\mu q_{2\mu} + m_e^2 = M_{\theta i}^2$$

$$p^\mu = (M_{\theta i}, \vec{0})$$

$$q_2^\mu = (E_2, -\vec{p})$$

$$p^\mu q_{2\mu} = \frac{1}{2} (M_{p_0}^2 - M_{\theta i}^2 + m_e^2) = M_{\theta i} \cdot E_2 - (\vec{0} \cdot (-\vec{p}))$$

$$\therefore E_2 = \frac{(M_{p_0}^2 - M_{\theta i}^2 + m_e^2)}{2M_{\theta i}} \quad \text{which is fixed}$$

Ex Compton Scattering $\gamma + e^- \rightarrow \gamma + e^-$

Before

λ, f

$\gamma \xrightarrow{\hat{z}} e^-$

at rest

$p_\gamma^\mu = (E, \vec{p})$

$p_e^\mu = (m_e, \vec{0})$

$hf = \frac{h}{\lambda} = \frac{2\pi}{\lambda}$

$\hbar = 1 \Rightarrow h = 2\pi$

$\frac{h}{\lambda} = \frac{2\pi}{\lambda}$

$$p_\gamma^\mu = \left(\frac{2\pi}{\lambda}, \frac{2\pi}{\lambda} \hat{z} \right)$$

Conservation of 4-momentum.

$$p_\gamma^\mu + p_e^\mu = p_\gamma'^\mu + p_e'^\mu$$

After

λ', f'

$\gamma' \xrightarrow{\hat{n}} e'^-$

$\hat{n} \cdot \hat{z} = \cos \theta$

$p_\gamma'^\mu = \left(\frac{2\pi}{\lambda'}, \frac{2\pi}{\lambda'} \hat{n} \right)$

$p_e'^\mu = (E', \vec{p}')$

$= (\gamma m_e, \gamma m_e \vec{v})$

$p_\gamma^\mu p_{\gamma\mu} = 0, p_\gamma'^\mu p_{\gamma'\mu} = 0$

$p_e^\mu p_{e\mu} = p_e'^\mu p_{e'\mu} = m_e^2$

Ex Compton Scattering $\gamma + e^- \rightarrow \gamma + e^-$

Before

λ, f

$\gamma \rightarrow \hat{z}$

$\vec{p}_\gamma = (E, \vec{p})$

at rest

$\vec{p}_e = (m_e, \vec{0})$

$hf = \frac{h}{\lambda} = \frac{2\pi}{\lambda}$

$\hbar = 1 \Rightarrow h = 2\pi$

$\frac{h}{\lambda} = \frac{2\pi}{\lambda}$

$$\vec{p}_\gamma = \left(\frac{2\pi}{\lambda}, \frac{2\pi}{\lambda} \hat{z} \right)$$

Conservation of 4-momentum.

$$\vec{p}_\gamma + \vec{p}_e = \vec{p}'_\gamma + \vec{p}'_e$$

$$\vec{p}_\gamma - \vec{p}'_\gamma = \vec{p}'_e - \vec{p}_e \quad \text{dot with itself}$$

$$(\vec{p}_\gamma - \vec{p}'_\gamma) \cdot (\vec{p}_\gamma - \vec{p}'_\gamma) = (\vec{p}'_e - \vec{p}_e) \cdot (\vec{p}'_e - \vec{p}_e)$$

$$\cancel{\vec{p}_\gamma \cdot \vec{p}_\gamma} - 2\vec{p}_\gamma \cdot \vec{p}'_\gamma + \cancel{\vec{p}'_\gamma \cdot \vec{p}'_\gamma} = \cancel{\vec{p}'_e \cdot \vec{p}'_e} - 2\vec{p}'_e \cdot \vec{p}_e + \cancel{\vec{p}_e \cdot \vec{p}_e}$$

$m_e^2 \quad m_e^2$

$$-2\vec{p}_\gamma \cdot \vec{p}'_\gamma = 2m_e^2 - 2\vec{p}'_e \cdot \vec{p}_e$$

$$\vec{p}_\gamma \cdot \vec{p}'_\gamma = \vec{p}'_e \cdot \vec{p}_e - m_e^2$$

$$\frac{2\pi}{\lambda} \cdot \frac{2\pi}{\lambda'} - \frac{2\pi}{\lambda} \cdot \frac{2\pi}{\lambda'} \hat{z} \cdot \hat{n} = E \cdot m_e - (\vec{0} \cdot \vec{p}) - m_e^2$$

$$\frac{(2\pi)^2}{\lambda \lambda'} (1 - \cos \theta) = \left(2\pi \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) + m_e \right) m_e - m_e^2$$

$$\frac{(2\pi)^2}{\lambda \lambda'} (1 - \cos \theta) = 2\pi \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) m_e$$

$$\frac{2\pi}{m_e} (1 - \cos \theta) = \lambda \lambda' \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) = \lambda' - \lambda$$

$$\therefore \Delta \lambda = \lambda' - \lambda = \frac{2\pi}{m_e} (1 - \cos \theta)$$

After

λ', f'

$\vec{p}'_\gamma = \left(\frac{2\pi}{\lambda'}, \frac{2\pi}{\lambda'} \hat{n} \right)$

$\hat{n} \cdot \hat{z} = \cos \theta$

$\vec{p}'_e = (E, \vec{p}) = (\gamma m_e, \gamma m_e \vec{v})$

$$\vec{p}_\gamma \cdot \vec{p}_{\gamma\mu} = 0, \quad \vec{p}'_\gamma \cdot \vec{p}'_{\gamma\mu} = 0$$

$$\vec{p}_e \cdot \vec{p}_{e\mu} = \vec{p}'_e \cdot \vec{p}'_{e\mu} = m_e^2$$

$$\vec{p}_\gamma = \left(\frac{2\pi}{\lambda}, \frac{2\pi}{\lambda} \hat{z} \right)$$

$$\vec{p}'_\gamma = \left(\frac{2\pi}{\lambda'}, \frac{2\pi}{\lambda'} \hat{n} \right)$$

$$\vec{p}_e = (m_e, \vec{0})$$

$$\vec{p}'_e = (E, \vec{p})$$

before

$$E + \frac{2\pi}{\lambda'} = m_e + \frac{2\pi}{\lambda}$$

after

$$E = 2\pi \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) + m_e$$

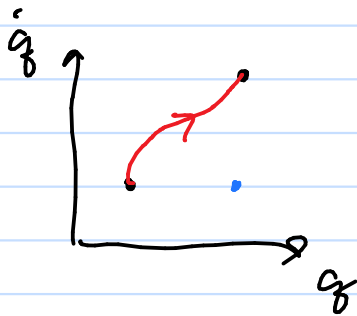
Field

Let's review the Lagrangian approach of Classical Dynamics

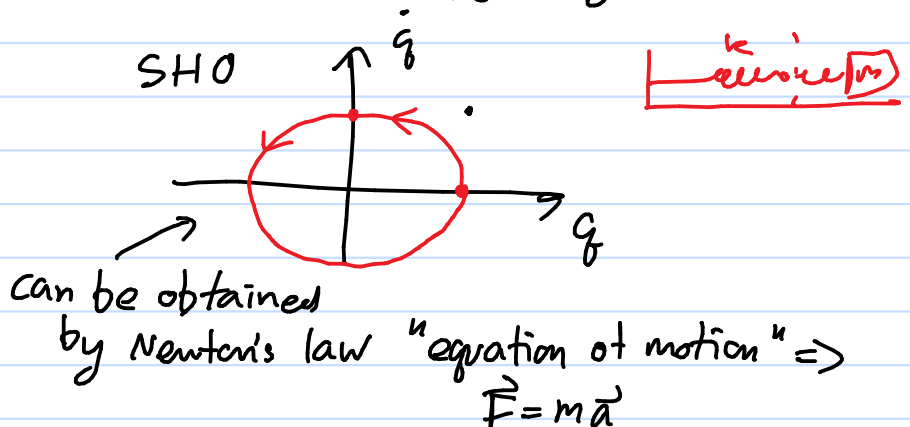
The most important principle in CM is the action principle

The action principle

Classical Path $\Rightarrow$ a trajectory (in $q, \dot{q}$ plane)



that a particle physically travels.

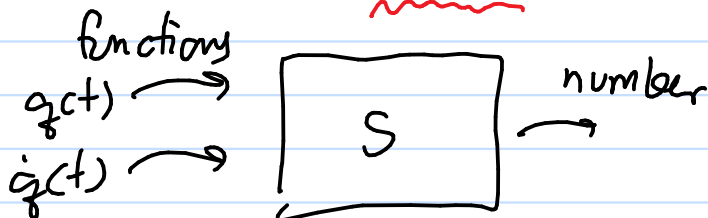


Classical Path $\Rightarrow$ Equation of motion ($\vec{F} = m\vec{a}$)

But in the Lagrangian (action) approach we can get this Classical path from the action principle.

There is a quantity called an action $S[q(t), \dot{q}(t)]$

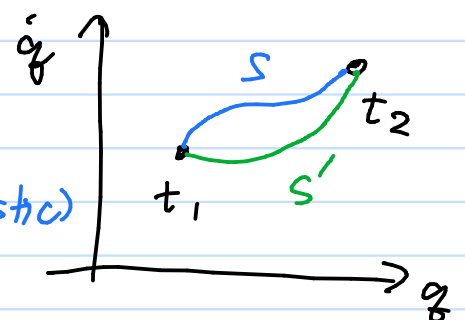
S is a scalar $\Rightarrow$ functional (its input is function)



$$S = \int_{t_1}^{t_2} dt L(q, \dot{q})$$

(non-relativistic)
1 particle

$$L(q, \dot{q}) = \text{Lagrangian} = \frac{1}{2}m\dot{q}^2 - V(q)$$



The classical path = the path that minimise the action

⇒ Variational calculus ⇒ calculus of functional

$$\delta S = 0 = \int_{t_1}^{t_2} dt \delta L(q, \dot{q}) = \int_{t_1}^{t_2} dt \left(\frac{\delta L}{\delta q(t)} \delta q(t) + \frac{\delta L}{\delta \dot{q}(t)} \delta \dot{q}(t) \right)$$

differential

$$df(x, y) = 0$$

$$df(x, y) = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

$$\delta \dot{q}(t) = \frac{d}{dt}(\delta q)$$

$$\therefore \frac{\delta L}{\delta q(t)} \frac{d}{dt}(\delta q) \rightarrow v$$

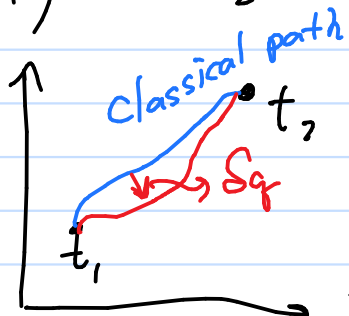
$$= \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \delta q \right) - \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \right) \delta q$$

$$\therefore u \frac{dv}{dt} = \frac{d(uv)}{dt} - \left(\frac{du}{dt} \right) v$$

$$\frac{d(uv)}{dt} = u \frac{dv}{dt} + \left(\frac{du}{dt} \right) v$$

$$\therefore \delta S = 0 = \int_{t_1}^{t_2} dt \left(\frac{\delta L}{\delta q(t)} \delta q(t) + \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \delta q \right) - \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \right) \delta q \right)$$

$$\int_{t_1}^{t_2} dt \frac{d}{dt}(X) = X(t_2) - X(t_1)$$



At the end points, the variation vanishes

$$\delta q(t_1) = \delta q(t_2) = 0$$

$$\therefore \delta S = 0 = \int_{t_1}^{t_2} dt \left(\frac{\delta L}{\delta q(t)} \delta q(t) - \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \right) \delta q(t) \right)$$

$$0 = \int_{t_1}^{t_2} dt \left(\frac{\delta L}{\delta q} - \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \right) \right) \delta q(t)$$

variation which is free to choose

= 0 Euler-Lagrange Equation

$$\vec{\nabla} f = (\partial_i f) \hat{e}_i \quad (\vec{\nabla} f)_i = \partial_i f \quad \vec{\nabla} \cdot \vec{A} = \partial_i A_i$$

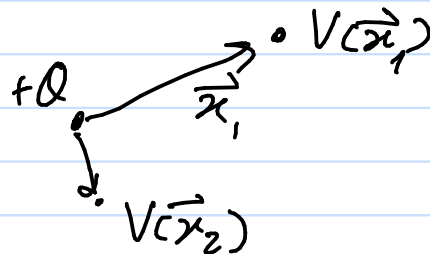
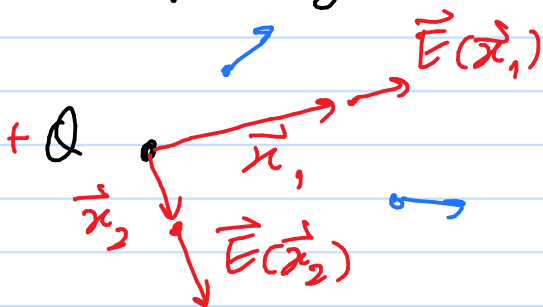
Equation of motion $\Rightarrow$ action principle

Euler Lagrange equation

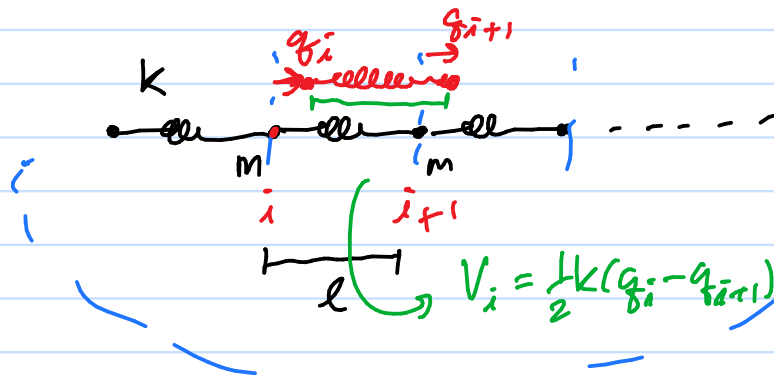
$$\frac{\delta L}{\delta q} - \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \right) = 0$$

Field

An entity that has a value for every point in space (and possibly in time), e.g., $\vec{E}(\vec{x})$, $V(\vec{x})$



Let's consider a 1-D system of N objects connected by spring



Periodic Boundary Condition

A 0th particle = Nth particle

Let's find the Lagrangian of the system

$$L = \sum_{i=1}^N \left(\frac{1}{2} m \dot{q}_i^2 \right) - V_{\text{system}}$$

Sum of pot. energy of every spring

$$\sum_{i=1}^{N-1} \frac{1}{2} k (q_i - q_{i+1})^2$$

Let's find the Lagrangian of the system

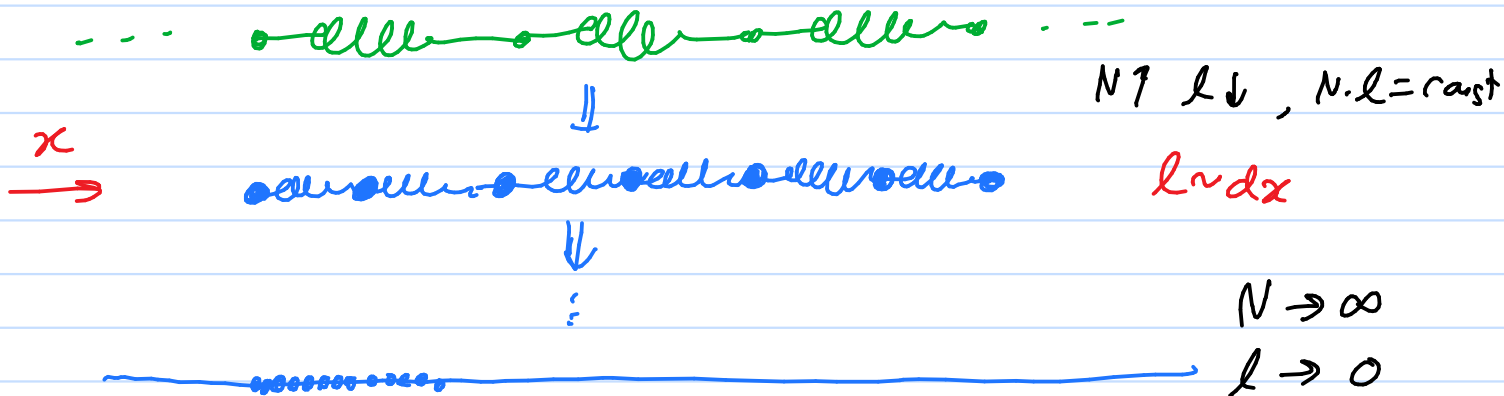
$$L = \sum_{i=1}^N \left(\frac{1}{2} m \dot{q}_i^2 \right) - V_{\text{system}}$$

Sum of pot. energy of every spring

$$\sum_{i=1}^{N-1} \frac{1}{2} k (q_i - q_{i+1})^2$$

$$L = \sum_{i=1}^N \left(\frac{1}{2} m \dot{q}_i^2 \right) - \sum_{i=1}^{N-1} \frac{1}{2} k (q_i - q_{i+1})^2$$

Now let's increase the number of particles.



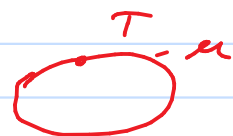
$$L = \frac{1}{2} \left(\frac{m}{l} \right) \sum_{i=1}^N \left(l \dot{q}_i^2 \right) - \frac{1}{2} k \sum_{i=1}^{N-1} \left(\frac{q_i - q_{i+1}}{l} \right)^2$$

$$= \frac{1}{2} \mu \sum_{i=1}^N l \cdot \dot{q}_i^2 - \frac{1}{2} k l \sum_{i=1}^{N-1} \left(\frac{q_i - q_{i+1}}{l} \right)^2$$

$$\therefore N \rightarrow \infty, l(dx) \rightarrow 0$$

$$L = \frac{1}{2} \mu \int dx \dot{q}^2 - \frac{1}{2} k l \int dx \left(\frac{dq}{dx} \right)^2$$

$$v = \sqrt{\frac{T}{\mu}}$$



$$L = \frac{1}{2} \int dx \left(\mu \dot{q}^2 - T \left(\frac{dq}{dx} \right)^2 \right)$$

Let's define $\tilde{q} = \sqrt{T} q$

$$L = \frac{1}{2} \int dx \left(\frac{\mu}{T} \dot{\tilde{q}}^2 - \left(\frac{d\tilde{q}}{dx} \right)^2 \right)$$

$$L = \frac{1}{2} \int dx \left(\underbrace{\frac{1}{\dot{q}^2}}_{\frac{1}{\dot{q}^2}} \dot{q}^2 - \left(\frac{dq}{dx} \right)^2 \right) \Rightarrow \frac{\partial}{\partial x} \text{ since } q(x, t)$$

$$L = \frac{1}{2} \int dx \left(\frac{1}{\dot{q}^2} \left(\frac{\partial \tilde{q}}{\partial t} \right)^2 - \left(\frac{\partial \tilde{q}}{\partial x} \right)^2 \right)$$

$$L = \frac{1}{2} \int dx \left(\underbrace{\left(\frac{\partial \tilde{q}}{\partial t} \right)^2}_{\partial^\mu \tilde{q} \partial_\mu \tilde{q}} - \left(\frac{\partial \tilde{q}}{\partial x} \right)^2 \right)$$

suppose that
we are dealing
with a system
such that the
velocity = 1

$$\Downarrow$$

$$1D \rightarrow 3D$$

$$\underbrace{\begin{pmatrix} \frac{\partial \tilde{q}}{\partial t} & \frac{\partial \tilde{q}}{\partial x} \end{pmatrix}}_{\partial^\mu \tilde{q}} \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}_{g_{\mu\nu}} \underbrace{\begin{pmatrix} \frac{\partial \tilde{q}}{\partial t} \\ \frac{\partial \tilde{q}}{\partial x} \end{pmatrix}}_{\partial_\mu \tilde{q}}$$

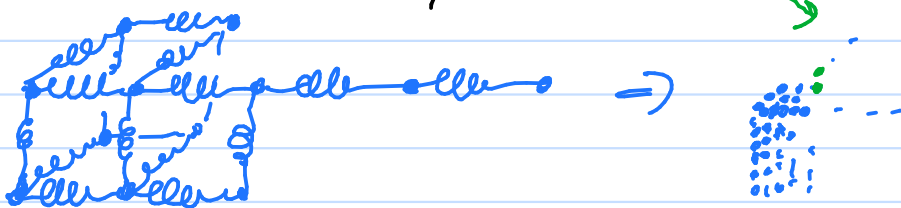
$$L = \frac{1}{2} \int d^3x \left(\left(\frac{\partial \tilde{q}}{\partial t} \right)^2 - \left(\frac{\partial \tilde{q}}{\partial x} \right)^2 - \left(\frac{\partial \tilde{q}}{\partial y} \right)^2 - \left(\frac{\partial \tilde{q}}{\partial z} \right)^2 \right)$$

$$= \frac{1}{2} \int d^3x \left(\partial^\mu \tilde{q} g_{\mu\nu} \partial^\nu \tilde{q} \right) = \frac{1}{2} \int d^3x \left(\partial^\mu \tilde{q} \partial_\mu \tilde{q} \right)$$

Normally we think of the field as $\phi(x)$

$$\boxed{L = \frac{1}{2} \int d^3x \left(\partial^\mu \phi \partial_\mu \phi \right)} \Rightarrow \text{Free field}$$

Since we are trying to describe particles in terms of
field $\Rightarrow$ we use this $\phi(x)$



Now we see that the general form of L

$$L = \int d^3x \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi \right) = \int d^3x \mathcal{L}(\phi(x), \partial_\mu \phi(x))$$

Now we see that the general form of L

$$L = \int d^3x \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi \right) = \underbrace{\int d^3x}_{\text{Volume}} \underbrace{\mathcal{L}(\phi(x), \partial_\mu \phi(x))}_{\text{Lagrangian density}}$$

Then the action

$$S = \int dt L = \int dt d^3x \mathcal{L}(\phi, \partial_\mu \phi)$$

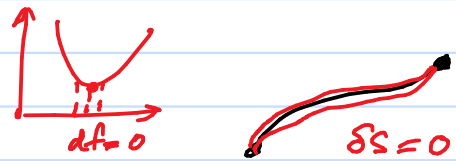
$$S = \int d^4x \mathcal{L}(\phi, \partial_\mu \phi) \quad \left| \quad \begin{array}{l} \text{The general form} \\ \text{of action for field theory} \end{array} \right.$$

Now let's discuss the equation of motion of $\phi(x)$

* Caution: Sometimes we call $\mathcal{L}$ simply Lagrangian

(density is omitted)

The action principle



The equation of motion $\Rightarrow$ The path that minimise action $\Rightarrow \delta S = 0$

$$\delta S = 0 = \int d^4x \delta \mathcal{L}(\phi, \partial_\mu \phi) \quad \left. \begin{array}{l} \text{d}f(x,y) = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \end{array} \right\}$$

$$0 = \int d^4x \left(\frac{\delta \mathcal{L}}{\delta \phi} \delta \phi + \frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \delta(\partial_\mu \phi) \right) \quad \left. \begin{array}{l} \delta(\partial_\mu \phi) \\ = \partial_\mu(\delta \phi) \end{array} \right\}$$

$$0 = \int d^4x \left(\frac{\delta \mathcal{L}}{\delta \phi} \delta \phi + \frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \partial_\mu(\delta \phi) \right)$$

$$\left(\partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \delta \phi \right) - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) \delta \phi \right)$$

$u \frac{dv}{dt} = \frac{d(uv)}{dt} - \left(\frac{du}{dt} \right) v$

$$0 = \int d^4x \left(\frac{\delta \mathcal{L}}{\delta \phi} \delta \phi + \underbrace{\left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \partial_\mu (\delta \phi) \right)}_{\text{divergence term}} \right)$$

$$\partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \delta \phi \right) = \frac{d}{dt} (uv) = \frac{d}{dt} (u \frac{v}{u}) = \frac{d}{dt} (u) \frac{v}{u} + u \frac{d}{dt} \left(\frac{v}{u} \right)$$

$$\rightarrow \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \delta \phi \right) = \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) \delta \phi + \frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \partial_\mu (\delta \phi)$$

$$0 = \int d^4x \left(\frac{\delta \mathcal{L}}{\delta \phi} \delta \phi + \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \delta \phi \right) - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) \delta \phi \right)$$

$$0 = \int d^4x \underbrace{\partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \delta \phi \right)}_{\text{boundary terms}} + \int d^4x \left(\frac{\delta \mathcal{L}}{\delta \phi} - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) \right) \delta \phi$$

$$\int d^4x \left(\frac{\partial}{\partial t} \left(\frac{\delta \mathcal{L}}{\delta(\partial_t \phi)} \delta \phi \right) - \frac{\partial}{\partial x} \left(\frac{\delta \mathcal{L}}{\delta(\partial_x \phi)} \delta \phi \right) - \dots \right) = 0$$

Boundary terms



$$\int d^3x \left(\frac{\delta \mathcal{L}}{\delta \dot{\phi}} \delta \phi \right) \Big|_{t=t_i}^{t_f} - \int dt dy dz \left(\frac{\delta \mathcal{L}}{\delta(\partial_x \phi)} \delta \phi \right) \Big|_{x=-\infty}^{x=\infty}$$

$\delta \phi$ is fixed at the boundaries

$\star \phi, \delta \phi \rightarrow 0$ at $-\infty$ & ∞ requirement

$\therefore$ We have

$$0 = \int d^4x \left(\frac{\delta \mathcal{L}}{\delta \phi} - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) \right) \delta \phi$$

$\hookrightarrow$ arbitrary

$\therefore = 0$

$\therefore$ The Euler-Lagrange equation for a field

$$\boxed{\frac{\delta \mathcal{L}}{\delta \phi} - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) = 0}$$

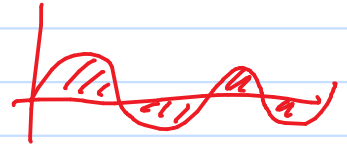
$$0 = \int d^4x \left(\underbrace{\frac{\delta \mathcal{L}}{\delta \phi} - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial_\mu \phi)} \right)}_{=0} \right) \underbrace{\delta \phi}_{\text{arbitrary}}$$

$$0 = \int_{-\infty}^{\infty} dt f(t) g(t) \quad \text{This is true for any function } g(t)$$

Choose $g(t) = 0$, The equation is true automatically
 $\Rightarrow$ no information on $f(t)$

$$g(t) = \sin(t), \quad f(t) = \text{const.}$$

$$g(t) = \text{const}, \quad f(t) = \sin(t)$$



$\Rightarrow$ we are left only $f(t) = 0$

$\therefore$ The Euler-Lagrange equation for a field

$$\boxed{\frac{\delta \mathcal{L}}{\delta \phi} - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial_\mu \phi)} \right) = 0}$$

Example

$$L = \int d^3x \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi \right) = \int d^3x \mathcal{L}(\phi(x), \partial_\mu \phi(x))$$

$$\therefore \mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi$$

$$\text{Let's put this in the EL eq.} \quad \frac{\delta \mathcal{L}}{\delta \phi} - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial_\mu \phi)} \right) = 0$$

$$1^{\text{st}} \text{ term: } \frac{\delta \mathcal{L}}{\delta \phi} = \frac{\delta}{\delta \phi} \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi \right) = 0$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{\partial}{\partial \phi} \left(\frac{1}{2} m \dot{\phi}^2 \right) = 0$$

$\hookrightarrow$ treat $\phi, \partial_\mu \phi$ as independent variables

$$2^{nd} \text{ term: } \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) = \partial_\mu \left(\frac{\delta \left(\frac{1}{2} \partial_\gamma \phi \partial^\gamma \phi \right)}{\delta(\partial_\mu \phi)} \right)$$

Note on how to take derivative:

$$1) \frac{\delta(\partial_\alpha \phi)}{\delta(\partial_\beta \phi)} = \delta_\alpha^\beta = \begin{cases} 1, & \alpha = \beta \\ 0, & \alpha \neq \beta \end{cases}$$

$$\frac{\delta(\partial_t \phi)}{\delta(\partial_t \phi)} = 1$$

$$\frac{\delta(\partial_x \phi)}{\delta(\partial_t \phi)} = 0$$

$$2) \frac{\delta(\partial^\alpha \phi)}{\delta(\partial_\beta \phi)} = \frac{\delta(g^{\alpha\gamma} \partial_\gamma \phi)}{\delta(\partial_\beta \phi)} = g^{\alpha\gamma} \frac{\delta(\partial_\gamma \phi)}{\delta(\partial_\beta \phi)} = g^{\alpha\gamma} \delta_\gamma^\beta = g^{\alpha\beta}$$

$$3) \frac{\delta(\partial^\alpha \phi)}{\delta(\partial^\beta \phi)} = \delta^\alpha_\beta = \begin{cases} 1, & \alpha = \beta \\ 0, & \alpha \neq \beta \end{cases}$$

$$4) \frac{\delta(\partial_\alpha \phi)}{\delta(\partial^\beta \phi)} = \frac{\delta(g_{\alpha\gamma} \partial^\gamma \phi)}{\delta(\partial^\beta \phi)} = g_{\alpha\gamma} \frac{\delta(\partial^\gamma \phi)}{\delta(\partial^\beta \phi)} = g_{\alpha\gamma} \delta^\gamma_\beta = g_{\alpha\beta}$$

$$\partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) = \partial_\mu \left(\frac{\delta \left(\frac{1}{2} \partial_\gamma \phi \partial^\gamma \phi \right)}{\delta(\partial_\mu \phi)} \right)$$

$$= \frac{1}{2} \partial_\mu \left[\frac{\delta(\partial_\gamma \phi)}{\delta(\partial_\mu \phi)} \cdot \partial^\gamma \phi + \partial_\gamma \phi \cdot \frac{\delta(\partial^\gamma \phi)}{\delta(\partial_\mu \phi)} \right]$$

$$= \frac{1}{2} \partial_\mu \left[\delta_\gamma^\mu \partial^\gamma \phi + \partial_\gamma \phi g^{\gamma\mu} \right]$$

$$= \frac{1}{2} \partial_\mu \left(\partial^\mu \phi + \partial^\mu \phi \right)$$

Recall $A^\mu = g^{\mu\nu} A_\nu$

$$= \partial_\mu \partial^\mu \phi$$

$$\therefore \frac{\delta \mathcal{L}}{\delta \phi} - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) = 0 - \partial_\mu \partial^\mu \phi = 0$$

Equation of motion of ϕ

$$\boxed{\partial_\mu \partial^\mu \phi = 0}$$

Euler-Lagrange eqⁿ.

What is $\partial_\mu \partial^\mu \phi = 0$?

$$\partial_\mu \partial^\mu = \partial^\nu g_{\nu\mu} \partial^\mu = \begin{pmatrix} \frac{\partial}{\partial t} & \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial t} \\ \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix}$$

$$= \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2}$$

$$\boxed{\partial_\mu \partial^\mu \phi = 0} \Rightarrow \frac{\partial^2}{\partial t^2} \phi - \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \phi = 0$$

$\square = D'Alembertian$

$\nabla^2 \equiv \vec{\nabla} \cdot \vec{\nabla}$ Laplacian

$$\boxed{\frac{1}{c^2} \frac{\partial^2}{\partial t^2} \phi = \nabla^2 \phi} \Rightarrow \text{Wave equation with velocity} = 1$$

$\phi = \text{scalar field} \Rightarrow$ There is "a number" at every point in space time.

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi \Rightarrow \text{massless scalar field}$$

$\hookrightarrow v=1$

Now let's consider another type of scalar field: massive scalar

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \quad \text{Klein-Gordon Lagrangian}$$

Equation of motion: $\frac{\delta \mathcal{L}}{\delta \phi} - \partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} \right) = 0$

1st term: $\frac{\delta \mathcal{L}}{\delta \phi} = \frac{\delta}{\delta \phi} \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right) = -\frac{1}{2} m^2 \frac{\delta \phi^2}{\delta \phi} = -m^2 \phi$

2nd term: $\partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} \right) = \partial^\mu \left(\frac{\delta \left(\frac{1}{2} \partial_\nu \phi \partial^\nu \phi - \frac{1}{2} m^2 \phi^2 \right)}{\delta (\partial^\mu \phi)} \right) = \partial^\mu \partial_\mu \phi$

$\partial_\mu \phi$

$$\frac{\delta \mathcal{L}}{\delta \phi} - \partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} \right) = 0 \Rightarrow -m^2 \phi - \partial^\mu \partial_\mu \phi = 0$$

$$\therefore \partial^\mu \partial_\mu \phi + m^2 \phi = 0$$

Klein-Gordon equation.

↳ First Relativistic version of Schrödinger's eqⁿ.

$$\left. \begin{array}{l} H = \sqrt{p^2 + m^2} \\ H^2 - p^2 = m^2 \\ \downarrow \frac{\partial}{\partial t} \quad \downarrow \vec{\nabla} \end{array} \right\} \text{Klein-Gordon}$$

Maxwell's equation.

$$c=1, \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 1, \mu_0 = 1, \epsilon_0 = 1$$

$$1). \vec{\nabla} \cdot \vec{E} = \rho \quad \text{Gauss Law} \quad \oint \vec{E} \cdot d\vec{a} = Q = \int \rho \cdot dV$$

$$2). \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\oint \vec{E} \cdot d\vec{a} = \int \vec{\nabla} \cdot \vec{E} dV$$

(Diagram of a circle with a dot in the center, representing a loop integral and Stokes' theorem)

$$3). \vec{\nabla} \cdot \vec{B} = 0$$

$$4). \vec{\nabla} \times \vec{B} = \vec{j} + \frac{\partial \vec{E}}{\partial t}$$

Electric potential

We can write $\vec{E}$ & $\vec{B}$ in terms of potential $\phi, \vec{A}$ ← Magnetic vector potential!

$$\vec{E} = -\vec{\nabla} \phi - \frac{\partial \vec{A}}{\partial t} \quad \& \quad \vec{B} = \vec{\nabla} \times \vec{A}$$

We can write 4-vector $A^\mu = (\phi, \vec{A}) = (A^0, A^i)$

$$\vec{E} = -\vec{\nabla} \phi - \frac{\partial \vec{A}}{\partial t} = -\left(\vec{\nabla} \phi + \frac{\partial \vec{A}}{\partial t}\right), \quad \partial_\mu = \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$$

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$$= \frac{\partial}{\partial x^\mu}$$

$$= (\partial_0, \partial_1, \partial_2, \partial_3)$$

Define the strength tensor

$$F^{\mu\nu} \equiv \partial^\mu A^\nu - \partial^\nu A^\mu$$

antisymmetric

$$1). F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu = -(\partial^\nu A^\mu - \partial^\mu A^\nu) = -F^{\nu\mu}$$

$$2). F^{11} = \partial^1 A^1 - \partial^1 A^1 = 0$$

Let's look at the components

$$\partial_\mu = \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \dots \right)$$

$$F^{0i} = \partial^0 A^i - \partial^i A^0 \quad A^i = (\vec{A})^i \quad \partial^\mu = \left(\frac{\partial}{\partial t}, -\frac{\partial}{\partial x}, -\frac{\partial}{\partial y}, \dots \right)$$

$$i=1,2,3 \quad = \frac{\partial (\vec{A})^i}{\partial t} + (\vec{\nabla})^i \phi \quad A^\mu = (\phi, \vec{A}) \quad \vec{\nabla}$$

Since $\vec{E}_i = -\vec{\nabla} \phi - \frac{\partial \vec{A}}{\partial t} = -(\vec{\nabla} \phi + \frac{\partial \vec{A}}{\partial t})$

$$F^{0i} = -(\vec{E})^i = -E^i, \quad F^{00} = 0$$

$$F^{i0} = +E^i$$

$$\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$$F^{ij} = \partial^i A^j - \partial^j A^i, \quad F^{23} = \partial^2 A^3 - \partial^3 A^2 = B^1$$

$$i \neq j$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial^1 & \partial^2 & \partial^3 \\ A^1 & A^2 & A^3 \end{vmatrix}$$

$$F^{12} = \partial^1 A^2 - \partial^2 A^1 = B^3$$

$$F^{13} = \partial^1 A^3 - \partial^3 A^1 = -B^2$$

$$\vec{B} = (\partial^2 A^3 - \partial^3 A^2) \hat{i} \rightarrow B^1$$

$$+ (\partial^3 A^1 - \partial^1 A^3) \hat{j} \rightarrow B^2$$

$$+ (\partial^1 A^2 - \partial^2 A^1) \hat{k} \rightarrow B^3$$

$\therefore$ If we write in terms of matrix

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & B^3 & -B^2 \\ E^2 & -B^3 & 0 & B^1 \\ E^3 & B^2 & -B^1 & 0 \end{pmatrix}$$

$$F^{0i} = -E^i$$

$$F^{i0} = -F^{0i} = E^i$$

Electric & Magnetic potential $\Rightarrow A^\mu$ 4-potential

Electric & Magnetic field $\Rightarrow F^{\mu\nu}$ field strength tensor

* Claim 2 of 4 Maxwell equations can be written as

$$\partial_\mu F^{\mu\nu} = j^\nu$$

$$j^\mu = 4\text{-current} = (\rho, \vec{j})$$

* Claim 2 of 4 Maxwell equations can be written as

$$\partial_\mu F^{\mu\nu} = j^\nu \Rightarrow \begin{array}{l} \nu = \text{free index} \\ \nu = 0, 1, 2, 3 \\ \hookrightarrow 4 \text{ equations} \end{array}$$

$$j^\mu = 4\text{-current} = (\rho, \vec{j})$$

1). $\nu = 0$, $\partial_\mu F^{\mu 0} = j^0 = \rho$ $F^{00} = 0$

$$\rho = \partial_\mu F^{\mu 0} = \cancel{\partial_0 F^{00}} + \partial_i F^{i0} = \partial_i E^i$$

$$\rho = \vec{\nabla} \cdot \vec{E} \quad \text{Gauss' law}$$

2). $\nu = i$, $\partial_\mu F^{\mu i} = j^i = (\vec{j})^i$
 $(i = 1, 2, 3)$

$$(\vec{j})^i = \partial_0 F^{0i} + \partial_j F^{ji}$$

$$(\vec{j})^1 = \partial_0 F^{01} + \partial_j F^{j1} = \partial_0 F^{01} + \cancel{\partial_1 F^{11}} + \partial_2 F^{21} + \partial_3 F^{31}$$

$$= -\frac{\partial E^1}{\partial t} + \partial_3 B^2 - \partial_2 B^3$$

$$= -\frac{\partial E^1}{\partial t} + \underbrace{\partial^2 B^3 - \partial^3 B^2}_{(\vec{\nabla} \times \vec{B})^1}$$

$\partial^\mu = (\partial_0, -\partial_1, -\partial_2, -\partial_3)$

$$\vec{\nabla} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial^1 & \partial^2 & \partial^3 \\ B^1 & B^2 & B^3 \end{vmatrix}$$

$$= \underbrace{(\partial^2 B^3 - \partial^3 B^2)}_{\hat{i}} + (\partial^3 B^1 - \partial^1 B^3) \hat{j} + (\partial^1 B^2 - \partial^2 B^1) \hat{k}$$

$$(\vec{j})^1 = -\frac{\partial (\vec{E})^1}{\partial t} + (\vec{\nabla} \times \vec{B})^1$$

$$(\vec{j})^2 = -\frac{\partial (\vec{E})^2}{\partial t} + (\vec{\nabla} \times \vec{B})^2, (\vec{j})^3 = \dots$$

$\therefore \vec{j} = -\frac{\partial \vec{E}}{\partial t} + \vec{\nabla} \times \vec{B}$ Maxwell equation!

How about the other two?

Exercise Show that the rest can be written as

$$\partial_\mu F_{\nu\lambda} + \partial_\lambda F_{\mu\nu} + \partial_\nu F_{\lambda\mu} = 0 \quad \text{Bianchi identity}$$

* can be proven from $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$

Maxwell equations $\Rightarrow$ Bianchi identity $\Rightarrow$ trivial

$$\Rightarrow \partial_\mu F^{\mu\nu} = j^\nu \text{ equation of motion}$$

Since $A^\mu, F^{\mu\nu}$ are fields $\Rightarrow A^\mu(\vec{x}, t), F^{\mu\nu}(\vec{x}, t)$

$\hookrightarrow \vec{E}, \vec{B}$

Can we do this in Lagrangian density approach?

Answer: Yes, $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_\mu j^\mu$ EM-Lagrangian (photon)

Let's try derive EL equation.

$\mathcal{L}(A^\mu, F^{\mu\nu})$ (previously, we have $\mathcal{L}(\phi, \partial^\mu \phi)$)

$$\hookrightarrow \mathcal{L}(A^\mu, \partial^\mu A^\nu)$$

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_\mu j^\mu = -\frac{1}{4} (\partial_\mu A_\nu - \partial_\nu A_\mu) (\partial^\mu A^\nu - \partial^\nu A^\mu) + A_\mu j^\mu$$

$$= -\frac{1}{4} (\underbrace{\partial_\mu A_\nu}_{\mu \leftrightarrow \nu} \underbrace{\partial^\mu A^\nu}_{\mu \leftrightarrow \nu} - \underbrace{\partial_\mu A_\nu}_{\mu \leftrightarrow \nu} \underbrace{\partial^\nu A^\mu}_{\mu \leftrightarrow \nu} - \underbrace{\partial_\nu A_\mu}_{\mu \leftrightarrow \nu} \underbrace{\partial^\mu A^\nu}_{\mu \leftrightarrow \nu} + \underbrace{\partial_\nu A_\mu}_{\mu \leftrightarrow \nu} \underbrace{\partial^\nu A^\mu}_{\mu \leftrightarrow \nu}) + A_\mu j^\mu$$

$$= -\frac{1}{2} (\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\nu A_\mu \partial^\mu A^\nu) + A_\mu j^\mu$$

$$\mathcal{L}(A^\mu, \partial^\nu A^\mu) = -\frac{1}{2} (\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\nu A_\mu \partial^\mu A^\nu) + A_\mu j^\mu$$

Comparing to $\mathcal{L}(\phi, \partial^\mu \phi)$, $\mathcal{L}(A^\mu, \partial^\nu A^\mu)$

$\hookrightarrow$ 4 scalar fields $\mu=0,1,2,3$

$$\frac{\delta \mathcal{L}}{\delta \phi} - \partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} \right) = 0$$

$$\Rightarrow \frac{\delta \mathcal{L}}{\delta A^\nu} - \partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu A^\nu)} \right) = 0$$

$\hookrightarrow$ 4 fields

$\hookrightarrow$ 4 fields

$$\mathcal{L}(A^\mu, \partial^\nu A^\mu) = -\frac{1}{2}(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\nu A_\mu \partial^\mu A^\nu) - A_\mu j^\mu$$

$$\frac{\delta \mathcal{L}}{\delta A^\nu} - \partial^\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial^\mu A^\nu)} \right) = 0 \quad A^\mu \& \partial^\nu A^\mu \text{ are inde.}$$

$$1^{\text{st}} \text{ term: } \frac{\delta \mathcal{L}}{\delta A^\nu} = \frac{\delta}{\delta A^\nu} \left(-\frac{1}{2}(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\nu A_\mu \partial^\mu A^\nu) - A^\mu j_\mu \right)$$

$$= - \left[\frac{\delta(A^\mu)}{\delta A^\nu} \right] j_\mu = -\delta^\mu_\nu j_\mu$$

$$\frac{\delta A^\sigma}{\delta A^\sigma} = 1$$

$$\frac{\delta A^\sigma}{\delta A^\sigma} = 0$$

$$= -j_\nu$$

$$(\mu \rightarrow \rho, \nu \rightarrow \rho)$$

$$2^{\text{nd}} \text{ term: } \partial^\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial^\mu A^\nu)} \right) = \partial^\mu \left(\frac{\delta}{\delta(\partial^\mu A^\nu)} \left(-\frac{1}{2}(\partial_\rho A_\sigma \partial^\rho A^\sigma - \partial_\sigma A_\rho \partial^\rho A^\sigma) - A^\mu j_\mu \right) \right)$$

Note on derivatives

$$1). \frac{\delta A^\mu}{\delta A^\nu} = \delta^\mu_\nu \quad 2). \frac{\delta A_\mu}{\delta A^\nu} = g_{\mu\sigma} \frac{\delta A^\sigma}{\delta A^\nu} = g_{\mu\sigma} \delta^\sigma_\nu = g_{\mu\nu}$$

$$3). \frac{\delta(\partial^\rho A^\sigma)}{\delta(\partial^\mu A^\nu)} = \delta^\sigma_\nu \delta^\rho_\mu \quad 4). \frac{\delta(\partial_\rho A_\sigma)}{\delta(\partial^\mu A^\nu)} = \dots = g_{\rho\mu} g_{\sigma\nu}$$

$$\partial^\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial^\mu A^\nu)} \right) = -\frac{1}{2} \left(\frac{\delta(\partial_\rho A_\sigma)}{\delta(\partial^\mu A^\nu)} \partial^\rho A^\sigma + \partial_\rho A_\sigma \frac{\delta(\partial^\rho A^\sigma)}{\delta(\partial^\mu A^\nu)} - \frac{\delta(\partial_\sigma A_\rho)}{\delta(\partial^\mu A^\nu)} \partial^\rho A^\sigma - \partial_\sigma A_\rho \frac{\delta(\partial^\rho A^\sigma)}{\delta(\partial^\mu A^\nu)} \right)$$

$$= -\frac{1}{2} \partial^\mu \left(g_{\rho\mu} g_{\sigma\nu} \partial^\rho A^\sigma + \partial_\rho A_\sigma \delta^\rho_\mu \delta^\sigma_\nu - g_{\rho\mu} g_{\sigma\nu} \partial^\rho A^\sigma - \partial_\sigma A_\rho \delta^\rho_\mu \delta^\sigma_\nu \right)$$

$$= -\frac{1}{2} \partial^\mu \left(\partial_\mu A_\nu + \partial_\mu A_\nu - \partial_\nu A_\mu - \partial_\nu A_\mu \right)$$

$$= -\frac{1}{2} \partial^\mu \left(\partial_\mu A_\nu + \partial_\mu A_\nu - \partial_\nu A_\mu - \partial_\nu A_\mu \right)$$

$$= -\partial^\mu (\partial_\mu A_\nu - \partial_\nu A_\mu) = -\partial^\mu F_{\mu\nu}$$

$\therefore$ EL equation: $-j_r - (-\partial^\mu F_{\mu r}) = 0$

$$\partial^\mu F_{\mu r} = j_r$$

Maxwell equation.

$$\partial_\mu F^{\mu r} = j^r$$

Field Hamiltonian

CM: $L(q, \dot{q}) \xrightarrow[\substack{p = \frac{\partial L}{\partial \dot{q}} \\ \hookrightarrow \text{momentum.}}]{\text{Legendre trans.}} H(q, p) = \dot{q}p - L(q, \dot{q})$

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = 0$$

$\hookrightarrow$ Euler-Lagrange equation

$$\frac{\partial H}{\partial p} = \dot{q}, \quad \frac{\partial H}{\partial q} = -\dot{p}$$

$\hookrightarrow$ Hamiltonian equation

Field Theory version?

π = momentum of field
"conjugate momentum"

$$L = \int d^3x \mathcal{L}(\phi, \partial^\mu \phi) \xrightarrow{\text{Legendre}} H = \int d^3x \mathcal{H}(\phi, \pi)$$

$$S = \int dt L = \int d^4x \mathcal{L}(\phi, \partial^\mu \phi)$$

$$\pi = \frac{\delta \mathcal{L}}{\delta \dot{\phi}} \Rightarrow \text{conjugate momentum.}$$

$$\frac{\delta \mathcal{L}}{\delta \phi} - \partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} \right) = 0$$

$$\frac{\delta \mathcal{H}}{\delta \pi} = \dot{\phi}, \quad \frac{\delta \mathcal{H}}{\delta \phi} = -\dot{\pi}$$

Hamiltonian equation

where

$$\mathcal{H}(\phi, \pi) = \dot{\phi}\pi - \mathcal{L}(\phi, \partial^\mu \phi)$$

Ex Klein-Gordon $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$
massive scalar

Ex Klein-Gordon $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$
 massive scalar

$$\pi(x) = \frac{\delta \mathcal{L}}{\delta \dot{\phi}} = \frac{\delta}{\delta \dot{\phi}} \left(\frac{1}{2} (\dot{\phi}^2 - (\vec{\nabla} \phi) \cdot (\vec{\nabla} \phi)) - \frac{1}{2} m^2 \phi^2 \right)$$

$= \dot{\phi}(x)$ * π is also a field.

Let's find Hamiltonian density

$$\mathcal{H} = \pi \dot{\phi} - \mathcal{L} = \pi^2(x) - \left[\frac{1}{2} (\pi^2(x) - (\vec{\nabla} \phi) \cdot (\vec{\nabla} \phi)) - \frac{1}{2} m^2 \phi^2 \right]$$

$$\mathcal{H} = \frac{1}{2} \pi^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + \frac{1}{2} m^2 \phi^2$$

Massive Scalar field
Hamiltonian density.

$$\mathcal{H} > 0 \quad \checkmark$$

Exercise Find $\mathcal{H}$ of $\mathcal{L}(A^\mu, \partial^\mu A^\nu)$

Symmetry

Lorentz transformation $\Rightarrow A_\mu A^\mu$ conserve under Lorentz transf.

Symmetry $\Rightarrow$ conserved quantity
 (momentum, E, charge)

$\hookrightarrow$ invariant quantity independent from inertia frame.

Transformation on a scalar field

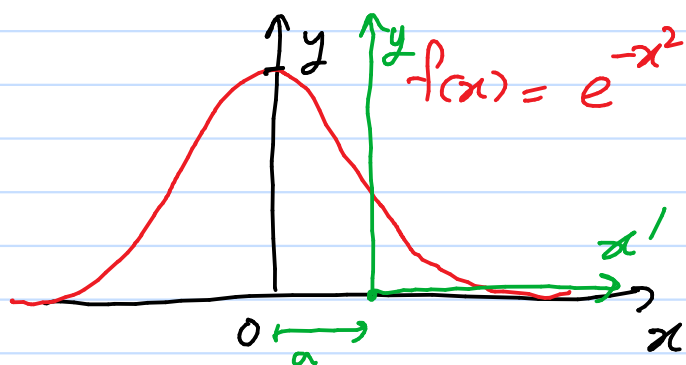
Let's consider a function of 1 variable first

$f(x)$ under translation in x , $x \rightarrow x' = x - a$

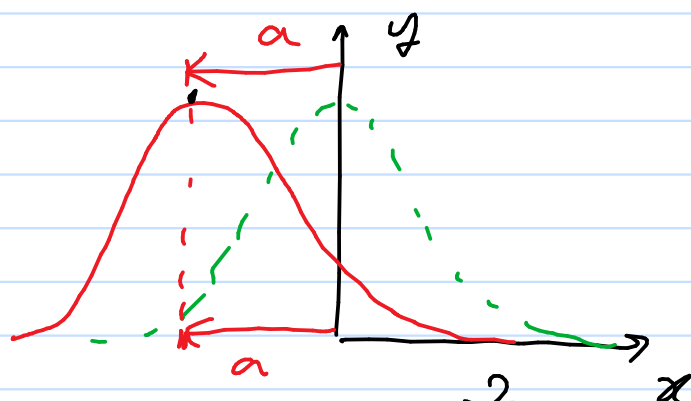
Let's consider a function of 1 variable first

$f(x)$ under translation in x , $x \rightarrow x' = x - a$

$$\hat{T}x = x' = x - a$$

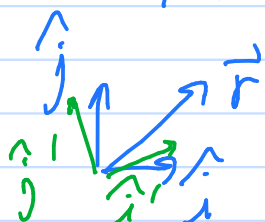


coordinate transt.

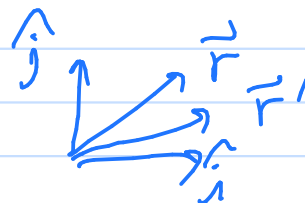


$$f'(x) = e^{-(x+a)^2}$$

Active



Passive transformation



$$f(x) = e^{-x^2} \quad f'(x) = e^{-(x+a)^2} = f(x+a) = f(\hat{T}^{-1}x)$$

$$\hat{T}x = x' = x - a$$



$$\hat{T}^{-1}x = x'' = x + a$$



$$\therefore \text{For a function, } f(\hat{T}^{-1}x) = f'(x)$$

$$\text{For a scalar } \phi(\hat{T}^{-1}\vec{x}) = \phi'(\vec{x})$$

where $\hat{T}$ is an operator on coordinates.

What about Lorentz transformation? $\Rightarrow \Lambda^\mu_{\nu} x^\nu$

$\hat{\Lambda}$ is an operator on coordinates

For a scalar $\phi(\hat{T}^{-1}\vec{x}) = \phi'(\vec{x})$

where $\hat{T}$ is an operator on coordinates.

What about Lorentz transformation? $\Rightarrow \Lambda^\mu{}_\nu x^\nu$

$\hat{\Lambda}$ is an operator on coordinates

$\therefore$ For a scalar, $\phi(\hat{\Lambda}^{-1} x^\mu, x^\nu) = \phi'(x^\mu)$

$$\phi(\Lambda^{-1}x) = \phi'(x)$$

We should be able to show that $\phi'(x)$ satisfies the same equation of motion (physical law)

$$\text{Consider } \mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$$

Under Lorentz transformation: $\partial^\mu \rightarrow \partial'^\mu = \Lambda^\mu{}_\nu \partial^\nu$

and also: $\phi \rightarrow \phi'(x) = \phi(\Lambda^{-1}x)$

$$\mathcal{L} = \frac{1}{2} \partial^\nu \phi g_{\mu\nu} \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$$

$$\mathcal{L} \rightarrow \mathcal{L}' = \frac{1}{2} \partial'^\nu \phi' g_{\mu\nu} \partial'^\mu \phi' - \frac{1}{2} m^2 \phi'^2$$

$$= \frac{1}{2} \Lambda^\nu{}_\sigma \partial^\sigma \phi' g_{\mu\nu} \Lambda^\mu{}_\rho \partial^\rho \phi' - \frac{1}{2} m^2 \phi'^2$$

$$= \frac{1}{2} \partial^\sigma \phi' g_{\sigma\rho} \partial^\rho \phi' - \frac{1}{2} m^2 \phi'^2$$

$$= \frac{1}{2} \partial_\sigma \phi' \partial^\sigma \phi'$$

$$\Lambda^\nu{}_\sigma g_{\mu\nu} \Lambda^\mu{}_\rho = g_{\sigma\rho}$$

$$\mathcal{L} \rightarrow \mathcal{L}' = \frac{1}{2} \partial^{\nu'} \phi' g_{\mu\nu} \partial^{\mu'} \phi' - \frac{1}{2} m^2 \phi'^2$$

$$\Lambda^\nu_\sigma g_{\mu\nu} \Lambda^\mu_\rho = \frac{1}{2} \Lambda^\nu_\sigma \partial^\sigma \phi' g_{\mu\nu} \Lambda^\mu_\rho \partial^\rho \phi' - \frac{1}{2} m^2 \phi'^2$$

$$= g_{\sigma\rho}$$

$$\phi(x)$$

$$= \phi(\Lambda^{-1}x)$$

$$= \frac{1}{2} \partial^\sigma \phi' g_{\sigma\rho} \partial^\rho \phi' - \frac{1}{2} m^2 \phi'^2$$

$$= \frac{1}{2} \partial_\mu \phi(\Lambda^{-1}x) \partial^\mu \phi(\Lambda^{-1}x) - \frac{1}{2} m^2 \phi^2(\Lambda^{-1}x)$$

$$= \frac{1}{2} \partial_\mu \phi(\Lambda^{-1}x) \partial^\mu \phi(\Lambda^{-1}x) - \frac{1}{2} m^2 \phi^2(\Lambda^{-1}x)$$

$$\boxed{\mathcal{L}'(x) = \mathcal{L}(\Lambda^{-1}x)}$$

How does the action transform?

$$S = \int d^4x \mathcal{L}(x)$$

$$\mathcal{L}'(x) = \mathcal{L}(\Lambda^{-1}x) \checkmark$$

$$S \rightarrow S'$$



measure = infinitesimal "volume"
 $dt dx dy dz$

in 4D $\Rightarrow$ Lorentz



$$dx dy dz$$

under rotation



$$dx' dy' dz'$$

=

$$dt dx dy dz = dt' dx' dy' dz'$$

should be multiplied by

$$\text{Jacobian} = 1$$

$$\det(R) = 1$$

$$\therefore S \rightarrow S'$$

$$S' = \int d^4x' \mathcal{L}'(x')$$

$$= \int d^4x \mathcal{L}(\Lambda^{-1}x) = \int d^4x'' \mathcal{L}(x'') = S$$

$$x'' = \Lambda^{-1}x$$

$\Rightarrow$ Euler Lagrange is invariant
 physical law is invariant.

Scalar fields : $\phi(x) \rightarrow \phi'(x) = \phi(\Lambda^{-1}x)$

Vector fields : $A^\mu(x) \rightarrow A'^\mu(x) = \Lambda^\mu_\nu A^\nu(\Lambda^{-1}x)$

Fermion fields : ?

representation of
Lorentz group.

Noether theorem

The connection between symmetries & conservation laws.

• Example rotational symmetry

$\Rightarrow$ Central force problem : $\vec{F} = F(r)\hat{r}$

$\Rightarrow$ rotational symmetry $\vec{r} \rightarrow \vec{r}' = \hat{R}\vec{r}$

$\Rightarrow$ torque $\vec{\tau} = \vec{r} \times \vec{F} = 0 = \frac{d\vec{L}}{dt}$
 $\vec{L}$ conserves.

Symmetry
= physical
invariant!

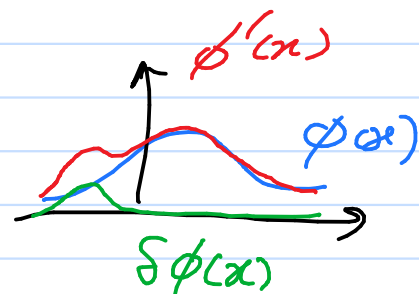
More examples : translational symmetry $\Rightarrow \vec{p}$ conserves

time translational symmetry $\Rightarrow E$ conserves

Noether theorem

Consider an infinitesimal on field

$$\phi(x) \rightarrow \phi'(x) = \phi(x) + \delta\phi(x)$$

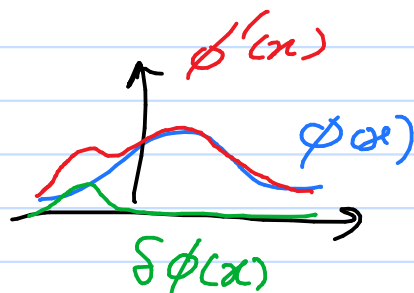


We say the transformation is a symmetry

Noether theorem

Consider an infinitesimal on field

$$\phi(x) \rightarrow \phi'(x) = \phi(x) + \delta\phi(x)$$



We say the transformation is a symmetry when $(\delta\phi(x) \equiv X(\phi))$ the physical law does not change

that is $\mathcal{L} \rightarrow \mathcal{L}' = \mathcal{L} + \delta\mathcal{L}$

$$\Rightarrow \boxed{\delta\mathcal{L} = \partial_\mu F^\mu} \rightarrow F^\mu \text{ can be anything}$$

★ Why is this equivalent to the invariant of physical law?

$$S = \int d^4x \mathcal{L} \rightarrow S' = \int d^4x \mathcal{L}'$$

$$= \int d^4x (\mathcal{L} + \delta\mathcal{L})$$

$$S' = \int d^4x \mathcal{L} + \int d^4x \delta\mathcal{L}$$

$$F^\mu \Big|_{t=-\infty}^{t=\infty} = \partial_0 F^0 + \partial_i F^i$$

boundary term

$$S' = S + \cancel{0} \rightarrow 0 \quad \hookrightarrow \text{required to be 0}$$

$$\delta S' = 0 \Rightarrow \delta S = 0 \Rightarrow \text{EL equation}$$

Noether theorem For a symmetry $\delta\phi \equiv X(\phi)$,

we can find the conserved current, j^μ

Consider the transformation $\delta\phi = X(\phi)$ on $\mathcal{L}$

$$\begin{aligned} \delta\mathcal{L}(\phi, \partial^\mu\phi) &= \frac{\delta\mathcal{L}}{\delta\phi} \delta\phi + \frac{\delta\mathcal{L}}{\delta(\partial^\mu\phi)} \underbrace{\delta(\partial^\mu\phi)}_{\partial^\mu(\delta\phi)} \\ &= \frac{\delta\mathcal{L}}{\delta\phi} \delta\phi + \partial^\mu \left(\frac{\delta\mathcal{L}}{\delta(\partial^\mu\phi)} \delta\phi \right) - \partial^\mu \left(\frac{\delta\mathcal{L}}{\delta(\partial^\mu\phi)} \right) \delta\phi \end{aligned}$$

$u dv = d(uv) - v du$
 $\hookrightarrow \partial^\mu(\delta\phi)$

$$\begin{aligned}\delta \mathcal{L} &= \frac{\delta \mathcal{L}}{\delta \phi} \delta \phi + \partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} \delta \phi \right) - \partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} \right) \delta \phi \\ &= \left(\frac{\delta \mathcal{L}}{\delta \phi} - \cancel{\partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} \right)} \right) \delta \phi + \partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} \delta \phi \right)\end{aligned}$$

EL eq^y $\Rightarrow$ equation of motion

$$\delta \mathcal{L} = \partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} \delta \phi \right)$$

transformation on ϕ leads to this change on $\mathcal{L}$

$$\delta \mathcal{L} = \partial^\mu F_\mu \quad \leftarrow \text{transformation on } \mathcal{L} \text{ directly.}$$

Symmetry = $X(\phi)$

Since we requires $\partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} \delta \phi \right) = \partial^\mu F_\mu$

$$\partial^\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} X(\phi) - F_\mu \right) = 0$$

$\rightarrow$ why?

$\equiv j_\mu = \text{Conserved current}$

$\partial^\mu j_\mu = 0$

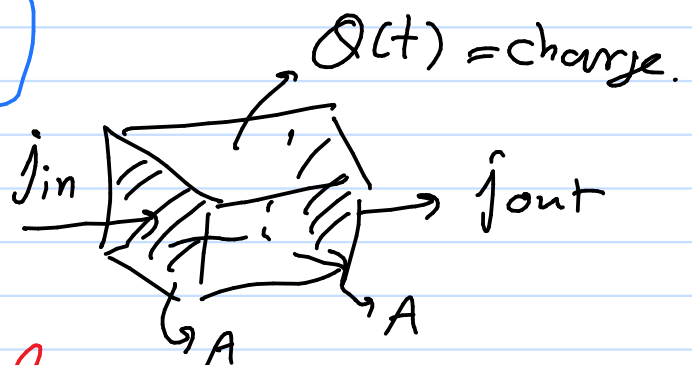
where

$j_\mu \equiv \frac{\delta \mathcal{L}}{\delta (\partial^\mu \phi)} X(\phi) - F_\mu$

We can justify the conserved current from the continuity equation.

$\frac{\partial \rho}{\partial t} = -\vec{\nabla} \cdot \vec{j}$

$$\frac{\partial Q}{\partial t} = -\int d\vec{a} \cdot \vec{j}$$



$$\frac{\partial Q}{\partial t} = -\int dV \vec{\nabla} \cdot \vec{j}$$

Green theorem?
Stoke?

$$\frac{\partial}{\partial t} \int \rho dV = \int dV (\vec{\nabla} \cdot \vec{j})$$

In current = $j_{in} \cdot A$

Out current = $j_{out} \cdot A$

$\therefore \Delta Q(t) = (j_{in} - j_{out}) A \cdot \Delta t$

$$\frac{\partial \rho}{\partial t} = -\vec{\nabla} \cdot \vec{j} \Rightarrow \frac{\partial}{\partial t} \rho + \vec{\nabla} \cdot \vec{j} = 0$$

$$\begin{pmatrix} \frac{\partial}{\partial t} & -\vec{\nabla} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \rho \\ \vec{j} \end{pmatrix} = 0$$

$$\boxed{\partial^\mu j_\mu = 0} \Rightarrow \text{continuity equation} \\ = \text{conserved current}$$

$$j^\mu = (\rho, \vec{j}) \checkmark$$

Since ρ is called charge density, $\therefore Q = \int \rho dV$

$Q = \text{conserved charge}$

$$\boxed{Q \equiv \int j^0 d^3x}$$

$-\infty \rightarrow \infty$

$$\frac{\partial Q}{\partial t} = \int \frac{\partial}{\partial t} j^0 d^3x$$

$$= \int (-\vec{\nabla} \cdot \vec{j}) d^3x$$

$$\frac{\partial Q}{\partial t} = \int d\vec{a} \cdot \vec{j} = 0$$

since the flux at ∞ should vanish

Example Translations (spacetime translation) $\Rightarrow$ Symmetry.

Consider $\mathcal{L}(\phi, \partial^\mu \phi)$

$\epsilon \ll 1$

Translation $x^\mu \rightarrow x'^\mu = x^\mu - \epsilon^\mu$

$$\begin{pmatrix} t \rightarrow t' = t - \epsilon^0 \\ x \rightarrow x' = x - \epsilon^i \end{pmatrix}$$

Field $\phi(x)$ changes under translation as

inverse transf. $\therefore$

$$\phi'(x^\mu) = \phi(x^\mu + \epsilon^\mu) = \phi(x^\mu) + (\partial_\mu \phi) \epsilon^\mu$$

$$f(x + \Delta x) = f(x) + \frac{df}{dx} \cdot \Delta x$$

$$\phi'(x^\mu) = \phi(x^\mu + \varepsilon^\mu) = \phi(x^\mu) + (\partial_\mu \phi) \varepsilon^\mu$$

$$f(x + \Delta x) = f(x) + \frac{df}{dx} \cdot \Delta x$$

$$f(x + \Delta x, y + \Delta y) = f(x, y) + \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y$$

$$\therefore \phi'(x^\mu) = \phi(x^\mu) + \boxed{(\partial_\mu \phi) \varepsilon^\mu} \rightarrow \delta\phi \equiv X(\phi)$$

$$\text{How about } \mathcal{L}'(x^\mu) = \mathcal{L}(x^\mu + \varepsilon^\mu)$$

$$= \mathcal{L}(x^\mu) + \boxed{(\partial_\mu \mathcal{L}) \varepsilon^\mu} = \delta\mathcal{L}$$

Since we need to write in the form of $\partial_\mu F^\mu$

$$\delta\mathcal{L} = \underbrace{\partial_\mu \mathcal{L}}_{\text{constant}} \varepsilon^\mu = \partial_\mu \left(\underbrace{\varepsilon^\nu \delta_\nu^\mu \mathcal{L}}_{\equiv F^\mu} \right)$$

$\therefore$ The conserved current

$$j^\mu = \frac{\delta\mathcal{L}}{\delta(\partial_\mu \phi)} X(\phi) - F^\mu$$

$$= \frac{\delta\mathcal{L}}{\delta(\partial_\mu \phi)} (\partial_\nu \phi \varepsilon^\nu) - \varepsilon^\nu \delta_\nu^\mu \mathcal{L}$$

$$= \varepsilon^\nu \left(\frac{\delta\mathcal{L}}{\delta(\partial_\nu \phi)} \partial_\nu \phi - \delta_\nu^\mu \mathcal{L} \right)$$

$$j^\mu = \varepsilon^\nu T^\mu{}_\nu$$

$T^\mu{}_\nu$ = energy-momentum tensor.

$$\varepsilon^0 \Rightarrow T^\mu{}_0 \rightarrow 1 \text{ current}$$

$$\varepsilon^i \Rightarrow T^\mu{}_i \rightarrow 3 \text{ currents.}$$

4 conserved currents

$$\text{Since } \partial_\mu j^\mu = 0 \Rightarrow \partial_\mu (\varepsilon^\mu T^\mu_\nu) = 0$$

$$\Rightarrow \varepsilon^\nu \partial_\mu T^\mu_\nu = 0$$

$$\therefore \boxed{\partial_\mu T^\mu_\nu = 0}$$

$\therefore T^\mu_\nu$ is conserved currents.
Energy-momentum tensor.

$\cdot \underbrace{T^\mu_0}_{j^\mu} \Rightarrow$ Energy as a conserved charge

$$\underbrace{E}_Q = \int \underbrace{T^0_0}_{j^0} d^3x \Rightarrow \text{Energy of the field } \phi(x)$$

$\underbrace{T^\mu_i}_{j^\mu} \Rightarrow$ Momentum as a conserved charge

$$\underbrace{P_i}_Q = \int \underbrace{T^0_i}_{j^0} d^3x \Rightarrow \text{momentum of } \phi(x)$$

Example Massive scalar field

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$$

$$T^\mu_\nu = \underbrace{\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)}}_{\downarrow} \partial_\nu \phi - \delta_\nu^\mu \mathcal{L}$$

$$= \partial^\mu \phi \partial_\nu \phi - \delta_\nu^\mu \left(\frac{1}{2} \partial_\rho \phi \partial^\rho \phi - \frac{1}{2} m^2 \phi^2 \right)$$

$$E = \int d^3x T^0_0 = \int d^3x \left(\underbrace{\dot{\phi}^2}_{\substack{\partial_0 \phi \partial^0 \phi = \dot{\phi}^2 \\ = 1}} - \left(\frac{1}{2} \partial_\rho \phi \partial^\rho \phi - \frac{1}{2} m^2 \phi^2 \right) \right)$$

$$= \int d^3x \left(\underbrace{\dot{\phi}^2}_{\frac{1}{2} \dot{\phi}^2} - \frac{1}{2} \dot{\phi}^2 + \frac{(\vec{\nabla} \phi)^2}{2} + \frac{1}{2} m^2 \phi^2 \right) = \int d^3x \mathcal{H}$$

which is consistent.

$$P_i = \int d^3x T^0_i = \int d^3x (\partial^0 \phi \partial_i \phi - 0) \quad \delta^0_i = 0$$

$$= \int d^3x \dot{\phi} \partial_i \phi$$

T^μ_ν is an important quantity (physical quantity)

$\Rightarrow$ EM, Gravity

Be careful! , sometimes we use the symmetric term of T^μ_ν as a physical quantity.

* T^μ_ν is not necessarily symmetric, $T^{\mu\nu} \neq T^{\nu\mu}$

but GR: $f(g^{\mu\nu}) = T^{\mu\nu} \Rightarrow$ symmetric
 $\uparrow$ symmetric $\uparrow$ not necessarily.

We can make $T^{\mu\nu}$ symmetric by add a term

$$T^{\mu\nu'} = T^{\mu\nu} + \partial_\rho K^{\rho\mu\nu}, \quad K^{\rho\mu\nu} \text{ is a tensor which is antisym. in } \rho\mu$$

$K^{\rho\mu\nu} = -K^{\mu\rho\nu}$

We can show that the conserved current equation for $T^{\mu\nu}$ & $T^{\mu\nu'}$ is the same.

$$\partial_\mu T^{\mu\nu} = 0 \quad \xrightarrow{\text{we show that}} \quad \partial_\mu T^{\mu\nu'} = 0$$

$$\partial_\mu T^{\mu\nu'} = \partial_\mu (T^{\mu\nu} + \partial_\rho K^{\rho\mu\nu}) = 0$$

$$= \cancel{\partial_\mu T^{\mu\nu}}^0 + \partial_\mu \partial_\rho K^{\rho\mu\nu} = 0$$

$$\partial_\mu T^{\mu\nu} = \partial_\mu (T^{\mu\nu} + \partial_\rho K^{\rho\mu\nu}) = 0$$

$$= \cancel{\partial_\mu T^{\mu\nu}} + \partial_\mu \partial_\rho K^{\rho\mu\nu} = 0$$

$$\partial_\mu \partial_\rho K^{\rho\mu\nu} = \frac{1}{2} \left(\partial_\mu \partial_\rho K^{\rho\mu\nu} + \partial_\rho \partial_\mu K^{\mu\rho\nu} \right)$$

$$= \frac{1}{2} \partial_\mu \partial_\rho \left(K^{\rho\mu\nu} + \underbrace{K^{\mu\rho\nu}}_{-K^{\rho\mu\nu}} \right) = 0$$

equal

$\therefore \boxed{\partial_\mu T^{\mu\nu} = 0} \Rightarrow T^{\mu\nu}$ can be energy-momentum tensor

Homework $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$

Find $T^{\mu\nu} \rightarrow$ make it symmetric

Lorentz symmetry

We know already that field theory has Lorentz symmetry. $\phi(x^\mu) \rightarrow \phi'(x^\mu) = \phi(\Lambda^{-1}{}^\mu{}_\nu x^\nu)$

$$\mathcal{L} \rightarrow \mathcal{L}' \quad S \rightarrow S' = S + \phi \rightarrow 0$$

Also true for the EM field

$$A^\mu(x^\nu) \rightarrow A'^\mu(x^\nu) = \Lambda^\mu{}_\nu A^\nu(\Lambda^{-1}x)$$

What is the conserved currents & conserved charges?

Since we have to do $\phi \rightarrow \phi' + \delta\phi$, $\rightarrow$ infinitesimal transformation of Lorentz

Lorentz transformation

Let's consider rotation $\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \approx \begin{pmatrix} 1 - \frac{\theta^2}{2!} & \theta \\ -\theta & 1 - \frac{\theta^2}{2!} \end{pmatrix}$

infinitesimal
 $\theta \ll 1$

$$\sim \begin{pmatrix} 1 & \theta \\ -\theta & 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{\substack{\mathbb{1} \\ \text{unchange}}} + \underbrace{\begin{pmatrix} 0 & \theta \\ -\theta & 0 \end{pmatrix}}_{\substack{\Delta \\ \text{infinitesimal} \\ \text{change}}}$$

$\therefore$ We start by looking for

$$\Lambda^\mu{}_\nu \cong \underbrace{\delta^\mu{}_\nu}_{\text{unchange}} + \underbrace{\omega^\mu{}_\nu}_{\text{change}}$$

$$x \rightarrow x' = x - a$$

unchange.

Let's consider the rotation matrix

We can always write a rotation matrix in terms of the exponential of a matrix

We can show that

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = e^{\theta J}, \text{ where } J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

? what is exponential of a matrix?

square matrix $\rightarrow$

$$e^X \equiv \mathbb{1} + X + \frac{X \cdot X}{2!} + \frac{X \cdot X \cdot X}{3!} + \dots$$

$$(e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots)$$

Let's check that the above relation is true

$$e^{\theta J} = \mathbb{1} + \theta J + \frac{\theta^2 J \cdot J}{2!} + \frac{\theta^3 J \cdot J \cdot J}{3!} + \dots$$

Let's check that the above relation is true

$$e^{\theta J} = 1 + \theta J + \frac{\theta^2}{2!} J \cdot J + \frac{\theta^3}{3!} J \cdot J \cdot J + \dots$$

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J \cdot J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -1$$

$$\therefore \underline{J \cdot J \cdot J} = -1 \cdot J = -J, \quad \underline{J \cdot J \cdot J \cdot J} = -1 \cdot (-1) = 1$$

$$e^{\theta J} = 1 + \theta J - \frac{\theta^2}{2!} 1 - \frac{\theta^3}{3!} J + \dots$$

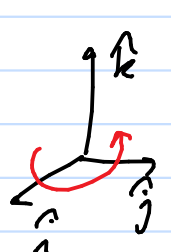
$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \theta \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} - \frac{\theta^2}{2!} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{\theta^3}{3!} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 - \frac{\theta^2}{2!} + \dots & -\theta + \frac{\theta^3}{3!} + \dots \\ \theta - \frac{\theta^3}{3!} + \dots & 1 - \frac{\theta^2}{2!} + \dots \end{pmatrix} \xrightarrow{\text{Taylor series of sin \& cos}} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$$\therefore e^{\theta J} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

parameter θ $\rightarrow$ generator J $\rightarrow$ element of rotation group $\rightarrow$ Lie group

Let's do the rotation in 2D of 3D vectors



$$\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = e^{\theta J}$$

$$J = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Try checking this by yourself!

How about the boost parts of Lorentz transf?.

Consider $\begin{pmatrix} \cosh \eta & \sinh \eta \\ \sinh \eta & \cosh \eta \end{pmatrix} = e^{\eta K}$

we can write in terms of a generator K

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K \cdot K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbb{1}$$

$$\therefore K \cdot K \cdot K = \mathbb{1} \cdot K = K, \quad K \cdot K \cdot K \cdot K = \mathbb{1} \cdot \mathbb{1} = \mathbb{1}$$

$$e^{\eta K} = \mathbb{1} + \eta K + \frac{\eta^2}{2!} K \cdot K + \frac{\eta^3}{3!} K \cdot K \cdot K$$

$$= \mathbb{1} + \eta K + \frac{\eta^2}{2!} \cdot \mathbb{1} + \frac{\eta^3}{3!} K$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \eta \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \frac{\eta^2}{2!} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{\eta^3}{3!} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 + \frac{\eta^2}{2!} & \eta + \frac{\eta^3}{3!} \\ \eta + \frac{\eta^3}{3!} & 1 + \frac{\eta^2}{2!} \end{pmatrix} \xrightarrow{\text{Taylor series of cosh, sinh}} = \begin{pmatrix} \cosh \eta & \sinh \eta \\ \sinh \eta & \cosh \eta \end{pmatrix}$$

We can extend this result (and rotation) into 4D

For rotation, we have 3 ways to rotate

1 - plane xy , 2 - plane yz , 3 - plane xz

$$J_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} t \\ x \\ y \\ z \end{matrix}, \quad J_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{matrix} t \\ x \\ y \\ z \end{matrix}, \quad J_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \begin{matrix} t \\ x \\ y \\ z \end{matrix}$$

rotation axis = 3 rotation axis = 1 rotation axis = 2

$\therefore$ Rotation matrix in 4D = $e^{\theta_1 J_1 + \theta_2 J_2 + \theta_3 J_3}$

$\underbrace{\theta_1 J_1 + \theta_2 J_2 + \theta_3 J_3}_{\text{generators}}$
3 parameters

We can do the same for boost! $\rightarrow$ 3 directions

boost in $\hat{x}$ boost in $\hat{y}$ boost in $\hat{z}$

$$K_1 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad K_2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad K_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

boost matrix = $e^{\eta_1 K_1 + \eta_2 K_2 + \eta_3 K_3}$

$\underbrace{\eta_1 K_1 + \eta_2 K_2 + \eta_3 K_3}_{\text{generators}}$
Parameters

$\therefore$ Full Lorentz transf. requires 6 parameters
(6 generators)

$(\Lambda)^\mu_\nu = \left(e^{\sum_i^3 \theta_i J_i + \sum_i^3 \eta_i K_i} \right)^\mu_\nu$

not the 4 vector language!

We can define the above expression in the 4-vector language!

$$\Lambda^\mu_\nu = \left(e^{\frac{1}{2} \theta_{\rho\sigma} M^{\rho\sigma}} \right)^\mu_\nu$$

$\theta_{\rho\sigma}$ contain θ_i & η_i
 $M^{\rho\sigma}$ contain J_i & K_i

$\theta_{\rho\sigma} = -\theta_{\sigma\rho}$ & $M^{\rho\sigma} = -M^{\sigma\rho}$ ← matrix

number antisymmetric in $\rho \leftrightarrow \sigma$

Where

$$(M^{\rho\sigma})^\mu_\nu \equiv g^{\rho\mu} g^\sigma_\nu - g^{\sigma\mu} g^\rho_\nu$$

$$\Lambda^\mu{}_\nu = \left(e^{\frac{1}{2} \theta_{\rho\sigma} M^{\rho\sigma}} \right)_\nu^\mu$$

$\theta_{\rho\sigma}$ contain θ_i & η_i

$M^{\rho\sigma}$ contain J_i & K_i

$\theta_{\rho\sigma} = -\theta_{\sigma\rho}$ & $M^{\rho\sigma} = -M^{\sigma\rho}$ ← matrix

number antisymmetric in $\rho \leftrightarrow \sigma$

Where

$$(M^{\rho\sigma})^\mu{}_\nu \equiv g^{\rho\mu} \delta^\sigma{}_\nu - g^{\sigma\mu} \delta^\rho{}_\nu$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$(M^{01})^\mu{}_\nu = \underbrace{g^{0\mu}}_{\mu=0, \nu=1} \delta^1{}_\nu - \underbrace{g^{1\mu}}_{\mu=1, \nu=0} \delta^0{}_\nu$$

$$g^{00} \delta^1{}_1 = 1 \cdot 1$$

$$-(g^{11}) \cdot \delta^0{}_0 = -1(-1) \cdot 1 = 1$$

$$= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = K^1$$

$$(M^{12})^\mu{}_\nu = \underbrace{g^{1\mu}}_{\mu=1, \nu=2} \delta^2{}_\nu - \underbrace{g^{2\mu}}_{\mu=2, \nu=1} \delta^1{}_\nu$$

$$= g^{11} \cdot \delta^2{}_2 = -1 \cdot 1 = -1$$

$$= -(g^{22}) \delta^1{}_1 = (-1)(-1)(1) = 1$$

$$= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = J_3$$

$\therefore e^{\frac{1}{2} \theta_{\rho\sigma} M^{\rho\sigma}}$ since $\theta_{\rho\sigma}$ & $M^{\rho\sigma}$ antisym $\theta_{00} = \theta_{11} = \dots = 0$

$$e^{\frac{1}{2} \theta_{\rho\sigma} M^{\rho\sigma}} = e^{\frac{1}{2} (\theta_{01} M^{01} + \theta_{02} M^{02} + \theta_{03} M^{03} + \dots)}$$

$e^{\frac{1}{2}\Theta_{\rho\sigma}M^{\rho\sigma}}$ since $\Theta_{\rho\sigma} \propto M^{\rho\sigma}$ antisym $\Theta_{00} = \Theta_{11} = \dots = 0$

$$\begin{aligned}
 e^{\frac{1}{2}\Theta_{\rho\sigma}M^{\rho\sigma}} &= e^{\frac{1}{2}(\Theta_{01}M^{01} + \Theta_{02}M^{02} + \Theta_{03}M^{03} + \Theta_{12}M^{12} + \Theta_{13}M^{13} \\
 &\quad + \Theta_{23}M^{23} + \Theta_{32}M^{32} + \Theta_{31}M^{31} + \Theta_{30}M^{30} \\
 &\quad + \Theta_{21}M^{21} + \Theta_{20}M^{20} + \Theta_{10}M^{10})} \\
 &\quad \quad \quad \begin{array}{c} \underbrace{-\Theta_{02}M^{02}}_{\text{blue}} \quad \underbrace{-\Theta_{01}M^{01}}_{\text{blue}} \\ \underbrace{-\Theta_{12}M^{12}}_{\text{blue}} \quad \underbrace{-\Theta_{13}M^{13}}_{\text{blue}} \end{array} \\
 &= e^{\underbrace{\Theta_{01}M^{01} + \Theta_{02}M^{02} + \Theta_{03}M^{03}}_{\eta_i K_i} + \underbrace{\Theta_{12}M^{12} + \Theta_{13}M^{13} + \Theta_{23}M^{23}}_{\Theta_i J_i}}
 \end{aligned}$$

which is the full Lorentz transf.

$$\therefore \Lambda^\mu{}_\nu = \left(e^{\frac{1}{2}\Theta_{\rho\sigma}M^{\rho\sigma}} \right)^\mu{}_\nu \quad \downarrow \text{infinitesimal}$$

$$\Lambda^\mu{}_\nu = \left(\mathbb{1} + \frac{1}{2}\Theta_{\rho\sigma}M^{\rho\sigma} + \dots \right)^\mu{}_\nu$$

$$= \delta^\mu{}_\nu + \frac{1}{2}\Theta_{\rho\sigma}(M^{\rho\sigma})^\mu{}_\nu$$

$$\Lambda^\mu{}_\nu \cong \delta^\mu{}_\nu + \omega^\mu{}_\nu$$

$$\therefore \boxed{\omega^\mu{}_\nu = \frac{1}{2}\Theta_{\rho\sigma}(M^{\rho\sigma})^\mu{}_\nu}$$

Notice that $\omega^{\mu\nu}$ is antisymmetric: $\omega^{\mu\nu} = -\omega^{\nu\mu}$

From, Lorentz transformation property:

$$\Lambda^\mu{}_\sigma g^{\sigma\rho} \Lambda^\nu{}_\rho = g^{\mu\nu}$$

$$(\delta^\mu{}_\sigma + \omega^\mu{}_\sigma) g^{\sigma\rho} (\delta^\rho{}_\nu + \omega^\rho{}_\nu) = g^{\mu\nu}$$

From, Lorentz transformation property:

$$\Lambda^\mu_\sigma g^{\sigma\rho} \Lambda^\nu_\rho = g^{\mu\nu} \quad \text{approx.}$$

$$(\delta^\mu_\sigma + \omega^\mu_\sigma) g^{\sigma\rho} (\delta^\nu_\rho + \omega^\nu_\rho) = g^{\mu\nu}$$

$$\delta^\mu_\sigma g^{\sigma\rho} \delta^\nu_\rho + \omega^\mu_\sigma g^{\sigma\rho} \delta^\nu_\rho + \delta^\mu_\sigma g^{\sigma\rho} \omega^\nu_\rho$$

$$+ \omega^\mu_\sigma g^{\sigma\rho} \omega^\nu_\rho + \dots = g^{\mu\nu}$$

$$\underline{g^{\mu\nu}} + \underbrace{\omega^{\mu\nu} + \omega^{\nu\mu}}_{=0} + \mathcal{O}(\omega^2) = \underline{g^{\mu\nu}}$$

$$\Rightarrow \omega^{\mu\nu} + \omega^{\nu\mu} = 0 \Rightarrow \omega^{\mu\nu} = -\omega^{\nu\mu}$$

Let's consider the inverse transform

$$\Lambda^\mu_\nu = \delta^\mu_\nu + \omega^\mu_\nu$$

we can see that the inverse is simply

$$\boxed{(\Lambda^{-1})^\mu_\nu = \delta^\mu_\nu - \omega^\mu_\nu}$$

$$(\Lambda)^\mu_\sigma (\Lambda^{-1})^\sigma_\nu = (\delta^\mu_\sigma + \omega^\mu_\sigma) (\delta^\sigma_\nu - \omega^\sigma_\nu)$$

$$= \delta^\mu_\sigma \delta^\sigma_\nu + \omega^\mu_\sigma \delta^\sigma_\nu - \delta^\mu_\sigma \omega^\sigma_\nu$$

$$+ \mathcal{O}(\omega^2)$$

$$= \delta^\mu_\nu + \cancel{\omega^\mu_\nu} - \cancel{\omega^\mu_\nu}$$

$$\Lambda \Lambda^{-1} = \mathbb{1}$$

Let's consider the effect of Lorentz transf. on $\phi(x)$

$$\begin{aligned}\phi(x^\mu) &\rightarrow \phi'(x^\mu) = \phi((\Lambda^{-1})^\mu{}_\nu x^\nu) \\ &= \phi((\delta^\mu{}_\nu - \omega^\mu{}_\nu) x^\nu) \\ &= \phi(\underbrace{x^\mu - \omega^\mu{}_\nu x^\nu}_{\equiv \vartheta^\mu}) \quad \text{Taylor expansion}\end{aligned}$$

$$= \phi(x^\mu) + \vartheta^\mu \partial_\mu \phi(x^\mu) + \dots$$

$$\phi'(x) = \phi(x^\mu) - \underbrace{\omega^\mu{}_\nu x^\nu \partial_\mu \phi}_{\equiv \delta\phi}$$

$$\boxed{\delta\phi = -\omega^\mu{}_\nu x^\nu \partial_\mu \phi}$$

What about the Lagrangian?

$$\begin{aligned}\mathcal{L}(x^\mu) &\rightarrow \mathcal{L}'(x^\mu) = \mathcal{L}((\Lambda^{-1})^\mu{}_\nu x^\nu) \\ &= \mathcal{L}((\delta^\mu{}_\nu - \omega^\mu{}_\nu) x^\nu) \\ &= \mathcal{L}(x^\mu - \omega^\mu{}_\nu x^\nu) \\ &= \mathcal{L}(x) - \underbrace{\omega^\mu{}_\nu x^\nu \partial_\mu \mathcal{L}(x)}_{\delta\mathcal{L}}\end{aligned}$$

but now we have

$$\delta\mathcal{L} = -\omega^\mu{}_\nu x^\nu \partial_\mu \mathcal{L}$$

which is not in the form $\partial_\mu F^\mu$

$$\delta \mathcal{L} = - \omega^\mu{}_\nu x^\nu \partial_\mu \mathcal{L}$$

which is not in the form $\partial_\mu F^\mu$

$\therefore$ we can rearrange a bit!

$$\delta \mathcal{L} = \underbrace{-\omega^\mu{}_\nu x^\nu}_{u} \underbrace{\partial_\mu \mathcal{L}}_v = \partial_\mu \underbrace{(-\omega^\mu{}_\nu x^\nu \mathcal{L})}_{uv} - \mathcal{L} \underbrace{\partial_\mu (-\omega^\mu{}_\nu x^\nu)}_v$$

$$= \partial_\mu \underbrace{(-\omega^\mu{}_\nu x^\nu \mathcal{L})}_{F^\mu} + \mathcal{L} \underbrace{\omega^\mu{}_\nu \partial_\mu x^\nu}_{\delta^\nu_\mu}$$

$\omega = \text{const.}$
 $\frac{\partial x^\nu}{\partial x^\mu} = \delta^\nu_\mu$

$$\delta \mathcal{L} = \partial_\mu \underbrace{(-\omega^\mu{}_\nu x^\nu \mathcal{L})}_{F^\mu}$$

$\hookrightarrow$ symmetry.

$\omega^\mu{}_\nu = \omega^{\mu\nu} g_{\nu\mu}$

$\omega^{\mu\nu}$ antisymmetric.
 $g_{\mu\nu}$ symmetric

$$= \frac{1}{2} (\omega^{\mu\nu} g_{\nu\mu} + \omega^{\nu\mu} g_{\mu\nu}) = 0$$

$-\omega^{\mu\nu} g_{\nu\mu}$

$\therefore$ The conserved currents

$$j^\mu = \frac{\delta \mathcal{L}}{\delta (\partial_\mu \phi)} \cdot \delta \phi - F^\mu$$

$$= \frac{\delta \mathcal{L}}{\delta (\partial_\mu \phi)} (-\omega^\mu{}_\nu x^\nu \partial_\mu \phi) + \omega^\mu{}_\nu x^\nu \mathcal{L}$$

Since $\omega^\mu{}_\nu$ has 6 non zero component!

$$\omega^{\mu\nu} = \text{antisymmetric} = \begin{pmatrix} 0 & \omega^{01} & \omega^{02} & \omega^{03} \\ -\omega^{01} & 0 & \omega^{12} & \omega^{13} \\ \omega^{02} & -\omega^{12} & 0 & \omega^{23} \\ -\omega^{03} & -\omega^{13} & -\omega^{23} & 0 \end{pmatrix}$$

$\Rightarrow$ We expect 6 currents

$$\therefore j^\mu = \omega^\mu{}_\nu \left[\frac{\delta \mathcal{L}}{\delta (\partial_\mu \phi)} (x^\nu \partial_\nu \phi) + \delta^\mu{}_\nu x^\nu \mathcal{L} \right]$$

$\omega^\mu{}_\nu$ constant

$$j^\mu = \omega^{\rho}{}_{\nu} \left[-\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} (\cancel{x^\nu} \partial_\rho \phi) + \delta^\mu_{\cancel{\rho}} \cancel{x^\nu} \mathcal{L} \right]$$

constant

$$= \omega^{\rho}{}_{\nu} \left[-\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \partial_\rho \phi + \delta^\mu_{\rho} \mathcal{L} \right] x^\nu$$

$$\left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \partial_\nu \phi - \delta^\mu_{\nu} \mathcal{L} \right)$$

III $T^\mu{}_{\nu}$ = energy-momentum tensor.

$-T^\mu{}_{\rho}$ = energy momentum tensor.

$$j^\mu = -\omega^{\rho}{}_{\nu} T^\mu{}_{\rho} x^\nu$$

$$\partial_\mu j^\mu = 0 \Rightarrow -\cancel{\omega^{\rho}{}_{\nu}} \partial_\mu (T^\mu{}_{\rho} x^\nu) = 0$$

$$\cancel{*} \partial_\mu (T^\mu{}_{\rho} x^\nu) = 0$$

↳ currents associated with Lorentz transf.

Since ρ & ν are not antisymmetric (not 6 currents)

∴ we can construct 6 currents as

$$J^{\mu\rho\nu} \equiv T^{\mu\rho} x^\nu - T^{\mu\nu} x^\rho$$

$$\partial_\mu J^{\mu\rho\nu} = \partial_\mu (\cancel{T^{\mu\rho}} x^\nu) - \partial_\mu (\cancel{T^{\mu\nu}} x^\rho) = 0$$

∴ $J^{\mu\rho\nu}$ is still a current!

$$J^{\mu\rho\nu} \equiv T^{\mu\rho}x^\nu - T^{\mu\nu}x^\rho$$

$J^{\mu\rho\nu}$ is antisymmetric in $\rho\nu$

$$J^{\mu\rho\nu} = -J^{\mu\nu\rho}$$

There are 6 currents in $J^{\mu\rho\nu}$

$$\underbrace{J^{\mu 01}, J^{\mu 02}, J^{\mu 03}}_{\text{boost}}, \quad \underbrace{J^{\mu 12}, J^{\mu 13}, J^{\mu 23}}_{\text{rotation}}$$

Conserved charge of rotation $\leadsto$ angular momentum.

$$\begin{aligned} Q^{12} &= \int d^3x J^{012} = \int d^3x \left(\underset{\sim p^1}{T^{01}} x^2 - \underset{\sim p^2}{T^{02}} x^1 \right) \\ &= p^1 x^2 - p^2 x^1 \sim (\vec{x} \times \vec{p})_3 = \vec{L}_3 \end{aligned}$$

$$Q^{13} \sim \vec{L}_2, \quad Q^{23} \sim \vec{L}_1$$

Conserved charge of boost?

$$\begin{aligned} Q^{01} &= \int d^3x J^{0\underline{0}1} = \int d^3x \left(\underset{\sim E}{T^{0\underline{0}}} \underline{x^1} - \underset{\sim p}{T^{01}} \underline{x^0} \right) \\ &\sim \underline{E} x^1 - p^1 t \end{aligned}$$

Set $t=0$, $\underline{\gamma_m x^1}$ = center of mass

center of mass is invariant.

Internal Symmetry

Some symmetry is not in the form of coordinate transformation. Symmetry which involves only with

the field transformation = internal symmetry

$$A^\mu = (V, \vec{A})$$
$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{pmatrix}$$

Example $\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu}$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

Notice that we can arbitrarily choose a function

$$\Lambda(x^\mu). \quad A^\mu \rightarrow A'^\mu = A^\mu + \partial^\mu \Lambda(x)$$

symmetry
internal

Note that the transformation involves only A^μ (not x^μ)

$$F^{\mu\nu} \rightarrow F'^{\mu\nu} = \partial^\mu A'^\nu - \partial^\nu A'^\mu$$

$$= \partial^\mu (A^\nu + \partial^\nu \Lambda(x)) - \partial^\nu (A^\mu + \partial^\mu \Lambda(x))$$

$$= \partial^\mu A^\nu + \cancel{\partial^\mu \partial^\nu \Lambda} - \partial^\nu A^\mu - \cancel{\partial^\nu \partial^\mu \Lambda}$$

$$= \partial^\mu A^\nu - \partial^\nu A^\mu = F^{\mu\nu} \quad \text{invariant}$$

$$\mathcal{L} \rightarrow \mathcal{L}' = -\frac{1}{4} F'^{\mu\nu} F'_{\mu\nu} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} = \mathcal{L}$$

$\therefore$ This transformation is the symmetry of the system.

This is called gauge symmetry

$$A^\mu \rightarrow A'^\mu = A^\mu + \partial^\mu \Lambda(x)$$

This is the shift in EM potential

$$V(x) \rightarrow V'(x) = V(x) + C \Rightarrow \vec{E} \rightarrow \vec{E}' = -\vec{\nabla} V' = -\vec{\nabla} V = \vec{E}$$

Another example: complex scalar fields

Consider

$$\mathcal{L}(\phi_1, \phi_2) = \underbrace{\frac{1}{2} \partial_\mu \phi_1 \partial^\mu \phi_1 - \frac{1}{2} m^2 \phi_1^2}_{\mathcal{L}_1} + \underbrace{\frac{1}{2} \partial_\mu \phi_2 \partial^\mu \phi_2 - \frac{1}{2} m^2 \phi_2^2}_{\mathcal{L}_2}$$

We can show that ϕ_1 & ϕ_2 can be written in terms of $\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$ Complex scalar field!

$$\mathcal{L}(\phi) = (\partial_\mu \phi)^* (\partial^\mu \phi) - m^2 \phi^* \phi$$

$$\begin{aligned} \mathcal{L}(\phi) &= \partial_\mu \left(\frac{1}{\sqrt{2}} \phi_1 - \frac{i}{\sqrt{2}} \phi_2 \right) \partial^\mu \left(\frac{1}{\sqrt{2}} \phi_1 + \frac{i}{\sqrt{2}} \phi_2 \right) \\ &\quad - m^2 \left(\frac{1}{\sqrt{2}} \phi_1 - \frac{i}{\sqrt{2}} \phi_2 \right) \left(\frac{1}{\sqrt{2}} \phi_1 + \frac{i}{\sqrt{2}} \phi_2 \right) \\ &= \frac{1}{2} \partial_\mu \phi_1 \partial^\mu \phi_1 - \cancel{\frac{i}{2} \partial_\mu \phi_2 \partial^\mu \phi_1} + \cancel{\frac{i}{2} \partial_\mu \phi_1 \partial^\mu \phi_2} \\ &\quad + \frac{1}{2} \partial_\mu \phi_2 \partial^\mu \phi_2 - \cancel{\frac{m^2}{2} \phi_1^2} - \cancel{\frac{i m^2}{2} \phi_2 \phi_1} \\ &\quad + \cancel{\frac{i m^2}{2} \phi_1 \phi_2} - \cancel{\frac{m^2}{2} \phi_2^2} \end{aligned}$$

$$= \frac{1}{2} \partial_\mu \phi_1 \partial^\mu \phi_1 + \frac{1}{2} \partial_\mu \phi_2 \partial^\mu \phi_2 - \frac{m^2}{2} \phi_1^2 - \frac{m^2}{2} \phi_2^2 \neq$$

$$\mathcal{L}(\phi) = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi \Rightarrow \text{complex massive scalar field}$$

The most general form

$$\mathcal{L}(\phi) = \partial_\mu \phi^* \partial^\mu \phi - V(\phi^* \phi) \quad \hookrightarrow \quad m^2 \phi^* \phi + \frac{\lambda}{4!} (\phi^* \phi)^2$$

The most general form

$$\mathcal{L}(\phi) = \partial_\mu \phi^* \partial^\mu \phi - V(\phi^* \phi)$$

Let's consider the transformation (phase transformation)

$$\phi \rightarrow \phi' = e^{i\theta} \phi \quad \theta = \text{const}$$

This is a symmetry of the system!

$$\begin{aligned} \mathcal{L} \rightarrow \mathcal{L}' &= \partial_\mu \phi'^* \partial^\mu \phi' - V(\phi'^* \phi') \\ &= \partial_\mu (\cancel{e^{-i\theta}} \phi) \partial^\mu (\cancel{e^{i\theta}} \phi) - V(\cancel{e^{-i\theta}} \phi \cancel{e^{i\theta}} \phi) \\ &= \partial_\mu \phi^* \partial^\mu \phi - V(\phi^* \phi) = \mathcal{L} \end{aligned}$$

$\therefore \mathcal{L}$ invariant $\Rightarrow$ equation of motion is also invariant.

Let's apply Noether theorem onto this symmetry.

$$\begin{aligned} \delta\phi \equiv X : \quad \phi \rightarrow \phi' = e^{i\theta} \phi &\approx (1 + i\theta + \dots) \phi \\ &= \underbrace{\phi + i\theta\phi}_{\phi + \delta\phi} \rightarrow \delta\phi = X \\ &\quad \delta\phi = i\theta\phi \end{aligned}$$

but there are 2 fields $\Rightarrow \phi, \phi^*$

$$\begin{aligned} \delta\phi^* \equiv X^* : \quad \phi^* \rightarrow \phi'^* &= (\phi')^* = e^{-i\theta} \phi^* \\ &= (1 - i\theta + \dots) \phi^* \\ &= \phi^* - i\theta\phi^* = \delta\phi^* = -i\theta\phi^* \end{aligned}$$

$$F^\mu = ? : \quad \mathcal{L} \rightarrow \mathcal{L}' = \mathcal{L} \quad \therefore \delta\mathcal{L} = 0 \quad (= \partial_\mu F^\mu)$$

No F^μ is needed! ($F^\mu = \text{const}$)

We gather all of this into Noether current

$$F^\mu = ? : \mathcal{L} \rightarrow \mathcal{L}' = \mathcal{L} \therefore \delta \mathcal{L} = 0 \quad (= \partial_\mu F^\mu)$$

No F^μ is needed! ($F^\mu = \text{const}$)

We gather all of this into Noether current

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \cdot \delta \phi - \cancel{F^\mu}^0 + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} \cdot \delta \phi^* - \cancel{F^\mu}^0$$

$$= \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} (i\theta \phi) + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} (-i\theta \phi^*)$$

If we use $\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - V(\phi^* \phi)$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi^*, \quad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} = \partial^\mu \phi$$

$$\therefore j^\mu = \partial^\mu \phi^* (i\theta \phi) - \partial^\mu \phi (i\theta \phi^*)$$

$$= \theta (i\phi \partial^\mu \phi^* - i\phi^* \partial^\mu \phi)$$

Since $\theta = \text{const.} \Rightarrow$ it doesn't matter in the current

$$\therefore j^\mu \equiv i\phi \partial^\mu \phi^* - i\phi^* \partial^\mu \phi$$

Noether current of phase symmetry.

$$Q = \int d^3x j^0 = \int d^3x (i\phi \dot{\phi}^* - i\phi^* \dot{\phi}) \quad \cdot \equiv \frac{\partial}{\partial t}$$

Midterm Exam coming up

Quantization of Field

We need to promote classical field $\rightarrow$ quantum field

In terms of structure, we can use what we have learnt in CM $\rightarrow$ QM

$$\text{CM: } \left\{ \overset{g(q,p), f(q,p)}{\cancel{f(q,p)}, \cancel{g(q,p)}} \right\}_{\text{P.B.}} = \frac{\partial f}{\partial p} \cdot \frac{\partial g}{\partial q} - \frac{\partial f}{\partial q} \cdot \frac{\partial g}{\partial p}$$

$$\text{Equation of motions} \Rightarrow \boxed{\frac{\partial H}{\partial p} = \dot{q}, \quad \frac{\partial H}{\partial q} = -\dot{p}}$$

Side note. $L = \frac{1}{2} m \dot{q}^2 - \frac{k}{2} q^2 \Rightarrow H(q, p)$

$$p = \frac{\partial L}{\partial \dot{q}} = m \dot{q}, \quad H = p \dot{q} - L = \frac{p^2}{m} - \frac{p^2}{2m} + \frac{k q^2}{2}$$

$$\therefore H = \frac{p^2}{2m} + \frac{k q^2}{2}$$

$$\frac{\partial H}{\partial p} = \frac{p}{m} = \dot{q}, \quad \frac{\partial H}{\partial q} = kq = -\dot{p} \Rightarrow F = -kq \text{ (spring force!)} \quad \frac{dp}{dt} \rightarrow$$

equation of motion + Poisson bracket

$$\left\{ \overset{f(p,q), H(p,q)}{\cancel{H(p,q)}, \cancel{f(p,q)}} \right\}_{\text{P.B.}} = \frac{\partial H}{\partial p} \frac{\partial f}{\partial q} - \frac{\partial f}{\partial p} \frac{\partial H}{\partial q} \\ = \dot{q} \frac{\partial f}{\partial q} + \frac{\partial f}{\partial p} \dot{p}$$

$$= \frac{\partial f}{\partial q} \frac{dq}{dt} + \frac{\partial f}{\partial p} \frac{dp}{dt} = \frac{df(q,p)}{dt}$$

$$\therefore \frac{df}{dt} = \left\{ \overset{f(p,q), H(p,q)}{\cancel{H(p,q)}, \cancel{f(p,q)}} \right\}_{\text{P.B.}} \Rightarrow \text{Hamiltonian is responsible for the dynamics}$$

$$\therefore f(q,p) = \text{invariant (no change in time)} \quad \text{if} \quad \left\{ \overset{f(p,q), H(p,q)}{\cancel{H(p,q)}, \cancel{f(p,q)}} \right\}_{\text{P.B.}} = 0$$

CM: $\{q, q\}_{P.B.} = 0, \{p, p\}_{P.B.} = 0$

$$\{q, p\}_{P.B.} = \frac{\partial q}{\partial q} \cdot \frac{\partial p}{\partial p} - \frac{\partial q}{\partial p} \cdot \frac{\partial p}{\partial q} = 1$$

In order to promote CM $\rightarrow$ QM

1). $q, p \rightarrow \hat{x}, \hat{p}$

2). $\{, \}_{P.B.} \rightarrow [,]$ commutator.

Quantization!

CM		QM
$\{q, p\}_{P.B.} = 1$	$\Rightarrow$	$[\hat{x}, \hat{p}] = i$
$\{q, q\}_{P.B.} = \{p, p\}_{P.B.} = 0$		$[\hat{x}, \hat{x}] = [\hat{p}, \hat{p}] = 0$
$\frac{df(q, p)}{dt} = \{f, H\}_{P.B.}$		$\frac{d\hat{A}(\hat{x}, \hat{p})}{dt} = [\hat{A}, \hat{H}]$
		Heisenberg picture

Field Theory version

We have $\phi(\vec{x})$ & $\pi(\vec{x})$

Classical field

conjugate momentum.

Quantization: $\phi(\vec{x}) \rightarrow \hat{\phi}(\vec{x})$
 $\pi(\vec{x}) \rightarrow \hat{\pi}(\vec{x})$ } operators

Impose $[\hat{\phi}(\vec{x}), \hat{\phi}(\vec{y})] = 0, [\hat{\pi}(\vec{x}), \hat{\pi}(\vec{y})] = 0$

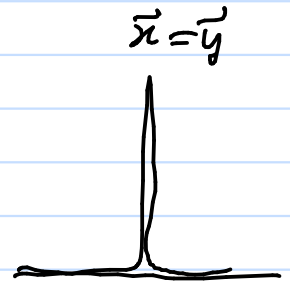
$$[\hat{\phi}(\vec{x}), \hat{\pi}(\vec{y})] = i\delta^{(3)}(\vec{x} - \vec{y})$$

Quantization: $\phi(\vec{x}) \rightarrow \hat{\phi}(\vec{x})$
 $\pi(\vec{x}) \rightarrow \hat{\pi}(\vec{x})$ } operators

Impose $[\hat{\phi}(\vec{x}), \hat{\phi}(\vec{y})] = 0, [\hat{\pi}(\vec{x}), \hat{\pi}(\vec{y})] = 0$

$[\hat{\phi}(\vec{x}), \hat{\pi}(\vec{y})] = i\delta^{(3)}(\vec{x}-\vec{y})$

$\delta^{(3)}(\vec{x}-\vec{y}) = \begin{cases} \infty, & \vec{x}-\vec{y}=0 \ (\vec{x}=\vec{y}) \\ 0, & \vec{x} \neq \vec{y} \end{cases}$



* Now, we only consider the Schrödinger picture

$\hat{\phi}(\vec{x})$ is time independent

state vector is time depends
 $|\psi(t)\rangle$ (dynamics)
 $\hat{H}|\psi\rangle = i\hbar \frac{\partial}{\partial t} |\psi\rangle$ schrödinger

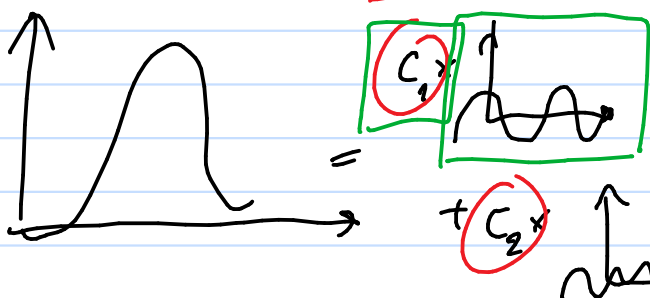
but operator is time inde.

The problem is that $\hat{H} = \frac{1}{2} \hat{\pi}^2 + \frac{1}{2} (\vec{\nabla} \hat{\phi})^2 + \frac{1}{2} m^2 \hat{\phi}^2$

↑ difficult!

contains infinite number of degree of freedom.

$\phi(\vec{x})$ has ∞ d.o.f. $\Rightarrow$ Fourier transform



$\Rightarrow$ coefficient in wave mode. $\rightarrow \infty$

* It's difficult to find the eigenvalues (vectors) of $\hat{H}$ for a system of ∞ d.o.f.

$\Rightarrow$ Solution $\Rightarrow$ Perform Fourier transform.

Let's consider the Fourier transform of Classical field

$$\phi(\vec{x}) = \underbrace{\int \frac{d^3p}{(2\pi)^3}}_{\text{Summation}} \underbrace{e^{i\vec{p}\cdot\vec{x}}}_{\text{basis}} \cdot \underbrace{\tilde{\phi}(\vec{p})}_{\text{component}} = \sum c_n \sin(\omega_n t)$$

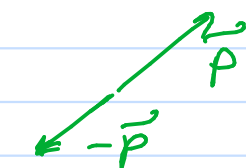
/ basis component (coefficient)

We consider the real scalar field

$$\phi(\vec{x})^* = \phi(\vec{x})$$

$$\phi(\vec{x})^* = \int \frac{d^3p}{(2\pi)^3} (e^{-i\vec{p}\cdot\vec{x}}) \tilde{\phi}^*(\vec{p})$$

$$= \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p}'\cdot\vec{x}} \tilde{\phi}^*(-\vec{p}')$$



$$\vec{p} \rightarrow -\vec{p}$$

$$\vec{p}' = -\vec{p}$$

$$\int_{-\infty}^{\infty} d^3\vec{p} = \int_{-\infty}^{\infty} d^3\vec{p}'$$

$$\vec{p} = \vec{p}'$$

$$= \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p}\cdot\vec{x}} \tilde{\phi}^*(-\vec{p})$$

must be equal

$$\phi(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p}\cdot\vec{x}} \tilde{\phi}(\vec{p})$$

$$\therefore \boxed{\tilde{\phi}^*(\vec{p}) = \tilde{\phi}(-\vec{p})}^*$$

Let's consider the equation of motion for $\tilde{\phi}(\vec{p})$

Massive real scalar field $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$

E.o.m: $\partial_\mu \partial^\mu \phi + m^2 \phi = 0$

Let's consider the equation of motion for $\tilde{\phi}(\vec{p})$

Massive real scalar field $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$

E.o.m: $\partial_\mu \partial^\mu \phi + m^2 \phi = 0$

$\downarrow$
 $\left(\frac{\partial^2}{\partial t^2} - \vec{\nabla}^2 + m^2 \right) \phi(\vec{x}) = 0$ } Fourier transform

$\left(\frac{\partial^2}{\partial t^2} - \vec{\nabla}^2 + m^2 \right) \int \frac{d^3 \vec{p}}{(2\pi)^3} e^{i\vec{p} \cdot \vec{x}} \tilde{\phi}(\vec{p}) = 0$

$\vec{\nabla} e^{i\vec{p} \cdot \vec{x}} = \hat{i} \partial_x e^{i(p_x x + p_y y + p_z z)} + \hat{j} \partial_y e^{i(p_x x + p_y y + p_z z)} + \hat{k} \partial_z e^{i(p_x x + p_y y + p_z z)}$

$= \hat{i} (i p_x) e^{i \dots} + \hat{j} (i p_y) e^{i \dots} + \hat{k} (i p_z) e^{i \dots}$

$\vec{\nabla} e^{i\vec{p} \cdot \vec{x}} = i\vec{p} e^{i\vec{p} \cdot \vec{x}}$ $\Rightarrow \vec{\nabla}^2 e^{i\vec{p} \cdot \vec{x}} = (i)^2 \vec{p} \cdot \vec{p} e^{i\vec{p} \cdot \vec{x}} = -\vec{p}^2 e^{i\vec{p} \cdot \vec{x}}$

$\therefore \left(\frac{\partial^2}{\partial t^2} + \vec{p}^2 + m^2 \right) \int \frac{d^3 \vec{p}}{(2\pi)^3} e^{i\vec{p} \cdot \vec{x}} \tilde{\phi}(\vec{p}) = 0$

$\int \frac{d^3 \vec{p}}{(2\pi)^3} e^{i\vec{p} \cdot \vec{x}} \underbrace{\left(\frac{\partial^2}{\partial t^2} + \vec{p}^2 + m^2 \right) \tilde{\phi}(\vec{p})}_{=0} = 0$ true for any $\vec{x}$

$\Rightarrow$ E.o.m: $\left(\frac{\partial^2}{\partial t^2} + (\vec{p}^2 + m^2) \right) \tilde{\phi}(\vec{p}) = 0$

$$\Rightarrow \text{E.O.m} : \left(\frac{\partial^2}{\partial t^2} + (\vec{p}^2 + m^2) \right) \tilde{\phi}(\vec{p}) = 0$$

• Einstein's dispersion relation, $E^2 = \vec{p}^2 + m^2$ (natural units)

• This equation is just a harmonic oscillator! $F = -kx = ma$

$$\frac{\partial^2}{\partial t^2} \tilde{\phi} = -(\underbrace{\vec{p}^2 + m^2}_{\substack{\equiv \omega_p^2 = \vec{p}^2 + m^2 \\ = E_p^2}}) \tilde{\phi}$$

$$\boxed{\begin{aligned} \ddot{x} &= -\omega^2 x \\ \ddot{x} + \omega^2 x &= 0 \end{aligned}} \quad \leftarrow a = -\frac{k}{m}x$$

$$\therefore \text{E.O.m. of } \tilde{\phi} : \left(\frac{\partial^2}{\partial t^2} + \omega_p^2 \right) \tilde{\phi}(\vec{p}) = 0$$

Quick Review on SHO (Simple Harmonic oscillator)

$$H = \underbrace{\frac{p^2}{2m}}_1 + \frac{1}{2} k q^2 \Rightarrow \text{H.M} = H = \frac{p^2}{2} + \frac{1}{2} \omega^2 q^2 \quad \leftarrow \text{Quantization} \quad \underbrace{k/m = \omega^2}$$

$$\hat{H} = \frac{\hat{p}^2}{2} + \frac{1}{2} \omega^2 \hat{x}^2, \quad [\hat{x}, \hat{p}] = i$$

We can solve this by changing variables $\hat{x}, \hat{p} \rightarrow \hat{a}, \hat{a}^+$

$$\hat{a} = \sqrt{\frac{\omega}{2}} \hat{x} + \frac{i}{\sqrt{2\omega}} \hat{p}, \quad \hat{a}^+ = \sqrt{\frac{\omega}{2}} \hat{x} - \frac{i}{\sqrt{2\omega}} \hat{p}$$

$$\text{also the inverse} \quad \hat{x} = \frac{1}{\sqrt{2\omega}} (\hat{a} + \hat{a}^+), \quad \hat{p} = -i \sqrt{\frac{\omega}{2}} (\hat{a} - \hat{a}^+)$$

We can find the commutator of $\hat{a}, \hat{a}^+$

$$\begin{aligned} [\hat{a}, \hat{a}^+] &= \left[\sqrt{\frac{\omega}{2}} \hat{x} + \frac{i}{\sqrt{2\omega}} \hat{p}, \sqrt{\frac{\omega}{2}} \hat{x} - \frac{i}{\sqrt{2\omega}} \hat{p} \right] \\ &= \frac{i}{\sqrt{2\omega}} \sqrt{\frac{\omega}{2}} [\hat{p}, \hat{x}] - \sqrt{\frac{\omega}{2}} \frac{i}{\sqrt{2\omega}} [\hat{x}, \hat{p}] = 1 \end{aligned}$$

$\frac{1}{2} \rightarrow -i$ $\frac{1}{2} \rightarrow i$

The Hamiltonian

$$\begin{aligned}\hat{H} &= \frac{\hat{p}^2}{2} + \frac{1}{2}\omega^2 \hat{x}^2 = \frac{1}{2} \left(-i\sqrt{\frac{\omega}{2}} (\hat{a} - \hat{a}^\dagger) \right)^2 + \frac{1}{2} \omega^2 \left(\frac{1}{\sqrt{2\omega}} (\hat{a} + \hat{a}^\dagger) \right)^2 \\ &= -\frac{1}{2} \cdot \frac{1}{2} \omega (\hat{a} - \hat{a}^\dagger)(\hat{a} - \hat{a}^\dagger) + \frac{1}{2} \omega^2 \cdot \frac{1}{2\omega} (\hat{a} + \hat{a}^\dagger)(\hat{a} + \hat{a}^\dagger) \\ &= \frac{\omega}{4} \left[-(\cancel{\hat{a}\hat{a}} - \hat{a}^\dagger\hat{a} - \hat{a}\hat{a}^\dagger + \cancel{\hat{a}^\dagger\hat{a}^\dagger}) + \cancel{\hat{a}\hat{a}} + \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger + \cancel{\hat{a}^\dagger\hat{a}^\dagger} \right] = \frac{\omega}{2} (\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger)\end{aligned}$$

We use $[\hat{a}, \hat{a}^\dagger] = 1 = \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} \Rightarrow \hat{a}\hat{a}^\dagger = 1 + \hat{a}^\dagger\hat{a}$

$$\begin{aligned}\therefore \hat{H} &= \frac{\omega}{2} (\hat{a}^\dagger\hat{a} + 1 + \hat{a}^\dagger\hat{a}) = \frac{\omega}{2} (2\hat{a}^\dagger\hat{a} + 1) \\ &= \omega (\hat{a}^\dagger\hat{a} + \frac{1}{2})\end{aligned}$$

We can start to interpret the meaning of $\hat{a}$ & $\hat{a}^\dagger$

$$[\hat{H}, \hat{a}^\dagger] = [\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2}), \hat{a}^\dagger] = [\omega\hat{a}^\dagger\hat{a}, \hat{a}^\dagger]$$

since $[\hat{A}\hat{B}, \hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B}$

$$\begin{aligned}&= \hat{A}\hat{B}\hat{C} - \hat{A}\hat{C}\hat{B} + \hat{A}\hat{C}\hat{B} - \hat{C}\hat{A}\hat{B} \\ &= \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}\end{aligned}$$

$$[\hat{H}, \hat{a}^\dagger] = \omega (\hat{a}^\dagger [\hat{a}, \hat{a}^\dagger] + [\hat{a}^\dagger, \hat{a}^\dagger] \hat{a})$$

$$[\hat{H}, \hat{a}^\dagger] = \omega \hat{a}^\dagger$$

Similarly, we can find

$$\begin{aligned}
 [\hat{H}, \hat{a}] &= [\omega(\hat{a}^\dagger \hat{a} + \frac{1}{2}), \hat{a}] = \omega[\hat{a}^\dagger \hat{a}, \hat{a}] \\
 &= \omega(\hat{a}^\dagger [\hat{a}, \hat{a}] + [\hat{a}^\dagger, \hat{a}] \hat{a}) \\
 &= -\omega \hat{a}
 \end{aligned}$$

$$\boxed{[\hat{H}, \hat{a}] = -\omega \hat{a}} \quad \boxed{[\hat{H}, \hat{a}^\dagger] = +\omega \hat{a}^\dagger}$$

We can interpret the meaning of $\hat{a}$ & $\hat{a}^\dagger$ as an operator that raise/lower energy.

• Suppose $|E\rangle$ is an eigenvector of $\hat{H}$

i.e., $\hat{H}|E\rangle = E|E\rangle$ (eigenvalue = E)

Consider the state $\hat{a}^\dagger |E\rangle$

We can check whether $\hat{a}^\dagger |E\rangle$ is also an eigenvector of $\hat{H}$?

↖ eigenvalue of $\hat{a}^\dagger |E\rangle$

$$\hat{H}(\hat{a}^\dagger |E\rangle) = \text{?} (\hat{a}^\dagger |E\rangle)$$

$$= \hat{H} \hat{a}^\dagger |E\rangle$$

$$= (\omega \hat{a}^\dagger + \hat{a}^\dagger \hat{H}) |E\rangle$$

$$\begin{aligned}
 & \left. \begin{aligned} [\hat{H}, \hat{a}^\dagger] &= \omega \hat{a}^\dagger \\ \hat{H} \hat{a}^\dagger &= \omega \hat{a}^\dagger + \hat{a}^\dagger \hat{H} \end{aligned} \right\}
 \end{aligned}$$

$$= \omega \hat{a}^\dagger |E\rangle + \hat{a}^\dagger (\hat{H} |E\rangle)$$

$$= (E + \omega) \hat{a}^\dagger |E\rangle$$

$$\boxed{\hat{a}^\dagger |E\rangle \propto |E + \omega\rangle}$$

Similarly $\hat{H}(\hat{a}|E\rangle) = \hat{H}\hat{a}|E\rangle$, $[\hat{H}, \hat{a}] = -\omega\hat{a}$

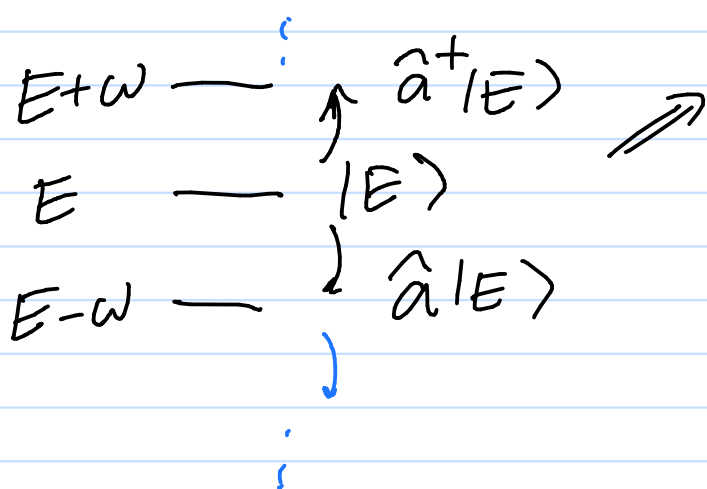
$$= (\hat{a}\hat{H} - \omega\hat{a})|E\rangle$$

$\hat{H}|E\rangle = E|E\rangle$

$$= (\hat{a}E - \omega\hat{a})|E\rangle$$

$$= \underbrace{(E - \omega)}_{\text{eigenvalue of } \hat{a}} \underbrace{\hat{a}|E\rangle}$$

$\therefore \hat{a}|E\rangle \propto |E - \omega\rangle$



except that we cannot go down infinitely

$\Rightarrow$ there must be the lowest energy state
ground state

$\hat{a}|0\rangle = 0 \quad \equiv |0\rangle$

$\nearrow$ ground state

$\nwarrow$ zero vector
null vector

The energy of the ground state

$$\hat{H}|0\rangle = \omega(\hat{a}^+\hat{a} + \frac{1}{2})|0\rangle$$

$$= \omega(\cancel{\hat{a}^+\hat{a}}|0\rangle + \frac{1}{2}|0\rangle)$$

$$= \frac{\omega}{2}|0\rangle$$

The energy of $|0\rangle$ is $\frac{\omega}{2}$

∴ Any energy state of $\hat{H}$ can be constructed from $|0\rangle$

$$\begin{aligned} & \text{--- } |E_n\rangle & |E_n\rangle \propto (a^\dagger)^n |0\rangle \\ & \vdots \\ & \text{--- } |E_2\rangle = (a^\dagger)^2 |0\rangle \\ & \text{--- } |E_1\rangle = a^\dagger |0\rangle \\ & \text{--- } |0\rangle \end{aligned}$$

Let's do this in field theory version!

Since we need to do Fourier transform $\rightarrow$ SHO
we can do Fourier transform + ladder operator.

$$\hat{\phi}(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left[\hat{a}_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + \hat{a}_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}} \right]$$

$$\hat{\pi}(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} (-i) \sqrt{\frac{\omega_p}{2}} \left[\hat{a}_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} - \hat{a}_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}} \right]$$

We need to find $[\hat{a}_{\vec{p}}, \hat{a}_{\vec{q}}^\dagger] = ?$

We will claim that

$$\begin{aligned} [\hat{\phi}(\vec{x}), \hat{\phi}(\vec{y})] &= 0, & [\hat{\pi}(\vec{x}), \hat{\pi}(\vec{y})] &= 0 \\ [\hat{\phi}(\vec{x}), \hat{\pi}(\vec{y})] &= i\delta^{(3)}(\vec{x} - \vec{y}) \end{aligned}$$



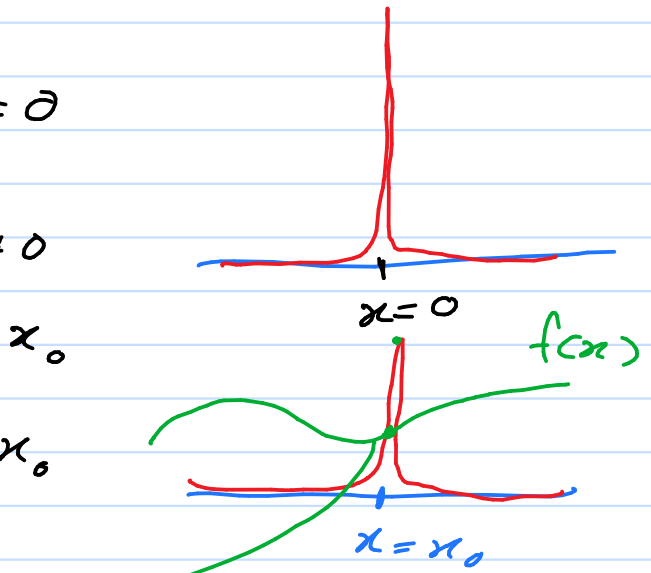
$$\begin{aligned} [\hat{a}_{\vec{p}}, \hat{a}_{\vec{q}}] &= 0, & [\hat{a}_{\vec{p}}^\dagger, \hat{a}_{\vec{q}}^\dagger] &= 0 \\ [\hat{a}_{\vec{p}}, \hat{a}_{\vec{q}}^\dagger] &= (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \end{aligned}$$

Dirac delta function

$$\delta(x) = \begin{cases} \infty, & x=0 \\ 0, & x \neq 0 \end{cases}$$

$$\delta(x-x_0) = \begin{cases} \infty, & x=x_0 \\ 0, & x \neq x_0 \end{cases}$$

$$f(x_0) = \int_{-\infty}^{\infty} f(x) \delta(x-x_0) dx$$



$$f(x) = \int_{-\infty}^{\infty} f(x_0) \delta(x_0-x) dx$$

Fourier transform of delta function

$$\delta(x-x_0) = \int_{-\infty}^{\infty} \frac{dk}{(2\pi)} e^{ik \cdot (x-x_0)}$$

Sum of all possible waves

Fourier transform

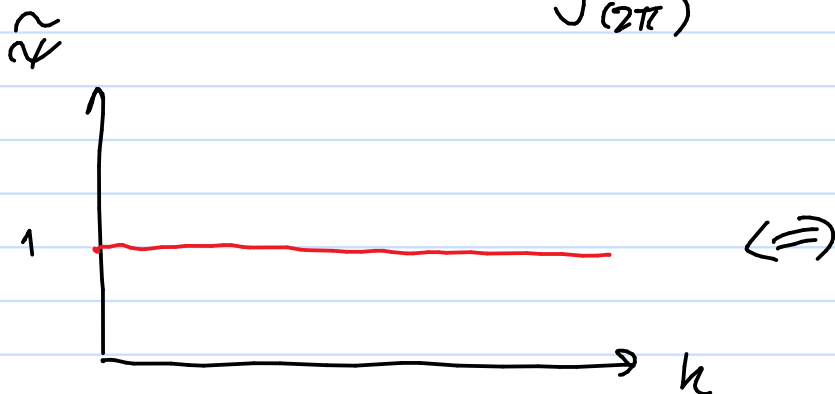
$$\int_{-\infty}^{\infty} e^{-ik \cdot (x-x_0)} \delta(x-x_0) dx = f(x_0) = 1$$

$\underbrace{\hspace{10em}}_{\psi(x)} \quad \underbrace{\hspace{10em}}_{\tilde{\psi}(k)}$

$$\psi(x) = \int_{-\infty}^{\infty} \frac{dk}{(2\pi)} \tilde{\psi}(k) e^{ik \cdot (x-x_0)}$$

Inverse transformation

$$\delta(x-x_0) = \int_{-\infty}^{\infty} \frac{dk}{(2\pi)} e^{ik \cdot (x-x_0)}$$



$\Leftrightarrow$



$$\delta(x-x_0) = \int_{-\infty}^{\infty} \frac{dk}{(2\pi)} e^{ik \cdot (x-x_0)} \Rightarrow \delta^{(3)}(\vec{x}-\vec{x}_0) = \int_{-\infty}^{\infty} \frac{d^3k}{(2\pi)^3} e^{i\vec{k} \cdot (\vec{x}-\vec{x}_0)}$$

Now let's check the claim

$$\begin{aligned} [\hat{\phi}(\vec{x}), \hat{\phi}(\vec{y})] &= 0, \quad [\hat{\pi}(\vec{x}), \hat{\pi}(\vec{y})] = 0 \\ [\hat{\phi}(\vec{x}), \hat{\pi}(\vec{y})] &= i \delta^{(3)}(\vec{x}-\vec{y}) \end{aligned} \Rightarrow \begin{aligned} [\hat{a}_{\vec{p}}, \hat{a}_{\vec{q}}] &= 0, \quad [\hat{a}_{\vec{p}}^{\dagger}, \hat{a}_{\vec{q}}^{\dagger}] = 0 \\ [\hat{a}_{\vec{p}}, \hat{a}_{\vec{q}}^{\dagger}] &= (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{q}) \end{aligned}$$

Check

$$\begin{aligned} [\hat{\phi}(\vec{x}), \hat{\pi}(\vec{y})] &= \left[\int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} (\hat{a}_{\vec{p}} e^{i\vec{p} \cdot \vec{x}} + \hat{a}_{\vec{p}}^{\dagger} e^{-i\vec{p} \cdot \vec{x}}), \int \frac{d^3q}{(2\pi)^3} (-i) \sqrt{\frac{\omega_{\vec{q}}}{2}} (\hat{a}_{\vec{q}} e^{i\vec{q} \cdot \vec{y}} - \hat{a}_{\vec{q}}^{\dagger} e^{-i\vec{q} \cdot \vec{y}}) \right] \\ &= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} (-i) \sqrt{\frac{\omega_{\vec{q}}}{2}} \times \end{aligned}$$

$$\left(-[\hat{a}_{\vec{p}}, \hat{a}_{\vec{q}}^{\dagger}] e^{i\vec{p} \cdot \vec{x} - i\vec{q} \cdot \vec{y}} + [\hat{a}_{\vec{p}}^{\dagger}, \hat{a}_{\vec{q}}] e^{-i\vec{p} \cdot \vec{x} + i\vec{q} \cdot \vec{y}} \right)$$

$$\int_{-\infty}^{\infty} f(q) \delta(p-q) dq = f(p) \quad \begin{array}{l} \text{remove } \int dq \\ \text{remove } \delta \\ \text{replace } q \rightarrow p \end{array}$$

$$[\hat{a}_{\vec{p}}, \hat{a}_{\vec{q}}^{\dagger}] = (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{q})$$

$$\begin{aligned} &= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} (-i) \sqrt{\frac{\omega_{\vec{q}}}{2}} \left(\cancel{(2\pi)^3 \delta^{(3)}(\vec{p}-\vec{q})} e^{i\vec{p} \cdot \vec{x} - i\vec{q} \cdot \vec{y}} \right. \\ &\quad \left. + (2\pi)^3 \delta^{(3)}(\vec{q}-\vec{p}) e^{-i\vec{p} \cdot \vec{x} + i\vec{q} \cdot \vec{y}} \right) \end{aligned}$$

$$= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2} \sqrt{\frac{\omega_{\vec{p}}}{\omega_{\vec{p}}}} \frac{(2\pi)^3}{(2\pi)^3} \left(e^{i\vec{p} \cdot \vec{x} - i\vec{p} \cdot \vec{y}} + e^{-i\vec{p} \cdot \vec{x} + i\vec{p} \cdot \vec{y}} \right)$$

$$= \frac{i}{2} \left(\int \frac{d^3p}{(2\pi)^3} e^{i\vec{p} \cdot (\vec{x}-\vec{y})} + \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p} \cdot (\vec{y}-\vec{x})} \right) = i \delta^{(3)}(\vec{x}-\vec{y})$$

consistent!

Now we know that

$$[\hat{a}_{\vec{p}}, \hat{a}_{\vec{q}}] = 0, \quad [\hat{a}_{\vec{p}}^{\dagger}, \hat{a}_{\vec{q}}^{\dagger}] = 0$$

$$[\hat{a}_{\vec{p}}, \hat{a}_{\vec{q}}^{\dagger}] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q})$$

Next step $\Rightarrow$ find $\hat{H}$ in terms of $\hat{a}$ & $\hat{a}^{\dagger}$

Recall that (for $\mathcal{L} = \frac{1}{2} \partial_{\mu} \phi \partial^{\mu} \phi - \frac{1}{2} m^2 \phi^2$)

$$H = \int d^3x \mathcal{H} = \int d^3x \left(\frac{1}{2} \pi^2(x) + \frac{1}{2} (\vec{\nabla} \phi(x))^2 + \frac{m^2}{2} \phi^2(x) \right)$$

$$\hat{H} = \frac{1}{2} \int d^3x \left(\underbrace{\pi^2(x)}_{\hat{\pi}(x)\hat{\pi}(x)} + \underbrace{(\vec{\nabla} \hat{\phi}(x))^2}_{\vec{\nabla} \hat{\phi} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (\hat{a}_{\vec{p}} \vec{e}^{i\vec{p}\cdot\vec{x}} + \dots)} + m^2 \hat{\phi}^2(x) \right)$$

$$\hat{H} = \frac{1}{2} \int d^3x \left(\int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (\hat{a}_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} - \hat{a}_{\vec{p}}^{\dagger} e^{-i\vec{p}\cdot\vec{x}}) \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2\omega_q}} (\hat{a}_{\vec{q}} e^{i\vec{q}\cdot\vec{x}} - \hat{a}_{\vec{q}}^{\dagger} e^{-i\vec{q}\cdot\vec{x}}) \right.$$

$$+ \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (\hat{a}_{\vec{p}} \underbrace{\vec{\nabla} e^{i\vec{p}\cdot\vec{x}}}_{i\vec{p} e^{i\vec{p}\cdot\vec{x}}} + \hat{a}_{\vec{p}}^{\dagger} \underbrace{\vec{\nabla} e^{-i\vec{p}\cdot\vec{x}}}_{-i\vec{p} e^{-i\vec{p}\cdot\vec{x}}}) \cdot \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2\omega_q}} (\hat{a}_{\vec{q}} \underbrace{\vec{\nabla} e^{i\vec{q}\cdot\vec{x}}}_{i\vec{q} e^{i\vec{q}\cdot\vec{x}}} + \hat{a}_{\vec{q}}^{\dagger} \underbrace{\vec{\nabla} e^{-i\vec{q}\cdot\vec{x}}}_{-i\vec{q} e^{-i\vec{q}\cdot\vec{x}}})$$

$$+ m^2 \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (\hat{a}_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + \hat{a}_{\vec{p}}^{\dagger} e^{-i\vec{p}\cdot\vec{x}}) \cdot \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2\omega_q}} (\hat{a}_{\vec{q}} e^{i\vec{q}\cdot\vec{x}} + \hat{a}_{\vec{q}}^{\dagger} e^{-i\vec{q}\cdot\vec{x}}) \Bigg)$$

$$= \frac{1}{2} \int d^3x \left(\int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} \left(\underbrace{(-2) \frac{\omega_p \omega_q}{2}}_{(2)} (\hat{a}_{\vec{p}} \hat{a}_{\vec{q}} e^{i(\vec{p}+\vec{q})\cdot\vec{x}})_{(2)} - \underbrace{\hat{a}_{\vec{p}}^{\dagger} \hat{a}_{\vec{q}} e^{i(\vec{q}-\vec{p})\cdot\vec{x}}}_{(1)} - \underbrace{\hat{a}_{\vec{p}} \hat{a}_{\vec{q}}^{\dagger} e^{i(\vec{p}-\vec{q})\cdot\vec{x}}}_{(1)} + \underbrace{\hat{a}_{\vec{p}}^{\dagger} \hat{a}_{\vec{q}}^{\dagger} e^{-i(\vec{p}+\vec{q})\cdot\vec{x}}}_{(2)} \right. \right.$$

$$+ \frac{1}{2\omega_p \omega_q} \left(\underbrace{\vec{p} \cdot \vec{q} \hat{a}_{\vec{p}} \hat{a}_{\vec{q}} e^{i(\vec{p}+\vec{q})\cdot\vec{x}}}_{(2)} - \underbrace{\vec{p} \cdot \vec{q} \hat{a}_{\vec{p}}^{\dagger} \hat{a}_{\vec{q}} e^{i(\vec{q}-\vec{p})\cdot\vec{x}}}_{(1)} - \underbrace{\vec{p} \cdot \vec{q} \hat{a}_{\vec{p}} \hat{a}_{\vec{q}}^{\dagger} e^{i(\vec{p}-\vec{q})\cdot\vec{x}}}_{(1)} + \underbrace{\vec{p} \cdot \vec{q} \hat{a}_{\vec{p}}^{\dagger} \hat{a}_{\vec{q}}^{\dagger} e^{-i(\vec{p}+\vec{q})\cdot\vec{x}}}_{(2)} \right)$$

$$+ \frac{m^2}{2\omega_p \omega_q} \left(\underbrace{\hat{a}_{\vec{p}} \hat{a}_{\vec{q}} e^{i(\vec{p}+\vec{q})\cdot\vec{x}}}_{(2)} + \underbrace{\hat{a}_{\vec{p}}^{\dagger} \hat{a}_{\vec{q}} e^{i(\vec{q}-\vec{p})\cdot\vec{x}}}_{(1)} + \underbrace{\hat{a}_{\vec{p}} \hat{a}_{\vec{q}}^{\dagger} e^{i(\vec{p}-\vec{q})\cdot\vec{x}}}_{(1)} + \underbrace{\hat{a}_{\vec{p}}^{\dagger} \hat{a}_{\vec{q}}^{\dagger} e^{-i(\vec{p}+\vec{q})\cdot\vec{x}}}_{(2)} \right) \Bigg)$$

Consider $\int \frac{d^3x}{(2\pi)^3} e^{i\vec{x}\cdot(\vec{p}-\vec{q})} = \delta^{(3)}(\vec{p}-\vec{q})$ (1), $\int \frac{d^3x}{(2\pi)^3} e^{i\vec{x}\cdot(\vec{p}+\vec{q})} = \delta^{(3)}(\vec{p}+\vec{q})$ (2)

$\int \frac{d^3x}{(2\pi)^3} e^{i\vec{x}\cdot(\vec{q}-\vec{p})} = \delta^{(3)}(\vec{q}-\vec{p})$, $\int \frac{d^3x}{(2\pi)^3} e^{-i\vec{x}\cdot(\vec{p}+\vec{q})} = \delta^{(3)}(-\vec{p}-\vec{q})$

$$= \frac{1}{2} \int d^3p \left(\int \frac{d^3q}{(2\pi)^3} \left(\frac{-1}{2} \frac{\omega_p \omega_q}{\omega_p \omega_q} (\hat{a}_{\vec{p}} \hat{a}_{\vec{q}} \delta^{(3)}(\vec{p}+\vec{q}) - \hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}} \delta^{(3)}(\vec{p}-\vec{q}) - \hat{a}_{\vec{p}} \hat{a}_{\vec{q}}^+ \delta^{(3)}(\vec{p}-\vec{q}) + \hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}}^+ \delta^{(3)}(\vec{p}+\vec{q}) \right) \right. \\ \left. + \frac{1}{2\sqrt{\omega_p \omega_q}} \left(\vec{p} \cdot \vec{q} \hat{a}_{\vec{p}} \hat{a}_{\vec{q}} \delta^{(3)}(\vec{p}+\vec{q}) - \vec{p} \cdot \vec{q} \hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}} \delta^{(3)}(\vec{p}-\vec{q}) - \vec{p} \cdot \vec{q} \hat{a}_{\vec{p}} \hat{a}_{\vec{q}}^+ \delta^{(3)}(\vec{p}-\vec{q}) + \vec{p} \cdot \vec{q} \hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}}^+ \delta^{(3)}(\vec{p}+\vec{q}) \right) \right. \\ \left. + \frac{m^2}{2\sqrt{\omega_p \omega_q}} (\hat{a}_{\vec{p}} \hat{a}_{\vec{q}} \delta^{(3)}(\vec{p}+\vec{q}) + \hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}} \delta^{(3)}(\vec{p}-\vec{q}) + \hat{a}_{\vec{p}} \hat{a}_{\vec{q}}^+ \delta^{(3)}(\vec{p}-\vec{q}) + \hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}}^+ \delta^{(3)}(\vec{p}+\vec{q})) \right)$$

$$\int d^3q \delta^{(3)}(\vec{p}-\vec{q}) f(\vec{q}) = f(\vec{p}), \quad \int d^3q \delta^{(3)}(\vec{p}+\vec{q}) f(\vec{q}) = f(-\vec{p}) \quad \left| \begin{array}{l} \omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2} \\ \omega_{-\vec{p}} = \omega_{\vec{p}} \end{array} \right.$$

$$\hat{H} = \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \left(-\frac{\omega_p}{2} (\hat{a}_{\vec{p}} \hat{a}_{-\vec{p}} - \hat{a}_{\vec{p}}^+ \hat{a}_{-\vec{p}} - \hat{a}_{\vec{p}} \hat{a}_{-\vec{p}}^+ + \hat{a}_{\vec{p}}^+ \hat{a}_{-\vec{p}}^+) \right. \\ \left. - \frac{1}{2\omega_p} (-\vec{p}^2 \hat{a}_{\vec{p}} \hat{a}_{-\vec{p}} - \vec{p}^2 \hat{a}_{\vec{p}}^+ \hat{a}_{-\vec{p}} - \vec{p}^2 \hat{a}_{\vec{p}} \hat{a}_{-\vec{p}}^+ - \vec{p}^2 \hat{a}_{\vec{p}}^+ \hat{a}_{-\vec{p}}^+) \right. \\ \left. + \frac{m^2}{2\omega_p} (\hat{a}_{\vec{p}} \hat{a}_{-\vec{p}} + \hat{a}_{\vec{p}}^+ \hat{a}_{-\vec{p}} + \hat{a}_{\vec{p}} \hat{a}_{-\vec{p}}^+ + \hat{a}_{\vec{p}}^+ \hat{a}_{-\vec{p}}^+) \right) \quad \omega_p^2 = \vec{p}^2 + m^2$$

$$= \frac{1}{4} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\omega_p} \left[(-\cancel{\omega_p^2} + \vec{p}^2 + m^2) \hat{a}_{\vec{p}} \hat{a}_{-\vec{p}} + (\omega_p^2 + \vec{p}^2 + m^2) \hat{a}_{\vec{p}}^+ \hat{a}_{-\vec{p}} \right. \\ \left. + (\omega_p^2 + \vec{p}^2 + m^2) \hat{a}_{\vec{p}} \hat{a}_{-\vec{p}}^+ + (-\cancel{\omega_p^2} + \vec{p}^2 + m^2) \hat{a}_{\vec{p}}^+ \hat{a}_{-\vec{p}}^+ \right]$$

$$= \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \omega_p \left[\hat{a}_{\vec{p}}^+ \hat{a}_{\vec{p}} + \hat{a}_{\vec{p}} \hat{a}_{\vec{p}}^+ \right]$$

Comparing to QHO, $\hat{H} = \frac{\omega}{2} (\hat{a}^+ \hat{a} + \hat{a} \hat{a}^+)$ $\hat{a} \hat{a}^+ - \hat{a}^+ \hat{a} = 1$
 ∞ d.o.f. $\downarrow [\hat{a}, \hat{a}^+] = 1$

$$\hat{H} = \frac{\omega}{2} (\hat{a}^+ \hat{a} + 1 + \hat{a} \hat{a}^+)$$

$$\hat{H} = \omega \left(\hat{a}^+ \hat{a} + \frac{1}{2} \right) \quad \hat{H} |0\rangle = \omega \hat{a}^+ \hat{a} |0\rangle + \frac{\omega}{2} |0\rangle$$

$$\hat{H} |0\rangle = \left(\frac{\omega}{2} \right) |0\rangle \quad \text{Vacuum energy}$$

We could do the same here

$$\hat{H} = \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \omega_p \left[\hat{a}_p^\dagger \hat{a}_p + \hat{a}_p \hat{a}_p^\dagger \right]$$

$$[\hat{a}_p, \hat{a}_q^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q})$$

$$\hat{a}_p \hat{a}_p^\dagger - \hat{a}_p^\dagger \hat{a}_p = (2\pi)^3 \delta^{(3)}(0)$$

$$\hat{H} = \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \omega_p \left[\hat{a}_p^\dagger \hat{a}_p + (2\pi)^3 \delta^{(3)}(0) + \hat{a}_p^\dagger \hat{a}_p \right]$$

$$= \int \frac{d^3 p}{(2\pi)^3} \omega_p \hat{a}_p^\dagger \hat{a}_p + \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \omega_p (2\pi)^3 \delta^{(3)}(0) \rightarrow \infty$$

Suppose that there is $|0\rangle$, such that $\hat{a}_p |0\rangle = 0, \forall \vec{p}$

$$\hat{H} |0\rangle = \int \frac{d^3 p}{(2\pi)^3} \omega_p \hat{a}_p^\dagger \hat{a}_p |0\rangle + \underbrace{\left(\frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \omega_p (2\pi)^3 \delta^{(3)}(0) \right)}_{\text{energy of } |0\rangle} |0\rangle$$

$\therefore$ Vacuum state has $E = \infty$! $\Rightarrow$ This is a problem?

What about the $\hat{a}_p^\dagger |0\rangle$? what is its energy then?

We can take a deeper look into $\delta^{(3)}(0)$

$$\delta^{(3)}(\vec{p} - \vec{q}) = \int_{-\infty}^{\infty} d^3 \vec{x} e^{i\vec{x} \cdot (\vec{p} - \vec{q})} \Rightarrow \delta^{(3)}(0) = \int_{-\infty}^{\infty} d^3 \vec{x} = V_\infty$$

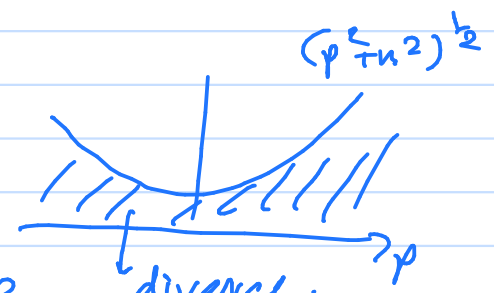
Large Volume limit, large distance $\Rightarrow$ low energy (momentum)

$$x \sim \frac{1}{p}, \quad [L] \propto \frac{1}{[E]} \quad (\text{IR}) \text{ "infrared divergence"}$$

$\uparrow \quad \downarrow$ $\lambda \rightarrow \max, E \rightarrow \min$

Another ∞ is in the integral (∞ d.o.f)

$$\hat{H} = \int \frac{d^3p}{(2\pi)^3} \omega_p \hat{a}_p^\dagger \hat{a}_p + \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \omega_p (2\pi)^3 \delta^{(3)}(0)$$

$\hookrightarrow \sim \int_{-\infty}^{\infty} d^3p (p^2 + m^2)^{1/2} \rightarrow \infty$

 ∞ with ∞ d.o.f $\Rightarrow$ large momentum limit

$\Rightarrow$ "Ultraviolet divergence" $E, p \rightarrow \infty$

(IR div $\Rightarrow E, p \rightarrow 0$)

We have to deal with these divergences a lot!

- The first and easy step toward ∞ free theory is simply subtraction.

IR divergence $\Rightarrow \delta^{(3)}(0) \rightarrow \infty$, energy of vacuum state diverges

We simply say that since we measure everything else respected with the vacuum

$\Rightarrow$ We set the energy level

$$\begin{array}{l} - |2\rangle \\ - \\ - |0\rangle \end{array} \left. \vphantom{\begin{array}{l} - |2\rangle \\ - \\ - |0\rangle \end{array}} \right\} \Delta E$$

of $|0\rangle$ to be zero (or finite value)

$\therefore$ We define $\hat{H}|0\rangle = 0$

normal ordering $\Rightarrow$ "set the lowest energy = 0"

- This is nothing new $\Rightarrow$ similar story in QM (QHG)

$\therefore$ We define ${}^x\hat{H}^x|0\rangle = 0$

normal ordering $\Rightarrow$ "set the lowest energy = 0"

• This is nothing new $\Rightarrow$ similar story in QM (QHO)

$\Rightarrow$ Ambiguity of ordering in QM

$$H_{cm} = \frac{p^2}{2} + \frac{\omega^2 q^2}{2} \Rightarrow \hat{H}_{QM} = \omega(\hat{a}^\dagger \hat{a} + \frac{1}{2})$$

Quantization

$$H_{cm} = \frac{1}{2}(\omega q - ip)(\omega q + ip) \Rightarrow \hat{H}_{QM} = \omega \hat{a}^\dagger \hat{a} \quad \text{no } \frac{1}{2}$$

factoring Quantization

★ We say the normal ordering operator $\Rightarrow$ order of Classical theory before quantization such that the

$${}^x\hat{H}^x|0\rangle = 0$$

$$\therefore {}^x\hat{H}^x = {}^x \int \frac{d^3p}{(2\pi)^3} \omega_p (\hat{a}_p^\dagger \hat{a}_p + \hat{a}_{-p} \hat{a}_{-p}^\dagger) {}^x$$

$$= \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \omega_p {}^x (\hat{a}_p^\dagger \hat{a}_p + \hat{a}_p \hat{a}_p^\dagger) {}^x$$

★★ normal ordering $\Rightarrow$ order in a way

$${}^x \hat{a}_p \hat{a}_p^\dagger {}^x = \hat{a}_p^\dagger \hat{a}_p$$

that $\hat{a}$ is always on the right

& $\hat{a}^\dagger$ is always on the left

$${}^x\hat{H}^x = \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \omega_p (\hat{a}_p^\dagger \hat{a}_p + \hat{a}_{-p}^\dagger \hat{a}_{-p}) = \int \frac{d^3p}{(2\pi)^3} \omega_p \hat{a}_p^\dagger \hat{a}_p$$

$${}^x\hat{H}^x|0\rangle = \int \frac{d^3p}{(2\pi)^3} \omega_p \hat{a}_p^\dagger \hat{a}_p |0\rangle = 0$$

What does the excited state mean in field theory?

$$\hat{a}_{\vec{p}}^+ |0\rangle$$

We can do the same as in QHO

$$\begin{aligned} [\hat{H}, \hat{a}_{\vec{p}}^+] &= \left[\int \frac{d^3\vec{q}}{(2\pi)^3} \omega_{\vec{q}} \hat{a}_{\vec{q}}^+ \hat{a}_{\vec{q}}, \hat{a}_{\vec{p}}^+ \right] \\ &= \int \frac{d^3\vec{q}}{(2\pi)^3} \omega_{\vec{q}} [\hat{a}_{\vec{q}}^+ \hat{a}_{\vec{q}}, \hat{a}_{\vec{p}}^+] \quad \begin{matrix} [\hat{A}\hat{B}, \hat{C}] \\ = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B} \end{matrix} \end{aligned}$$

$$= \int \frac{d^3\vec{q}}{(2\pi)^3} \omega_{\vec{q}} \left(\hat{a}_{\vec{q}}^+ [\hat{a}_{\vec{q}}, \hat{a}_{\vec{p}}^+] + [\hat{a}_{\vec{q}}^+, \hat{a}_{\vec{p}}^+] \hat{a}_{\vec{q}} \right)$$

$$[\hat{H}, \hat{a}_{\vec{p}}^+] = \int d^3\vec{q} \omega_{\vec{q}} \hat{a}_{\vec{q}}^+ \delta^{(3)}(\vec{q} - \vec{p})$$

$$(2\pi)^3 \delta^{(3)}(\vec{q} - \vec{p})$$

$$\vec{q} = \vec{p}$$

$$[\hat{H}, \hat{a}_{\vec{p}}^+] = \omega_{\vec{p}} \hat{a}_{\vec{p}}^+ = \sqrt{\vec{p}^2 + m^2} \hat{a}_{\vec{p}}^+$$

Similarly $[\hat{H}, \hat{a}_{\vec{p}}] = \dots = -\omega_{\vec{p}} \hat{a}_{\vec{p}} = -\sqrt{\vec{p}^2 + m^2} \hat{a}_{\vec{p}}$

$\therefore$ We can calculate energy of $\hat{a}_{\vec{p}}^+ |0\rangle$

$$\begin{aligned} \hat{H}(\hat{a}_{\vec{p}}^+ |0\rangle) &= \hat{H} \hat{a}_{\vec{p}}^+ |0\rangle = (\hat{H} \hat{a}_{\vec{p}}^+ - \hat{a}_{\vec{p}}^+ \hat{H} + \hat{a}_{\vec{p}}^+ \hat{H}) |0\rangle \\ &= ([\hat{H}, \hat{a}_{\vec{p}}^+] + \hat{a}_{\vec{p}}^+ \hat{H}) |0\rangle \quad \begin{matrix} [\hat{A}, \hat{B}] \\ \hat{A}\hat{B} - \hat{B}\hat{A} \end{matrix} \end{aligned}$$

$$\omega_{\vec{p}} \hat{a}_{\vec{p}}^+$$

$$\hat{H}|0\rangle = 0$$

$$= \omega_{\vec{p}} \hat{a}_{\vec{p}}^+ |0\rangle$$

$\hat{a}_{\vec{p}}^+ |0\rangle$ is eigenstate of $\hat{H}$
with eigenvalue = $\omega_{\vec{p}}$

$$\hat{H}(\hat{a}_{\vec{p}}^+ |0\rangle) = \omega_{\vec{p}} \hat{a}_{\vec{p}}^+ |0\rangle$$

$\hat{a}_{\vec{p}}^+ |0\rangle$ is eigenstate of $\hat{H}$ with eigenvalue = $\omega_{\vec{p}}$

What does $\hat{a}_{\vec{p}}^+ |0\rangle$ represent?

(relativistic)

$$\text{Its energy} = \omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2} = E_{\vec{p}} = \text{energy of 1 particle}$$

$\therefore$ We conclude that

$$\hat{a}_{\vec{p}}^+ |0\rangle = \text{a state of 1 particle.} \quad (\text{with momentum } \vec{p})$$

$$|0\rangle \xrightarrow{E=0} \hat{a}_{\vec{p}}^+ |0\rangle \xRightarrow{E=\sqrt{\vec{p}^2+m^2}} \hat{a}_{\vec{p}}^+ = \text{creation operator}$$

(raising)

Create a particle

no particle 1 particle $\hat{a}_{\vec{p}} = \text{annihilation operator}$

(lowering)

destroy a particle

Momentum operator

Let's check $\hat{a}_{\vec{p}}^+ |0\rangle$ is also the eigenstate of $\hat{P}$

Now, since we know $\hat{H}$ is conserved charge of time translation symmetry, we can use the conserved charge of spatial (x, y, z) translation for $\hat{P}$

$$\text{Recall that } p^i = \int d^3x \dot{\phi} \partial^i \phi \Rightarrow \vec{P} = \int d^3x \dot{\phi} \vec{\nabla} \phi$$

$$\hat{P} = - \int d^3x \dot{\phi} \vec{\nabla} \phi = - \int d^3x \hat{\pi} \vec{\nabla} \phi$$

$$\frac{\partial}{\partial x_{\mu}} = g^{\mu\nu} \frac{\partial}{\partial x^{\nu}}$$

$\downarrow$

$(-1) \quad \vec{\nabla}$

$$\hat{\vec{p}} = - \int d^3x \hat{\vec{\phi}} \vec{\nabla} \phi = - \int d^3x \pi \vec{\nabla} \phi$$

$$= - \int d^3x \int \frac{d^3p}{(2\pi)^3} (-i) \frac{\omega_p}{2} \left[\hat{a}_p e^{i\vec{p}\cdot\vec{x}} - \hat{a}_p^\dagger e^{-i\vec{p}\cdot\vec{x}} \right] \\ \times \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2\omega_q}} \left[i\vec{q} \hat{a}_q e^{i\vec{q}\cdot\vec{x}} + (-i\vec{q}) \hat{a}_q^\dagger e^{-i\vec{q}\cdot\vec{x}} \right]$$

$$= \frac{i}{2} \int \frac{d^3p d^3q}{(2\pi)^3 (2\pi)^3} \frac{\omega_p}{\omega_q} \int d^3x \left[i\vec{q} \hat{a}_p \hat{a}_q e^{i(\vec{p}+\vec{q})\cdot\vec{x}} \right. \\ \left. - i\vec{q} \hat{a}_p^\dagger \hat{a}_q e^{i(\vec{q}-\vec{p})\cdot\vec{x}} - i\vec{q} \hat{a}_p \hat{a}_q^\dagger e^{i(\vec{p}-\vec{q})\cdot\vec{x}} + i\vec{q} \hat{a}_p^\dagger \hat{a}_q^\dagger e^{-i(\vec{p}+\vec{q})\cdot\vec{x}} \right]$$

We use $\delta(\vec{p}-\vec{q}) = \int \frac{d^3x}{(2\pi)^3} e^{i(\vec{p}-\vec{q})\cdot\vec{x}}$, $\delta(\vec{p}+\vec{q}) = \int \frac{d^3x}{(2\pi)^3} e^{i(\vec{p}+\vec{q})\cdot\vec{x}}$

$$\hat{\vec{p}} = -\frac{1}{2} \int \frac{d^3p d^3q}{(2\pi)^3} \frac{\omega_p}{\omega_q} \left(\vec{q} \hat{a}_p \hat{a}_q \delta(\vec{p}+\vec{q}) - \vec{q} \hat{a}_p^\dagger \hat{a}_q \delta(\vec{q}-\vec{p}) \right. \\ \left. - \vec{q} \hat{a}_p \hat{a}_q^\dagger \delta(\vec{p}-\vec{q}) + \vec{q} \hat{a}_p^\dagger \hat{a}_q^\dagger \delta(\vec{p}+\vec{q}) \right)$$

$\omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2}$
 $\omega_{-\vec{p}} = \omega_{\vec{p}}$

$$= -\frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \vec{p} \left(\hat{a}_{\vec{p}} \hat{a}_{-\vec{p}} + \hat{a}_{\vec{p}}^\dagger \hat{a}_{\vec{p}} + \hat{a}_{\vec{p}} \hat{a}_{\vec{p}}^\dagger + \hat{a}_{\vec{p}}^\dagger \hat{a}_{-\vec{p}}^\dagger \right)$$

Consider $\int \frac{d^3p}{(2\pi)^3} \vec{p} \hat{a}_{\vec{p}} \hat{a}_{-\vec{p}} = \int \frac{d^3\vec{p}'}{(2\pi)^3} \vec{p}' \hat{a}_{\vec{p}'} \hat{a}_{-\vec{p}'} = \int \frac{d^3\vec{p}}{(2\pi)^3} (-\vec{p}) \hat{a}_{-\vec{p}} \hat{a}_{\vec{p}}$

$\vec{p} \rightarrow \vec{p}'$ $\vec{p}' \rightarrow -\vec{p}$ invariant

Consider $\int \frac{d^3 p}{(2\pi)^3} \vec{p} \hat{a}_{\vec{p}} \hat{a}_{-\vec{p}} = \int \frac{d^3 \vec{p}'}{(2\pi)^3} \vec{p}' \hat{a}_{\vec{p}'} \hat{a}_{-\vec{p}'} = \int \frac{d^3 \vec{p}}{(2\pi)^3} (-\vec{p}) \hat{a}_{-\vec{p}} \hat{a}_{\vec{p}}$

Annotations: $\vec{p} \rightarrow \vec{p}'$, $\vec{p}' \rightarrow -\vec{p}$ (invariant), commute $[\hat{a}_{\vec{p}}, \hat{a}_{\vec{q}}] = 0$

$$\therefore \int \frac{d^3 p}{(2\pi)^3} \vec{p} \hat{a}_{\vec{p}} \hat{a}_{-\vec{p}} = - \int \frac{d^3 p}{(2\pi)^3} \vec{p} \hat{a}_{\vec{p}} \hat{a}_{-\vec{p}} = 0$$

Similarly $\int \frac{d^3 p}{(2\pi)^3} \vec{p} \hat{a}_{\vec{p}}^+ \hat{a}_{-\vec{p}}^+ = 0$

$$\therefore \hat{p} = \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \vec{p} (\cancel{\hat{a}_{\vec{p}} \hat{a}_{-\vec{p}}} + \hat{a}_{\vec{p}}^+ \hat{a}_{\vec{p}} + \hat{a}_{\vec{p}} \hat{a}_{\vec{p}}^+ + \cancel{\hat{a}_{\vec{p}}^+ \hat{a}_{-\vec{p}}^+})$$

$$\hat{p} = \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \vec{p} (\hat{a}_{\vec{p}}^+ \hat{a}_{\vec{p}} + \hat{a}_{\vec{p}} \hat{a}_{\vec{p}}^+)$$

Similar to $\hat{H}$
contains $\delta^{(3)}(\vec{0})$

$$\hat{p}_x = \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \vec{p}_x (\hat{a}_{\vec{p}}^+ \hat{a}_{\vec{p}} + \hat{a}_{\vec{p}} \hat{a}_{\vec{p}}^+)$$

$$\boxed{\hat{p}_x = \int \frac{d^3 p}{(2\pi)^3} \vec{p}_x \hat{a}_{\vec{p}}^+ \hat{a}_{\vec{p}}} \quad \text{Momentum operator.}$$

Let's check $\hat{p} (\hat{a}_{\vec{p}}^+ |0\rangle) = \left(\int \frac{d^3 q}{(2\pi)^3} \vec{q} \hat{a}_{\vec{q}}^+ \hat{a}_{\vec{q}} \right) \hat{a}_{\vec{p}}^+ |0\rangle$

$$= \int \frac{d^3 q}{(2\pi)^3} \vec{q} \hat{a}_{\vec{q}}^+ \hat{a}_{\vec{q}} \hat{a}_{\vec{p}}^+ |0\rangle$$

$$[\hat{a}_{\vec{q}}, \hat{a}_{\vec{p}}^+] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q})$$

$$\hat{a}_{\vec{q}} \hat{a}_{\vec{p}}^+ = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) + \hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}}$$

$$\hat{p} (\hat{a}_{\vec{p}}^+ |0\rangle) = \int \frac{d^3 q}{(2\pi)^3} \vec{q} \hat{a}_{\vec{q}}^+ \left((2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) + \hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}} \right) |0\rangle$$

$$\begin{aligned}
 \hat{p}(\hat{a}_{\vec{p}}^+ | 0\rangle) &= \int \frac{d^3 q}{(2\pi)^3} \vec{q} \hat{a}_{\vec{q}}^+ \left((2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) + \cancel{\hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}}} \right) | 0\rangle \\
 &= \int d^3 q \vec{q} \hat{a}_{\vec{q}}^+ \delta^{(3)}(\vec{p} - \vec{q}) | 0\rangle \\
 &\quad \quad \quad \vec{q} = \vec{p} \\
 &= \vec{p} \hat{a}_{\vec{p}}^+ | 0\rangle \\
 &\quad \quad \quad \uparrow \text{eigenvalue of } \hat{a}_{\vec{p}}^+ | 0\rangle
 \end{aligned}$$

$\therefore$ The conclusion is that

$\hat{a}_{\vec{p}}^+ | 0\rangle$ is a state represent a particle

$$\text{with } E = \omega_p = \sqrt{\vec{p}^2 + m^2}$$

$$\& \vec{p} = \vec{p}$$

It makes sense to define

$$\hat{a}_{\vec{p}}^+ | 0\rangle \propto |\vec{p}\rangle$$

What about $\hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}}^+ | 0\rangle = ? \Rightarrow 2 \text{ particles with momentum } \vec{p} \& \vec{q}$

$$\hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}}^+ | 0\rangle \propto |\vec{p}, \vec{q}\rangle$$

$$\hat{a}_{\vec{p}_n}^+ \dots \hat{a}_{\vec{p}_1}^+ | 0\rangle \propto |\vec{p}_1, \dots, \vec{p}_n\rangle$$

Since we are considering boson (scalar field) the order of $\hat{a}^+$'s doesn't matter.

$$\text{Fermion : } \psi(\vec{r}_1, \vec{r}_2) = -\psi(\vec{r}_2, \vec{r}_1) \Rightarrow |\psi\rangle = -|\psi\rangle$$

$$\text{Boson : } \psi(\vec{r}_1, \vec{r}_2) = +\psi(\vec{r}_2, \vec{r}_1) \Rightarrow |\psi\rangle = |\psi\rangle$$

$$\Rightarrow |\vec{p}, \vec{q}\rangle = |\vec{q}, \vec{p}\rangle \text{ for boson}$$

$$\text{Consistent with } \hat{a}^+ ? \quad \hat{a}_{\vec{p}}^+ \hat{a}_{\vec{q}}^+ | 0\rangle = \hat{a}_{\vec{q}}^+ \hat{a}_{\vec{p}}^+ | 0\rangle \checkmark$$

Notice that if we keep applying $\hat{a}_{\vec{p}}^{\dagger} \Rightarrow$ states with arbitrary number of particles

- $|0\rangle \Rightarrow$ Vacuum = 1 state $\vec{p}_i \in (-\infty, \infty)$
- $\hat{a}_{\vec{p}}^{\dagger}|0\rangle \Rightarrow$ 1 particle with momentum $\vec{p} = \infty$ states
- $\hat{a}_{\vec{p}}^{\dagger}\hat{a}_{\vec{q}}^{\dagger}|0\rangle \Rightarrow$ 2 particles with momenta $\vec{p}$ & $\vec{q} = \infty \times \infty$ states
- $\vdots$
- $\vdots$

$\therefore$ The vectors space of states in quantum field theory (Hilbert space) has basis

$$\{|0\rangle, \hat{a}_{\vec{p}}^{\dagger}|0\rangle, \hat{a}_{\vec{p}_1}^{\dagger}\hat{a}_{\vec{p}_2}^{\dagger}|0\rangle, \dots\}$$

$\Rightarrow$ Fock space = infinite sum of n-particle Hilbert space
 $\neq$ Vector space in QM. co vector space

We define the number operator

$$\hat{N} = \int \frac{d^3p}{(2\pi)^3} \hat{a}_{\vec{p}}^{\dagger} \hat{a}_{\vec{p}}$$

$$\begin{aligned} \hat{N}(\hat{a}_{\vec{p}}^{\dagger}|0\rangle) &= \int \frac{d^3q}{(2\pi)^3} \hat{a}_{\vec{q}}^{\dagger} \hat{a}_{\vec{q}} \hat{a}_{\vec{p}}^{\dagger}|0\rangle \\ &= \int \frac{d^3q}{(2\pi)^3} \hat{a}_{\vec{q}}^{\dagger} \underbrace{\delta^{(3)}(\vec{p}-\vec{q})}_{\vec{q}=\vec{p}} \hat{a}_{\vec{p}}^{\dagger}|0\rangle \\ &= \hat{a}_{\vec{p}}^{\dagger}|0\rangle \rightarrow \text{eigenvalue of } \hat{N} = 1 \end{aligned}$$

$\hat{a}_{\vec{q}}|0\rangle = 0$

$$\therefore \hat{N}(\hat{a}_{p_1}^\dagger \hat{a}_{p_2}^\dagger \dots \hat{a}_{p_n}^\dagger |0\rangle) = n(\hat{a}_{p_1}^\dagger \hat{a}_{p_2}^\dagger \dots \hat{a}_{p_n}^\dagger |0\rangle)$$

We can sketch this

$$\int d^3q \hat{a}_{\vec{q}}^\dagger \hat{a}_{\vec{q}} \underbrace{(\hat{a}_{p_1}^\dagger \dots \hat{a}_{p_n}^\dagger |0\rangle)}_n = n \text{ terms} + \cancel{\hat{a}_{\vec{q}} |0\rangle}^0$$

• In QM, we know that $\hat{N}$ is conserved

$$[\hat{N}, \hat{H}] = i \frac{d\hat{N}}{dt} = 0$$

• In QFT, the number of operator can be changed

: applying $\hat{a}^\dagger, \hat{a}$ to states

$\therefore$ In QFT, $[\hat{N}, \hat{H}] \neq 0$ ($\hat{N}$ is not conserved)
interaction

Normalisation of $\hat{a}_{\vec{p}}^\dagger |0\rangle$

We have $\hat{a}_{\vec{p}}^\dagger |0\rangle \propto |\vec{p}\rangle$

Let's check what we have got for

$$\langle \vec{p} | \vec{q} \rangle = \langle 0 | \hat{a}_{\vec{p}} \hat{a}_{\vec{q}}^\dagger | 0 \rangle$$

$$= \langle 0 | ((2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) + \cancel{\hat{a}_{\vec{q}}^\dagger \hat{a}_{\vec{p}}}) | 0 \rangle$$

$$= (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \underbrace{\langle 0 | 0 \rangle}_1 \text{ set normalisation of } |0\rangle$$

$$= (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \rightarrow \text{This is not a good number since it is not invariant under Lorentz transformation}$$

The claim: The quantity $\delta^{(3)}(\vec{p}-\vec{q})$ is not invariant
under Lorentz transformation

but $2E_{\vec{p}}\delta^{(3)}(\vec{p}-\vec{q})$ is invariant.

$\therefore$ The normalisation is $\sqrt{2E_{\vec{p}}}$

$$\boxed{|\vec{p}\rangle = \sqrt{2E_{\vec{p}}} \hat{a}_{\vec{p}}^{\dagger} |0\rangle}$$

such that $\langle \vec{p} | \vec{q} \rangle = \sqrt{2E_{\vec{p}}} \sqrt{2E_{\vec{q}}} (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{q})$

$$= (2\pi)^3 \underbrace{2E_{\vec{p}}}_{\text{invariant}} \underbrace{\delta^{(3)}(\vec{p}-\vec{q})}_{\text{invariant under Lorentz transformation}}$$

invariant under
Lorentz transformation